

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T08:05:20

Source: data/json/noise_experiments/v2/noise8i_fixed_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_fixed/noise8i_fixed_memprimary_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_fixed
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

ROUND 1

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$3.5; payoffs Agent_1=\$5.5, Agent_2=\$5.5;
noise/visible: sent visible=\$2.34, received visible=\$7.02, returned visible=\$2.71
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$5; payoffs Agent_3=\$7, Agent_4=\$4;
noise/visible: sent visible=\$3.36, received visible=\$10.08, returned visible=\$4.89
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.5; payoffs Agent_5=\$6.5, Agent_6=\$4.5;
noise/visible: sent visible=\$3.03, received visible=\$9.09, returned visible=\$5.3
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$4.5; payoffs Agent_7=\$6.5, Agent_8=\$4.5;
noise/visible: sent visible=\$3.01, received visible=\$9.03, returned visible=\$4.55
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.5,
"Agent_2": 5.5,
"Agent_3": 7.0,
"Agent_4": 4.0,
"Agent_5": 6.5,
"Agent_6": 4.5,
"Agent_7": 6.5,
"Agent_8": 4.5

}

Visible balances after round: {

"Agent_1": 4.71,
"Agent_2": 3.5199999999999996,
"Agent_3": 6.89,
"Agent_4": 5.08,
"Agent_5": 7.3,
"Agent_6": 4.59,
"Agent_7": 6.55,
"Agent_8": 4.5299999999999999

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:51:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send ****\$3****. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.


```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
 - Show some willingness to cooperate
 - Test the waters
 - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:51:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    }
  },
```

```

    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

Assistant output:

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return **\$3.50**:

- Sender ends with: $\$2.66 + \$3.50 = \$6.16$ (profit of $\$1.16$)
- I end with: $\$7.02 - \$3.50 = \$3.52$ (profit of $\$3.52$)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

Reasoning:

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is $\$2.34 \times 3$)
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep: $\$5 - \$2.34 + \text{returned} = \$2.66 + \text{returned}$
4. I will keep: $\$7.02 - \text{returned}$

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return $\$6.01 - \$2.66 = \$3.35$
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received = $0.468 \times \$7.02 = \3.29
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with $\$2.66 + \$3.50 = \$6.16$ (up \$1.16 from initial \$5), and I keep $\$7.02 - \$3.50 = \$3.52$
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:51:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
```

```

"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the receiver reciprocates. If I send \$3:

- The receiver gets \$9 (3 × \$3)
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```

'''json
{"send": 3}
'''

```

Reasoning:

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = (3 × sent) - returned

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
 - Show some trust
 - Create value through the multiplier
 - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

...

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 626,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:52:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",
```

```
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

Assistant output:

Looking at this situation:

****Analysis:****

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

****My reasoning:****

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with: $\$1.64 + \$5.00 = \$6.64$ (profit of \$1.64)
- I end with: $\$10.08 - \$5.00 = \$5.08$ (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

{'return': 5.00}

Reasoning:

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received $\$3.36 \times 3 = \10.08
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created: $\$5$ (sender's initial) + $\$10.08$ (what I received) = $\$15.08$
- Equal split would be $\$7.54$ each
- Sender currently has: $\$5 - \$3.36 = \$1.64$
- To give sender $\$7.54$, I'd need to return: $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of $\$10.08$? $\$6.77$

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have: $\$1.64 + \$5 = \$6.64$ (profit of \$1.64)
- I would keep: $\$10.08 - \$5 = \$5.08$ (profit of \$5.08)

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep \$6.72

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground-it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains \$1.64 while I gain \$5.08. If I go higher to \$6, they'd gain \$2.64 but I'd only gain \$4.08-ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating \$6.72 in surplus.

Dividing this equally means each of us should gain \$3.36 from that surplus. Since the sender started with \$5 and kept \$1.64, they'd end with \$6.64 total. That leaves me with \$5.08 after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the \$6.72 surplus split evenly. If I return \$3.36, the sender gets their \$1.64 kept plus \$3.36 back, totaling \$5-their original amount. I'd keep \$6.72, which is my \$3.36 share of the surplus plus the original \$3.36 sent. This creates an equal split where we each benefit \$3.36 from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the \$6.72 surplus equally. That would mean returning \$6.72 to give them \$8.36 total, while I keep \$6.72. Let me check if this math works-\$8.36 plus \$6.72 equals \$15.08, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a \$3.36 gain, which feels genuinely fair. The sender's gain also happens to equal what they ori

ginally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received (\$5.04), return the original amount (\$3.36), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 2102,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:52:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
 - If I send \$5, receiver gets \$15
 - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
 - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates

- ****Partial trust****: Send some amount (\$2-3) - tests the waters
- ****Zero trust****: Send nothing (\$0) - avoid risk but miss cooperative gains
- ****Tit-for-tat****: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:53:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with: $\$1.97 + \$4.50 = \$6.47$ (gain of \$1.47)
- I will end with: $\$9.09 - \$4.50 = \$4.59$ (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

Reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has: $\$5 - \$3.03 = \$1.97$
- Will end up with: $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with: $\$9.09 - (\text{what I return})$

Key considerations:

1. ****Fairness**:** The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. ****Equal split consideration**:** If we were to split the total money equally:
 - Total available: $\$5$ (sender's original) + $\$9.09$ (what I received) = $\$14.09$

- Equal split: \$7.045 each
- Sender currently has: \$1.97
- To give sender \$7.045: I'd need to return $\$7.045 - \$1.97 = \$5.075$

3. ****Proportional reciprocity****: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:

- 60.6% of \$9.09 = \$5.51

4. ****Strategic considerations****:

- This is a multi-round game with multiple agents
- Building trust could be beneficial for future rounds
- However, there's communication noise, so exact amounts may not be perceived correctly
- Agent names are hidden, so reputation effects are limited to behavioral patterns

5. ****Common reciprocity patterns****: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1332,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:53:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit

- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:53:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has \$5 - \$3.01 = \$1.99

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with: \$1.99 + \$4.50 = \$6.49 (gain of \$1.49)
- I end with: \$9.03 - \$4.50 = \$4.53 (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is \$3.01 × 3)
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. ****The sender showed significant trust****: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. ****Payoff analysis****:
 - If I return \$0: Sender gets \$5 - \$3.01 + \$0 = \$1.99, I get \$9.03
 - If I return half (\$4.52): Sender gets \$5 - \$3.01 + \$4.52 = \$6.51, I get \$4.51
 - If I return \$3.01: Sender gets \$5 - \$3.01 + \$3.01 = \$5.00, I get \$6.02
 - If I return \$4.50: Sender gets \$5 - \$3.01 + \$4.50 = \$6.49, I get \$4.53
 - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. ****Fairness considerations****:
 - The sender's initial endowment was \$5, and they sent \$3.01
 - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
 - If I want us to split the gains equally, I should consider that the total is now \$5 - \$3.01 + \$9.03 = \$11.02
 - Equal split would give us each about \$5.51
 - For sender to get \$5.51: they need \$5 - \$3.01 + return = \$5.51, so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1131,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

```
dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_2 sent $3; received $9; Agent_1 returned $4; payoffs Agent_2=$6, Agent_1=$5;
noise/visible: sent visible=$2.71, received visible=$8.13, returned visible=$4.04
  Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 sent $3.5; received $10.5; Agent_3 returned $5.25; payoffs Agent_4=$6.75,
Agent_3=$5.25; noise/visible: sent visible=$3.49, received visible=$10.47, returned visible=$5.65
  Action validation: {"returned": {"amount": 5.25, "clamped": false, "max": 10.5, "min": 0, "raw": 5.25}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $4.4; payoffs Agent_6=$6.4, Agent_5=$4.6;
noise/visible: sent visible=$2.92, received visible=$8.76, returned visible=$3.55
  Action validation: {"returned": {"amount": 4.4, "clamped": false, "max": 9.0, "min": 0, "raw": 4.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $3.4; payoffs Agent_8=$5.4, Agent_7=$5.6;
noise/visible: sent visible=$2.27, received visible=$6.81, returned visible=$3.78
  Action validation: {"returned": {"amount": 3.4, "clamped": false, "max": 9.0, "min": 0, "raw": 3.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 10.5,
"Agent_2": 11.5,
"Agent_3": 12.25,
"Agent_4": 10.75,
"Agent_5": 11.1,
"Agent_6": 10.9,
"Agent_7": 12.1,
"Agent_8": 9.9
```

Visible balances after round: {

```
"Agent_1": 8.84,
"Agent_2": 9.559999999999999,
"Agent_3": 12.11,
"Agent_4": 12.23,
```



```

"Agent_5": 11.66,
"Agent_6": 10.14,
"Agent_7": 9.96,
"Agent_8": 10.309999999999999
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:53:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return ****\$3.50****:

- Sender ends with: $\$2.66 + \$3.50 = \$6.16$ (profit of \$1.16)
- I end with: $\$7.02 - \$3.50 = \$3.52$ (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

reasoning:

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is $\$2.34 \times 3$)
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep: $\$5 - \$2.34 + \text{returned} = \$2.66 + \text{returned}$
4. I will keep: $\$7.02 - \text{returned}$

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02
2. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return $\$6.01 - \$2.66 = \$3.35$
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received = $0.468 \times \$7.02 = \3.29
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with $\$2.66 + \$3.50 = \$6.16$ (up \$1.16 from initial \$5), and I keep $\$7.02 - \$3.50 = \$3.52$
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send ****\$3.00**** (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1: \$9.0 - \$3.50 = \$5.50

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was \$2.71/\$9.0 = 30.1%
4. This might make them think I wasn't very generous, even though I actually returned about 39% (\$3.50/\$9.0)

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

Usage: {

```
"input_tokens": 872,  
"output_tokens": 670,  
"reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:54:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send ****\$3****. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
 - Show some willingness to cooperate
 - Test the waters
 - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is 2.34×3)
3. They returned \$3.50
4. My visible payoff was \$4.71, which means: $\$5 - \$3 \text{ (sent)} + \$3.50 \text{ (returned)} = \5.50 ... but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)

- I received \$8.13 (which is $\$2.71 \times 3$)
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

```
Usage: {
  "input_tokens": 806,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:54:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

****My reasoning:****

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with: $\$1.64 + \$5.00 = \$6.64$ (profit of \$1.64)
- I end with: $\$10.08 - \$5.00 = \$5.08$ (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

{'return': 5.00}

reasoning:

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received $\$3.36 \times 3 = \10.08
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created: $\$5$ (sender's initial) + $\$10.08$ (what I received) = $\$15.08$
- Equal split would be $\$7.54$ each
- Sender currently has: $\$5 - \$3.36 = \$1.64$
- To give sender $\$7.54$, I'd need to return: $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of $\$10.08 = \6.77

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have: $\$1.64 + \$5 = \$6.64$ (profit of $\$1.64$)
- I would keep: $\$10.08 - \$5 = \$5.08$ (profit of $\$5.08$)

Option 4: Return enough so sender gets their \$5 back

- Sender needs $\$3.36$ more to get back to $\$5$
- If I return $\$3.36$, sender gets their initial endowment back and I keep $\$6.72$

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning $\$5$ seems like a solid middle ground-it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at $\$5$, the sender gains $\$1.64$ while I gain $\$5.08$. If I go higher to $\$6$, they'd gain $\$2.64$ but I'd only gain $\$4.08$ -ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating $\$6.72$ in surplus. Dividing this equally means each of us should gain $\$3.36$ from that surplus. Since the sender started with $\$5$ and kept $\$1.64$, they'd end with $\$6.64$ total. That leaves me with $\$5.08$ after sending back $\$5$.

Actually, let me think about this differently. The sender's initial $\$5$ should be restored to them, and the $\$6.72$ surplus split evenly. If I return $\$3.36$, the sender gets their $\$1.64$ kept plus $\$3.36$ back, totaling $\$5$ -their original amount. I'd keep $\$6.72$, which is my $\$3.36$ share of the surplus plus the original $\$3.36$ sent. This creates an equal split where we each benefit $\$3.36$ from the tripling effect.

But there's another way to frame it: the sender should recover their full $\$5$, then we divide the $\$6.72$ surplus equally. That would mean returning $\$6.72$ to give them $\$8.36$ total, while I keep $\$6.72$. Let me check if this math works- $\$8.36$ plus $\$6.72$ equals $\$15.08$, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a $\$3.36$ gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received ($\$5.04$), return the original amount ($\$3.36$), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2102,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$4.89$ returned, visible payoff $\$6.89$.

Your total visible earnings across all rounds are $\$5.08$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Analysis:****

- My co-player sent $\$3.0$ (60% of their endowment) in Round 1
- They saw a return of $\$4.89$, giving them a total payoff of $\$6.89$
- This represents good cooperation from their perspective

****My decision:****

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

```
{'send': 3.50}
```

Reasoning:

Let me analyze this situation:

****What I know:****

- Round 1: I was the RECEIVER
 - The sender sent \$3.36 (67.2% of their endowment)
 - I received \$10.08
 - I returned \$5.00
 - I ended with \$5.08

****Now in Round 2, the visible history shows:****

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff: $1.64 + \$5.00 = \6.64 ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

****Now I'm the SENDER:****

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1. From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

****Decision:****

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

```
Usage: {  
  "input_tokens": 892,  
  "output_tokens": 1059,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:54:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",
```

```

"move_index": 3,
"opponent_id": "Agent_4",
"pairing": {
  "agents": [
    "Agent_4",
    "Agent_3"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the receiver reciprocates. If I send \$3:

- The receiver gets \$9 ($3 \times \3)
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = $(3 \times \text{sent}) - \text{returned}$

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
 - Show some trust
 - Create value through the multiplier
 - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 626,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```
'''json
{"return": 5.25}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ($3 \times \3.36)
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be $\$5 - \$3 + \$5 = \7 , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ($3 \times \3.49)
- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with $\$5 - \$3.49 + \$5-6 = \$6.51-7.51$

...

```
Usage: {
  "input_tokens": 780,
  "output_tokens": 697,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:55:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with: $\$1.97 + \$4.50 = \$6.47$ (gain of \$1.47)
- I will end with: $\$9.09 - \$4.50 = \$4.59$ (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has: $\$5 - \$3.03 = \$1.97$
- Will end up with: $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with: $\$9.09 - (\text{what I return})$

Key considerations:

1. ****Fairness****: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. ****Equal split consideration****: If we were to split the total money equally:
 - Total available: $\$5$ (sender's original) + $\$9.09$ (what I received) = $\$14.09$
 - Equal split: $\$7.045$ each
 - Sender currently has: $\$1.97$
 - To give sender $\$7.045$: I'd need to return $\$7.045 - \$1.97 = \$5.075$
3. ****Proportional reciprocity****: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
 - 60.6% of $\$9.09 = \5.51
4. ****Strategic considerations****:
 - This is a multi-round game with multiple agents
 - Building trust could be beneficial for future rounds
 - However, there's communication noise, so exact amounts may not be perceived correctly
 - Agent names are hidden, so reputation effects are limited to behavioral patterns
5. ****Common reciprocity patterns****: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1332,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

****Round 1 Analysis:****

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

****Round 2 Decision:****

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

```
{'send': 3.00}
```

Reasoning:

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity. If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

```
Usage: {
  "input_tokens": 856,
  "output_tokens": 628,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:55:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
 - If I send \$5, receiver gets \$15
 - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
 - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.

5. ****Communication noise****: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. ****Multi-round consideration****: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. ****Multi-agent setting****: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- ****Full trust****: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- ****Partial trust****: Send some amount (\$2-3) - tests the waters
- ****Zero trust****: Send nothing (\$0) - avoid risk but miss cooperative gains
- ****Tit-for-tat****: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
 - The receiver saw \$3.03 (slight noise)
 - This became \$9.09 for them
 - They returned \$4.5
 - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible

was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was $\$2 + \$4.5 = \$6.5$

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER

- My co-player (the same person from Round 1) sent me \$2.92
- I received \$8.76 (which is $3 \times \$2.92$)
- I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
 - Give them enough to make their total payoff reasonable
 - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

Usage: {

```
"input_tokens": 824,  
"output_tokens": 873,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:55:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_4",  
"move_index": 6,  
"opponent_id": "Agent_7",  
"pairing": {  
  "agents": [  
    "Agent_8",  
    "Agent_7"  
  ],  
  "investor": "Agent_8",  
  "roles": {  
    "Agent_7": "trustee",  
    "Agent_8": "investor"  
  },  
  "trustee": "Agent_7"  
},  
"role": "investor",  
"role_label": "Sender",  
"round": 2,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has $\$5 - \$3.01 = \$1.99$

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with: $\$1.99 + \$4.50 = \$6.49$ (gain of \$1.49)
- I end with: $\$9.03 - \$4.50 = \$4.53$ (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is $\$3.01 \times 3$)
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. ****The sender showed significant trust****: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. ****Payoff analysis****:
 - If I return \$0: Sender gets $\$5 - \$3.01 + \$0 = \1.99 , I get \$9.03
 - If I return half (\$4.52): Sender gets $\$5 - \$3.01 + \$4.52 = \6.51 , I get \$4.51
 - If I return \$3.01: Sender gets $\$5 - \$3.01 + \$3.01 = \5.00 , I get \$6.02
 - If I return \$4.50: Sender gets $\$5 - \$3.01 + \$4.50 = \6.49 , I get \$4.53
 - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. ****Fairness considerations****:
 - The sender's initial endowment was \$5, and they sent \$3.01
 - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
 - If I want us to split the gains equally, I should consider that the total is now $\$5 - \$3.01 + \$9.03 = \11.02
 - Equal split would give us each about \$5.51
 - For sender to get \$5.51: they need $\$5 - \$3.01 + \text{return} = \$5.51$, so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1131,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.5299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

****Round 1 Analysis:****

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

****Round 2 Strategy:****

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that at trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```json

```
{"send": 3.00}
'''
```

#### Reasoning:

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings:  $\$9.03 - \$4.50 = \$4.53$

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

```
Usage: {
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Receiver - dyad\_4

Timestamp: 2026-08-12T07:55:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```json



```
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received see

ms fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was $\$5 - \$3 + \$4.5 = \6.5 (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them: $\$2.73 + \$3.40 = \$6.13$ total

And I would keep: $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

Usage: {

```
"input_tokens": 765,
"output_tokens": 522,
"reasoning_tokens": 0
```

}

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

```
dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $3.25; payoffs Agent_1=$5.25,
Agent_2=$5.75; noise/visible: sent visible=$2.15, received visible=$6.45, returned visible=$3.23
  Action validation: {"returned": {"amount": 3.25, "clamped": false, "max": 9.0, "min": 0, "raw": 3.25},
    "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent $3.5; received $10.5; Agent_4 returned $6.5; payoffs Agent_3=$8, Agent_4=$4;
noise/visible: sent visible=$4.27, received visible=$12.81, returned visible=$6.64
  Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5},
    "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_3: Agent_5 sent $3; received $9; Agent_6 returned $5.25; payoffs Agent_5=$7.25,
Agent_6=$3.75; noise/visible: sent visible=$3.5, received visible=$10.5, returned visible=$5.87
  Action validation: {"returned": {"amount": 5.25, "clamped": false, "max": 9.0, "min": 0, "raw": 5.25},
    "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent $3; received $9; Agent_8 returned $5.8; payoffs Agent_7=$7.8, Agent_8=$3.2;
noise/visible: sent visible=$3.82, received visible=$11.46, returned visible=$5.23
  Action validation: {"returned": {"amount": 5.8, "clamped": false, "max": 9.0, "min": 0, "raw": 5.8},
    "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

```

Balances after round: {
  "Agent_1": 15.75,
  "Agent_2": 17.25,
  "Agent_3": 20.25,
  "Agent_4": 14.75,
  "Agent_5": 18.35,
  "Agent_6": 14.65,
  "Agent_7": 19.9,
  "Agent_8": 13.100000000000001
}
Visible balances after round: {
  "Agent_1": 14.07,
  "Agent_2": 12.759999999999998,
  "Agent_3": 20.25,
  "Agent_4": 18.54,
  "Agent_5": 19.53,
  "Agent_6": 15.39,
  "Agent_7": 17.19,
  "Agent_8": 15.969999999999999
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:56:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send **\$3**. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should pro

bably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
 - Show some willingness to cooperate
 - Test the waters
 - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is 2.34×3)
3. They returned \$3.50
4. My visible payoff was \$4.71, which means: $5 - 3 \text{ (sent)} + 3.50 \text{ (returned)} = 5.50...$ but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02

- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is $\$2.71 \times 3$)
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate: $\$3.50 / \$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's \$5.50 + \$4.13 = \$9.63 actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

```
Usage: {
  "input_tokens": 1100,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:56:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return **\$3.50**:

- Sender ends with: \$2.66 + \$3.50 = \$6.16 (profit of \$1.16)
- I end with: \$7.02 - \$3.50 = \$3.52 (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

reasoning:

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is \$2.34 × 3)
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep: \$5 - \$2.34 + returned = \$2.66 + returned
4. I will keep: \$7.02 - returned

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return \$6.01 - \$2.66 = \$3.35
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received = $0.468 \times \$7.02 = \3.29
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with \$2.66 + \$3.50 = \$6.16 (up \$1.16 from initial \$5), and I keep \$7.02 - \$3.50 = \$3.52
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send **\$3.00** (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1: $\$9.0 - \$3.50 = \$5.50$
But the history shows my visible earnings as $\$3.52$ (approximately $\$3.50$ due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% ($\$3.0$)
2. Due to noise, they saw less returned ($\$2.71$) than I actually sent ($\3.50)
3. From their perspective, the return rate was $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ($\$3.50/\9.0)

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send $\$2.75$ (55%).

...

usage:

```
{
  "input_tokens": 872,
  "output_tokens": 670,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$2.71$ returned, visible payoff $\$4.71$.
- Round 2 against you, as RECEIVER: saw $\$2.71$ sent, it became $\$8.129999999999999$, returned $\$4.0$, visible payoff $\$4.129999999999999$.

Your total visible earnings across all rounds are $\$9.559999999999999$.

This round, you are the RECEIVER against your current co-player. They sent you $\$2.15$, so you received $\$6.449999999999999$. How much do you return? ($0-6.449999999999999$)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

Round 1: Co-player sent $\$3.0$ (60%), saw lower return due to noise
Round 2: Co-player returned $\$4.0$ when receiving (showing fairness)
Round 3: Co-player sent $\$2.15$ (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise, but they're still showing moderate trust.

Current situation:

- Co-player has $\$2.85$ left
- I received $\$6.45$

I'll return $\$3.25$ (approximately 50% of what I received):

- Sender ends with: $\$2.85 + \$3.25 = \$6.10$ (profit of $\$1.10$)
- I end with: $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent $\$3.0$ (60% of endowment)

- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)
- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff: $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with: $\$2.85 + \$3.00 = \$5.85$ (profit of \$0.85)
- I end with: $\$6.45 - \$3.00 = \$3.45$ (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

```
Usage: {
  "input_tokens": 1223,
  "output_tokens": 819,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:56:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      NOTE: There may be communication noise in this game.
      The amounts you see may differ slightly from what was actually sent/returned.
      This is part of the experimental design.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - Agent names are hidden in this run.
      - Your paired opponent is referred to as your current co-player.
  Message 2 role=user
    content:
      Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
  Message 3 role=assistant
    content:
      Looking at this first round, I need to balance trust with caution since I don't know my co
      -player yet.

      Sending money creates value through the 3x multiplier, which benefits both of us if the re
      ceiver reciprocates. If I send $3:
      - The receiver gets $9 (3 × $3)
      - If they return even $4-5, we both end up better off than the initial $5
      - I risk at most $3

      This shows willingness to cooperate while not being overly vulnerable in this first intera
      ction.

      '''json
      {"send": 3}
      '''
    reasoning:
      This is Round 1, and I'm the SENDER. Let me think about this strategically.

```

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = (3 × sent) - returned

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
 - Show some trust
 - Create value through the multiplier
 - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 626,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

ed.

```
'''json
{"return": 5.25}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ($3 \times \3.36)
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be $\$5 - \$3 + \$5 = \7 , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ($3 \times \3.49)
- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with $\$5 - \$3.49 + \$5-6 = \$6.51-7.51$

...

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 697,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: $\$5 - \$3 + \$5 = \7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff: $\$5 - \$3.5 + \$5.65 = \7.15

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

```
Usage: {
  "input_tokens": 1080,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:56:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

****My reasoning:****

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with: \$1.64 + \$5.00 = \$6.64 (profit of \$1.64)
- I end with: \$10.08 - \$5.00 = \$5.08 (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

```
{'return': 5.00}
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received $\$3.36 \times 3 = \10.08
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created: $\$5$ (sender's initial) + $\$10.08$ (what I received) = $\$15.08$
- Equal split would be $\$7.54$ each
- Sender currently has: $\$5 - \$3.36 = \$1.64$
- To give sender $\$7.54$, I'd need to return: $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of $\$10.08$? $\$6.77$

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have: $\$1.64 + \$5 = \$6.64$ (profit of $\$1.64$)
- I would keep: $\$10.08 - \$5 = \$5.08$ (profit of $\$5.08$)

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep $\$6.72$

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground—it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains \$1.64 while I gain \$5.08. If I go higher to \$6, they'd gain \$2.64 but I'd only gain \$4.08—ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating \$6.72 in surplus. Dividing this equally means each of us should gain \$3.36 from that surplus. Since the sender started with \$5 and kept \$1.64, they'd end with \$6.64 total. That leaves me with \$5.08 after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the \$6.72 surplus split evenly. If I return \$3.36, the sender gets their \$1.64 kept plus \$3.36 back, totaling \$5—their original amount. I'd keep \$6.72, which is my \$3.36 share of the surplus plus the original \$3.36 sent. This creates an equal split where we each benefit \$3.36 from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the \$6.72 surplus equally. That would mean returning \$6.72 to give them \$8.36 total, while I keep \$6.72. Let me check if this math works—\$8.36 plus \$6.72 equals \$15.08, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a \$3.36 gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received (\$5.04), return the original amount (\$3.36), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response

seems appropriate....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2102,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Analysis:****

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They saw a return of \$4.89, giving them a total payoff of \$6.89
- This represents good cooperation from their perspective

****My decision:****

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

{'send': 3.50}

reasoning:

Let me analyze this situation:

****What I know:****

- Round 1: I was the RECEIVER
 - The sender sent \$3.36 (67.2% of their endowment)
 - I received \$10.08
 - I returned \$5.00
 - I ended with \$5.08

****Now in Round 2, the visible history shows:****

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff: $\$1.64 + \$5.00 = \$6.64$? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

****Now I'm the SENDER:****

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0

- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

****Decision:****

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

usage:

```
{
  "input_tokens": 892,
  "output_tokens": 1059,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

****Analysis:****

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

****Current situation:****

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

****My decision:****

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%). They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

Reasoning:

Let me analyze the situation:

****Visible history:****

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became

\$10.47, returned \$5.25, their payoff \$5.22

****Current Round 3:****

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

****Analysis:****

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with: $\$0.73 + \$6.50 = \$7.23$ (profit of \$2.23)
- I end with: $\$12.81 - \$6.50 = \$6.31$

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with: $\$0.73 + \$7.00 = \$7.73$ (profit of \$2.73)
- I end with: $\$12.81 - \$7.00 = \$5.81$

This is even more generous and rewards their very high trust level.

...

```
Usage: {  
  "input_tokens": 1260,  
  "output_tokens": 864,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:56:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
 - If I send \$5, receiver gets \$15
 - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
 - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- **Partial trust**: Send some amount (\$2-3) - tests the waters
- **Zero trust**: Send nothing (\$0) - avoid risk but miss cooperative gains
- **Tit-for-tat**: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
 - The receiver saw \$3.03 (slight noise)
 - This became \$9.09 for them
 - They returned \$4.5
 - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
 - My co-player (the same person from Round 1) sent me \$2.92
 - I received \$8.76 (which is 3 × \$2.92)
 - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
 - Give them enough to make their total payoff reasonable
 - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 873,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

****Round 1****: I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

****Round 2****: They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

```
Usage: {
  "input_tokens": 1160,
  "output_tokens": 419,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:57:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with: $\$1.97 + \$4.50 = \$6.47$ (gain of \$1.47)
- I will end with: $\$9.09 - \$4.50 = \$4.59$ (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has: $\$5 - \$3.03 = \$1.97$
- Will end up with: $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with: $\$9.09 - (\text{what I return})$

Key considerations:

1. ****Fairness****: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. ****Equal split consideration****: If we were to split the total money equally:
 - Total available: $\$5$ (sender's original) + $\$9.09$ (what I received) = $\$14.09$
 - Equal split: $\$7.045$ each
 - Sender currently has: $\$1.97$
 - To give sender $\$7.045$: I'd need to return $\$7.045 - \$1.97 = \$5.075$
3. ****Proportional reciprocity****: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
 - 60.6% of $\$9.09 = \5.51
4. ****Strategic considerations****:
 - This is a multi-round game with multiple agents
 - Building trust could be beneficial for future rounds
 - However, there's communication noise, so exact amounts may not be perceived correctly
 - Agent names are hidden, so reputation effects are limited to behavioral patterns
5. ****Common reciprocity patterns****: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1332,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 Analysis:****

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

****Round 2 Decision:****

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

{'send': 3.00}

reasoning:

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.

If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

usage:

```
{
  "input_tokens": 856,
  "output_tokens": 628,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Co-player's behavior:****

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - ****increased trust!****

****Current state:****

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level. I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with: $\$1.5 + \$5.25 = \$6.75$ (gain of \$1.75)
- I end with: $\$10.5 - \$5.25 = \$5.25$ (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

{'return': 5.25}

Reasoning:

Let me analyze the visible history:

****Round 1:****

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

****Round 2:****

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

****Round 3 (Current):****

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:
 - Round 1: sent 60% (\$3.0)
 - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have \$5 - \$3.5 = \$1.5 left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with: $\$1.5 + \$5.0 = \$6.5$ (gain of \$1.5)
- I end with: $\$10.5 - \$5.0 = \$5.5$ (gain of \$5.5)

If I return \$5.25:
- Sender ends with: $\$1.5 + \$5.25 = \$6.75$ (gain of \$1.75)
- I end with: $\$10.5 - \$5.25 = \$5.25$ (gain of \$5.25)

If I return \$5.50:
- Sender ends with: $\$1.5 + \$5.50 = \$7.0$ (gain of \$2.0)
- I end with: $\$10.5 - \$5.50 = \$5.0$ (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

```
Usage: {  
  "input_tokens": 1230,  
  "output_tokens": 964,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:57:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was \$5 - \$3 + \$4.5 = \$6.5 (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept \$5 - \$2.27 = \$2.73

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1

2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.
 This would give them: \$2.73 + \$3.40 = \$6.13 total
 And I would keep: \$6.81 - \$3.40 = \$3.41

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

usage:

```
{
  "input_tokens": 765,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

- Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)
- Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

Usage: {


```
"input_tokens": 1075,
"output_tokens": 424,
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:57:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has $\$5 - \$3.01 = \$1.99$

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with: $\$1.99 + \$4.50 = \$6.49$ (gain of \$1.49)
- I end with: $\$9.03 - \$4.50 = \$4.53$ (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is $\$3.01 \times 3$)
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. ****The sender showed significant trust****: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. ****Payoff analysis****:
 - If I return \$0: Sender gets $\$5 - \$3.01 + \$0 = \1.99 , I get \$9.03
 - If I return half (\$4.52): Sender gets $\$5 - \$3.01 + \$4.52 = \6.51 , I get \$4.51
 - If I return \$3.01: Sender gets $\$5 - \$3.01 + \$3.01 = \5.00 , I get \$6.02
 - If I return \$4.50: Sender gets $\$5 - \$3.01 + \$4.50 = \6.49 , I get \$4.53
 - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. ****Fairness considerations****:
 - The sender's initial endowment was \$5, and they sent \$3.01
 - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
 - If I want us to split the gains equally, I should consider that the total is now $\$5 - \$3.01 + \$9.03 = \11.02
 - Equal split would give us each about \$5.51
 - For sender to get \$5.51: they need $\$5 - \$3.01 + \text{return} = \$5.51$, so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1131,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.5299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 Analysis:****

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

****Round 2 Strategy:****

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings: \$9.03 - \$4.50 = \$4.53

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

usage:

```
{
  "input_tokens": 901,
  "output_tokens": 668,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Round 3 Analysis:****

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

****Co-player's pattern:****

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with: \$1.18 + \$5.80 = \$6.98 (gain of \$1.98)
- I end with: \$11.46 - \$5.80 = \$5.66 (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

Reasoning:

Let me analyze the situation:

****Round 1:****

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

****Round 2:****

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!
- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is 2.27×3)
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff: $6.81 - 3.40 = 3.41$?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff: $2.00 + 3.40 = 5.40$

My total earnings should be: 4.53 (Round 1) + 5.40 (Round 2) = 9.93

But visible history says 10.31 ... There might be some rounding or noise in the visible earnings too.

****Round 3:****

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have: $5 - 3.82 = 1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

```
Usage: {
  "input_tokens": 1299,
  "output_tokens": 1178,
  "reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_1 returned \$3.9; payoffs Agent_2=\$5.9, Agent_1=\$5.1; noise/visible: sent visible=\$2.6, received visible=\$7.8, returned visible=\$4.61
Action validation: {"returned": {"amount": 3.9, "clamped": false, "max": 9.0, "min": 0, "raw": 3.9}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_4 sent \$3.75; received \$11.25; Agent_3 returned \$4.5; payoffs Agent_4=\$5.75, Agent_3=\$6.75; noise/visible: sent visible=\$3.02, received visible=\$9.06, returned visible=\$3.87
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 11.25, "min": 0, "raw": 4.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$4.4; payoffs Agent_6=\$6.4, Agent_5=\$4.6; noise/visible: sent visible=\$2.94, received visible=\$8.82, returned visible=\$5.08
Action validation: {"returned": {"amount": 4.4, "clamped": false, "max": 9.0, "min": 0, "raw": 4.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3; received \$9; Agent_7 returned \$5; payoffs Agent_8=\$7, Agent_7=\$4; noise/visible: sent visible=\$3.3, received visible=\$9.9, returned visible=\$5.18

```

    Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 20.85,
  "Agent_2": 23.15,
  "Agent_3": 27.0,
  "Agent_4": 20.5,
  "Agent_5": 22.950000000000003,
  "Agent_6": 21.05,
  "Agent_7": 23.9,
  "Agent_8": 20.1
}
Visible balances after round: {
  "Agent_1": 17.970000000000002,
  "Agent_2": 19.369999999999997,
  "Agent_3": 24.810000000000002,
  "Agent_4": 23.66,
  "Agent_5": 23.950000000000003,
  "Agent_6": 22.47,
  "Agent_7": 22.09,
  "Agent_8": 23.15
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:57:50
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {

```

    "agent_type": "standard",
    "dyad_id": "dyad_1",
    "move_index": 0,
    "opponent_id": "Agent_1",
    "pairing": {
      "agents": [
        "Agent_2",
        "Agent_1"
      ],
      "investor": "Agent_2",
      "roles": {
        "Agent_1": "trustee",
        "Agent_2": "investor"
      },
      "trustee": "Agent_1"
    },
    "role": "investor",
    "role_label": "Sender",
    "round": 4,
    "task": "game",
    "task_index": 0
  }

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return **\$3.50**:

- Sender ends with: \$2.66 + \$3.50 = \$6.16 (profit of \$1.16)
- I end with: \$7.02 - \$3.50 = \$3.52 (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

reasoning:

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is \$2.34 × 3)
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep: \$5 - \$2.34 + returned = \$2.66 + returned
4. I will keep: \$7.02 - returned

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.0

2. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return \$6.01 - \$2.66 = \$3.35

- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received = $0.468 \times \$7.02 = \3.29
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with $\$2.66 + \$3.50 = \$6.16$ (up \$1.16 from initial \$5), and I keep $\$7.02 - \$3.50 = \$3.52$
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send **\$3.00** (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was $\$4.71 = \2 (what they kept) + $\$2.71$ (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1: $\$9.0 - \$3.50 = \$5.50$

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ($\$3.50/\9.0)

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

usage:

```
{
  "input_tokens": 872,
  "output_tokens": 670,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise
Round 2: Co-player returned \$4.0 when receiving (showing fairness)
Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return **\$3.25** (approximately 50% of what I received):

- Sender ends with: $\$2.85 + \$3.25 = \$6.10$ (profit of \$1.10)
- I end with: $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)

- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff: $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with: $\$2.85 + \$3.00 = \$5.85$ (profit of \$0.85)
- I end with: $\$6.45 - \$3.00 = \$3.45$ (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

usage:

```
{
  "input_tokens": 1223,
  "output_tokens": 819,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise

Round 2: Co-player returned \$4.0 (generous)

Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving.

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)
 Round 2: They received and returned \$4.0 (generous return)
 Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)
 Round 2: I sent \$3.00, they returned \$4.0
 Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

```
Usage: {
  "input_tokens": 1674,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:57:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send ****\$3****. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round

- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
 - Show some willingness to cooperate
 - Test the waters
 - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is 2.34×3)

3. They returned \$3.50
4. My visible payoff was \$4.71, which means: $\$5 - \$3 \text{ (sent)} + \$3.50 \text{ (returned)} = \$5.50\dots$ but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is $\$2.71 \times 3$)
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```json

```
 {"send": 3}
 '''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50/\$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received  $\$7.8000000000000001$ . How much do you return? ( $0-7.8000000000000001$ )

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
```

'''

### Reasoning:

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

```
Usage: {
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_4 - Sender - dyad\_2

Timestamp: 2026-08-12T07:58:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned



- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

**\*\*My reasoning:\*\***

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with:  $1.64 + \$5.00 = \$6.64$  (profit of \$1.64)
- I end with:  $\$10.08 - \$5.00 = \$5.08$  (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

{'return': 5.00}

**reasoning:**

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received  $\$3.36 \times 3 = \$10.08$
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only  $\$5 - \$3.36 = \$1.64$

3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created: \$5 (sender's initial) + \$10.08 (what I received) = \$15.08
- Equal split would be \$7.54 each
- Sender currently has: \$5 - \$3.36 = \$1.64
- To give sender \$7.54, I'd need to return: \$7.54 - \$1.64 = \$5.90

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of \$10.08 = \$6.77

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have: \$1.64 + \$5 = \$6.64 (profit of \$1.64)
- I would keep: \$10.08 - \$5 = \$5.08 (profit of \$5.08)

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep \$6.72

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground-it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains \$1.64 while I gain \$5.08. If I go higher to \$6, they'd gain \$2.64 but I'd only gain \$4.08-ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating \$6.72 in surplus. Dividing this equally means each of us should gain \$3.36 from that surplus. Since the sender started with \$5 and kept \$1.64, they'd end with \$6.64 total. That leaves me with \$5.08 after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the \$6.72 surplus split evenly. If I return \$3.36, the sender gets their \$1.64 kept plus \$3.36 back, totaling \$5-their original amount. I'd keep \$6.72, which is my \$3.36 share of the surplus plus the original \$3.36 sent. This creates an equal split where we each benefit \$3.36 from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the \$6.72 surplus equally. That would mean returning \$6.72 to give them \$8.36 total, while I keep \$6.72. Let me check if this math works-\$8.36 plus \$6.72 equals \$15.08, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a \$3.36 gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received (\$5.04), return the original amount (\$3.36), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 2102,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They saw a return of \$4.89, giving them a total payoff of \$6.89
- This represents good cooperation from their perspective

**\*\*My decision:\*\***

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

{'send': 3.50}

**reasoning:**

Let me analyze this situation:

**\*\*What I know:\*\***

- Round 1: I was the RECEIVER
  - The sender sent \$3.36 (67.2% of their endowment)
  - I received \$10.08
  - I returned \$5.00
  - I ended with \$5.08

**\*\*Now in Round 2, the visible history shows:\*\***

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff:  $\$1.64 + \$5.00 = \$6.64$  ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

**\*\*Now I'm the SENDER:\*\***

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

**\*\*Decision:\*\***

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

usage:

```
{
 "input_tokens": 892,
 "output_tokens": 1059,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

**\*\*My decision:\*\***

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%). They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

reasoning:

Let me analyze the situation:

**\*\*Visible history:\*\***

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became \$10.47, returned \$5.25, their payoff \$5.22

**\*\*Current Round 3:\*\***

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

**\*\*Analysis:\*\***

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with:  $\$0.73 + \$6.50 = \$7.23$  (profit of \$2.23)
- I end with:  $\$12.81 - \$6.50 = \$6.31$

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with:  $\$0.73 + \$7.00 = \$7.73$  (profit of \$2.73)
- I end with:  $\$12.81 - \$7.00 = \$5.81$

This is even more generous and rewards their very high trust level.

...

**usage:**

```
{
 "input_tokens": 1260,
 "output_tokens": 864,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.2200000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.

Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

**\*\*Pattern observed:\*\***

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

**\*\*My decision:\*\***

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

**Reasoning:**

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending a round \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

```
Usage: {
 "input_tokens": 1769,
 "output_tokens": 533,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_3 - Receiver - dyad\_2

Timestamp: 2026-08-12T07:58:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the receiver reciprocates. If I send \$3:

- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```
'''json
{"send": 3}
'''
```

#### reasoning:

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = ( $3 \times \text{sent}$ ) - returned

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly.

However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
  - Show some trust
  - Create value through the multiplier
  - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 626,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```
'''json
{"return": 5.25}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ( $3 \times \$3.36$ )
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be  $\$5 - \$3 + \$5 = \$7$ , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ( $3 \times \$3.49$ )
- I need to decide how much to return

Analysis of their behavior:



- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with \$5 - \$3.49 + \$5-6 = \$6.51-\$7.51
- ...

usage:

```
{
 "input_tokens": 780,
 "output_tokens": 697,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: \$5 - \$3 + \$5 = \$7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)

- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff:  $\$5 - \$3.5 + \$5.65 = \$7.15$

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

**usage:**

```
{
 "input_tokens": 1080,
 "output_tokens": 477,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:

50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

```
Usage: {
 "input_tokens": 1454,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_6 - Sender - dyad\_3

Timestamp: 2026-08-12T07:58:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with:  $\$1.97 + \$4.50 = \$6.47$  (gain of \$1.47)
- I will end with:  $\$9.09 - \$4.50 = \$4.59$  (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

**reasoning:**

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has:  $\$5 - \$3.03 = \$1.97$
- Will end up with:  $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with:  $\$9.09 - (\text{what I return})$

Key considerations:

1. **\*\*Fairness\*\***: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. **\*\*Equal split consideration\*\***: If we were to split the total money equally:
  - Total available: \$5 (sender's original) + \$9.09 (what I received) = \$14.09
  - Equal split: \$7.045 each
  - Sender currently has: \$1.97
  - To give sender \$7.045: I'd need to return  $\$7.045 - \$1.97 = \$5.075$
3. **\*\*Proportional reciprocity\*\***: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
  - 60.6% of \$9.09 = \$5.51
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with multiple agents

- Building trust could be beneficial for future rounds
- However, there's communication noise, so exact amounts may not be perceived correctly
- Agent names are hidden, so reputation effects are limited to behavioral patterns

5. **\*\*Common reciprocity patterns\*\***: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1332,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

**\*\*Round 2 Decision:\*\***

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)

- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.  
If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

**usage:**

```
{
 "input_tokens": 856,
 "output_tokens": 628,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - **\*\*increased trust!\*\***

**\*\*Current state:\*\***

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level.

I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with: \$1.5 + \$5.25 = \$6.75 (gain of \$1.75)
- I end with: \$10.5 - \$5.25 = \$5.25 (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

```
{'return': 5.25}
```

**reasoning:**

Let me analyze the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

**\*\*Round 3 (Current):\*\***

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:
  - Round 1: sent 60% (\$3.0)
  - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have \$5 - \$3.5 = \$1.5 left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with:  $\$1.5 + \$5.0 = \$6.5$  (gain of \$1.5)
- I end with:  $\$10.5 - \$5.0 = \$5.5$  (gain of \$5.5)

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

If I return \$5.50:

- Sender ends with:  $\$1.5 + \$5.50 = \$7.0$  (gain of \$2.0)
- I end with:  $\$10.5 - \$5.50 = \$5.0$  (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

**usage:**

```
{
 "input_tokens": 1230,
 "output_tokens": 964,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Co-player's behavior pattern:\*\***

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

**\*\*Analysis:\*\***

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue matching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{ 'send': 3.00 }

**Reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

Usage: {

  "input\_tokens": 1702,

  "output\_tokens": 665,

  "reasoning\_tokens": 0

}

**GAME - Agent\_5 - Receiver - dyad\_3**

Timestamp: 2026-08-12T07:58:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

  "agent\_type": "standard",

  "dyad\_id": "dyad\_3",



```

"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
  - If I send \$5, receiver gets \$15
  - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
  - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- **Partial trust**: Send some amount (\$2-3) - tests the waters
- **Zero trust**: Send nothing (\$0) - avoid risk but miss cooperative gains
- **Tit-for-tat**: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense—it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

**usage:**

```
{
```

```
"input_tokens": 500,
"output_tokens": 981,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
  - The receiver saw \$3.03 (slight noise)
  - This became \$9.09 for them
  - They returned \$4.5
  - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
  - My co-player (the same person from Round 1) sent me \$2.92
  - I received \$8.76 (which is  $3 \times \$2.92$ )
  - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
  - Give them enough to make their total payoff reasonable

- Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

usage:

```
{
 "input_tokens": 824,
 "output_tokens": 873,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

content:

Looking at our history together:

**\*\*Round 1\*\*:** I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

**\*\*Round 2\*\*:** They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

usage:

```
{
 "input_tokens": 1160,
```

```
"output_tokens": 419,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our history:

```
Round 1: I sent ~$3, they returned $4.5 (50% of received)
Round 2: They sent $3, I returned $4.4 (49% of received)
Round 3: I sent $3, they returned $5.25 (exactly 50% of received)
Round 4: They sent $2.94, I received $8.82
```

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relationship.

```
'''json
{"return": 4.4}
'''
```

**Reasoning:**

Looking at our history:

```
Round 1: I sent ~$3, they received ~$9, returned $4.5
Round 2: They sent $3, I received $9, returned $4.4 (but they saw $3.55)
Round 3: I sent $3 (they saw $3.5), they received ~$10.5, returned $5.25
Round 4: They sent $2.94, I received $8.82
```

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them:  $\$5 - \$2.94 + \$4.41 = \$6.47$

And leave me with:  $\$8.82 - \$4.41 = \$4.41$

That seems fair and continues our mutually beneficial cooperation.

```
Usage: {
 "input_tokens": 1549,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-08-12T07:58:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has  $\$5 - \$3.01 = \$1.99$

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with:  $\$1.99 + \$4.50 = \$6.49$  (gain of \$1.49)
- I end with:  $\$9.03 - \$4.50 = \$4.53$  (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

**reasoning:**

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is  $\$3.01 \times 3$ )
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. **\*\*The sender showed significant trust\*\***: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. **\*\*Payoff analysis\*\***:
  - If I return \$0: Sender gets  $\$5 - \$3.01 + \$0 = \$1.99$ , I get \$9.03
  - If I return half (\$4.52): Sender gets  $\$5 - \$3.01 + \$4.52 = \$6.51$ , I get \$4.51
  - If I return \$3.01: Sender gets  $\$5 - \$3.01 + \$3.01 = \$5.00$ , I get \$6.02
  - If I return \$4.50: Sender gets  $\$5 - \$3.01 + \$4.50 = \$6.49$ , I get \$4.53
  - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. **\*\*Fairness considerations\*\***:
  - The sender's initial endowment was \$5, and they sent \$3.01
  - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
  - If I want us to split the gains equally, I should consider that the total is now  $\$5 - \$3.01 + \$9.03 = \$11.02$
  - Equal split would give us each about \$5.51
  - For sender to get \$5.51: they need  $\$5 - \$3.01 + \text{return} = \$5.51$ , so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1131,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.5299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

**\*\*Round 2 Strategy:\*\***

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings: \$9.03 - \$4.50 = \$4.53

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send



around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

**usage:**

```
{
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

**\*\*Co-player's pattern:\*\***

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with: \$1.18 + \$5.80 = \$6.98 (gain of \$1.98)
- I end with: \$11.46 - \$5.80 = \$5.66 (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\***

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

**\*\*Round 2:\*\***

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!
- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is  $2.27 \times 3$ )
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff:  $\$6.81 - \$3.40 = \$3.41$  ?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff:  $\$2.00 + \$3.40 = \$5.40$

My total earnings should be:  $\$4.53$  (Round 1) +  $\$5.40$  (Round 2) =  $\$9.93$

But visible history says \$10.31... There might be some rounding or noise in the visible earnings too.

**\*\*Round 3:\*\***

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have:  $\$5 - \$3.82 = \$1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

**usage:**

```
{
 "input_tokens": 1299,
 "output_tokens": 1178,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.810000000000005, returned \$3.4, visible payoff \$3.410000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)
- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

**\*\*Round 4 Strategy:\*\***

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our coop

erative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

#### Reasoning:

Looking at the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

**\*\*Round 3:\*\***

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

```
Usage: {
 "input_tokens": 1833,
 "output_tokens": 737,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Receiver - dyad\_4

Timestamp: 2026-08-12T07:58:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
```

```

"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```

'''json
{"send": 3}
'''

```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

**Key considerations:**

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 498,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.5299999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was  $\$5 - \$3 + \$4.5 = \$6.5$  (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept  $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them:  $\$2.73 + \$3.40 = \$6.13$  total

And I would keep:  $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

**usage:**

```
{
 "input_tokens": 765,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 8 role=user****content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history with my current co-player:

- Round 1: They returned \$4.5 (~50%)
- Round 2: They sent \$3, I returned \$3.4
- Round 3: They returned \$5.8 (~51%)
- Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

**Reasoning:**

Looking at the history with my current co-player:

- Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)
- Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to

```

 o noise)
Round 3: I sent $3, they received $11.46 (visible as $3.82 sent due to noise), returned $5.8 (
~51%)
Round 4: They sent $3.3, I received $9.9

This co-player has been very cooperative throughout:
- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, wh
ich would be around $5.

Let me calculate:
$9.9 * 0.5 = $4.95

I'll return $5 to keep the pattern going.
Usage: {
 "input_tokens": 1433,
 "output_tokens": 351,
 "reasoning_tokens": 0
}

=====
ROUND 5
=====
Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender,
Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver
Pairings:
 dyad_1: Agent_1=Sender, Agent_2=Receiver
 dyad_2: Agent_3=Sender, Agent_4=Receiver
 dyad_3: Agent_5=Sender, Agent_6=Receiver
 dyad_4: Agent_7=Sender, Agent_8=Receiver
Parsed game decisions:
 dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $3.5; payoffs Agent_1=$5.5, Agent_2=$5.5;
 noise/visible: sent visible=$2.32, received visible=$6.96, returned visible=$3.3
 Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw":
 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_3 sent $3.5; received $10.5; Agent_4 returned $4.5; payoffs Agent_3=$6, Agent_4=$6;
 noise/visible: sent visible=$3.02, received visible=$9.06, returned visible=$4.19
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 10.5, "min": 0, "raw":
 4.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
 dyad_3: Agent_5 sent $3; received $9; Agent_6 returned $4.2; payoffs Agent_5=$6.2, Agent_6=$4.8;
 noise/visible: sent visible=$2.8, received visible=$8.4, returned visible=$3.6
 Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_7 sent $3; received $9; Agent_8 returned $6; payoffs Agent_7=$8, Agent_8=$3;
 noise/visible: sent visible=$3.92, received visible=$11.76, returned visible=$6.98
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw":
 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 26.35,
 "Agent_2": 28.65,
 "Agent_3": 33.0,
 "Agent_4": 26.5,
 "Agent_5": 29.150000000000002,
 "Agent_6": 25.85,
 "Agent_7": 31.9,
 "Agent_8": 23.1
}
Visible balances after round: {
 "Agent_1": 23.270000000000003,
 "Agent_2": 22.83,
 "Agent_3": 30.500000000000004,
 "Agent_4": 28.22,
 "Agent_5": 29.550000000000004,
 "Agent_6": 26.669999999999998,
 "Agent_7": 31.07,
 "Agent_8": 28.909999999999997
}

```



## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T07:59:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send \*\*\$3\*\*. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
  - Show some willingness to cooperate
  - Test the waters
  - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

**usage:**

```
{
```

```
"input_tokens": 500,
"output_tokens": 547,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
```json  
{"return": 4.00}  
```
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is  $2.34 \times 3$ )
3. They returned \$3.50
4. My visible payoff was \$4.71, which means:  $\$5 - \$3 \text{ (sent)} + \$3.50 \text{ (returned)} = \$5.50...$  but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is  $2.71 \times 3$ )
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 909,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50/\$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.8000000000000001. How much do you return? (0-7.8000000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

Your total visible earnings across all rounds are \$17.970000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)

Round 2: They sent \$3.00, I returned ~\$4.00

Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)

Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%
- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

```
Usage: {
 "input_tokens": 1816,
 "output_tokens": 284,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T07:59:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
```

```

"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return \*\*\$3.50\*\*:

- Sender ends with: \$2.66 + \$3.50 = \$6.16 (profit of \$1.16)
- I end with: \$7.02 - \$3.50 = \$3.52 (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is  $2.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep:  $\$5 - \$2.34 + \text{returned} = \$2.66 + \text{returned}$
4. I will keep:  $\$7.02 - \text{returned}$

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return  $\$6.01 - \$2.66 = \$3.35$
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received =  $0.468 \times \$7.02 = \$3.29$
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with  $\$2.66 + \$3.50 = \$6.16$  (up \$1.16 from initial \$5), and I keep  $\$7.02 - \$3.50 = \$3.52$
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 731,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send \*\*\$3.00\*\* (60% of my endowment), matching their previous level of trust.



```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1:  $\$9.0 - \$3.50 = \$5.50$

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was  $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ( $\$3.50/\$9.0$ )

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

**usage:**

```
{
 "input_tokens": 872,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise

Round 2: Co-player returned \$4.0 when receiving (showing fairness)

Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise.

ion noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return \*\*\$3.25\*\* (approximately 50% of what I received):

- Sender ends with:  $\$2.85 + \$3.25 = \$6.10$  (profit of \$1.10)
- I end with:  $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)
- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff:  $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with:  $\$2.85 + \$3.00 = \$5.85$  (profit of \$0.85)
- I end with:  $\$6.45 - \$3.00 = \$3.45$  (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

**usage:**

```
{
 "input_tokens": 1223,
 "output_tokens": 819,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.  
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.  
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.  
Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise  
Round 2: Co-player returned \$4.0 (generous)  
Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)  
Round 2: They received and returned \$4.0 (generous return)  
Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)  
Round 2: I sent \$3.00, they returned \$4.0  
Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 426,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.  
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.  
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)

Round 3: Co-player sent \$3.0

Round 4: Co-player returned \$3.9 (about 50% of received)

Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left

- I received \$6.96

I'll return \*\*\$3.50\*\* (about 50% of received):

- Sender ends with: \$2.68 + \$3.50 = \$6.18

- I end with: \$6.96 - \$3.50 = \$3.46

```
'''json
{"return": 3.50}
'''
```

**Reasoning:**

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)

Round 3: Co-player sent \$3.0 (60% trust)

Round 4: Co-player received, returned \$3.9 (about 50% of what they got)

Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left (\$5 - \$2.32)

- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50

- Sender ends with: \$2.68 + \$3.50 = \$6.18 (profit of \$1.18)

- I end with: \$6.96 - \$3.50 = \$3.46

This seems fair and should maintain cooperation.

```
Usage: {
 "input_tokens": 2059,
 "output_tokens": 524,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Sender - dyad\_2**

Timestamp: 2026-08-12T07:59:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
]
 }
}
```

```

],
"investor": "Agent_3",
"roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
},
"trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the re

- ceiver reciprocates. If I send \$3:
- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```
'''json
{"send": 3}
'''
```

#### reasoning:

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff =  $(3 \times \text{sent}) - \text{returned}$

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
  - Show some trust
  - Create value through the multiplier
  - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

...

#### usage:

```
{
 "input_tokens": 500,
 "output_tokens": 626,
 "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```
'''json
{"return": 5.25}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ( $3 \times \$3.36$ )
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be  $\$5 - \$3 + \$5 = \$7$ , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ( $3 \times \$3.49$ )
- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with  $\$5 - \$3.49 + \$5-6 = \$6.51-7.51$

...

**usage:**

```
{
 "input_tokens": 780,
 "output_tokens": 697,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible

le payoff \$5.08.  
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.  
Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff:  $\$5 - \$3 + \$5 = \$7$  (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff:  $\$5 - \$3.5 + \$5.65 = \$7.15$

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

**usage:**

```
{
 "input_tokens": 1080,
 "output_tokens": 477,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.



This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:

50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

**usage:**

```
{
 "input_tokens": 1454,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our history:

Round 2: They sent \$3.5, I returned well

Round 3: I sent ~\$4.27, they returned \$6.5

Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this

s potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json
{"send": 3.5}
'''
```

#### Reasoning:

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)  
Round 3: I sent ~\$4.27 (they saw), they returned \$6.5  
Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?
3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

```
Usage: {
 "input_tokens": 1812,
 "output_tokens": 554,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_4 - Receiver - dyad\_2

Timestamp: 2026-08-12T07:59:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

**\*\*My reasoning:\*\***

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation

### 3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with:  $\$1.64 + \$5.00 = \$6.64$  (profit of \$1.64)
- I end with:  $\$10.08 - \$5.00 = \$5.08$  (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

```
{'return': 5.00}
```

#### reasoning:

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received  $\$3.36 \times 3 = \$10.08$
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only  $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created:  $\$5$  (sender's initial) +  $\$10.08$  (what I received) =  $\$15.08$
- Equal split would be  $\$7.54$  each
- Sender currently has:  $\$5 - \$3.36 = \$1.64$
- To give sender  $\$7.54$ , I'd need to return:  $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of  $\$10.08$  ?  $\$6.77$

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have:  $\$1.64 + \$5 = \$6.64$  (profit of \$1.64)
- I would keep:  $\$10.08 - \$5 = \$5.08$  (profit of \$5.08)

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep \$6.72

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground—it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains \$1.64 while I gain \$5.08. If I go higher to \$6, they'd gain \$2.64 but I'd only gain \$4.08—ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating \$6.72 in surplus. Dividing this equally means each of us should gain \$3.36 from that surplus. Since the sender started with \$5 and kept \$1.64, they'd end with \$6.64 total. That leaves me with \$5.08 after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the \$6.72 surplus split evenly. If I return \$3.36, the sender gets their \$1.64 kept plus \$3.36 back, totaling \$5—their original amount. I'd keep \$6.72, which is my \$3.36 share of the surplus plus the original \$3.36 sent. This creates an equal split where we each benefit \$3.36 from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the \$6.72 surplus equally. That would mean returning \$6.72 to give them \$8.36 total, while I keep \$6.72. Let me check if this math works—\$8.36 plus \$6.72 equals \$15.08, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a \$3.36 gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received (\$5.04), return the original amount (\$3.36), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 2102,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They saw a return of \$4.89, giving them a total payoff of \$6.89
- This represents good cooperation from their perspective

**\*\*My decision:\*\***

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

{'send': 3.50}

**reasoning:**

Let me analyze this situation:

**\*\*What I know:\*\***

- Round 1: I was the RECEIVER
  - The sender sent \$3.36 (67.2% of their endowment)
  - I received \$10.08
  - I returned \$5.00
  - I ended with \$5.08

**\*\*Now in Round 2, the visible history shows:\*\***

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff: \$1.64 + \$5.00 = \$6.64 ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actual

ly did.

**\*\*Now I'm the SENDER:\*\***

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

**\*\*Decision:\*\***

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

**usage:**

```
{
 "input_tokens": 892,
 "output_tokens": 1059,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

**\*\*My decision:\*\***

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%).

They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

```
{'return': 6.50}
reasoning:
Let me analyze the situation:

Visible history:
- Round 1: Co-player sent $3.0 as SENDER, saw $4.89 returned, payoff $6.89
- Round 2: Co-player as RECEIVER, saw $3.49 sent (I actually sent $3.50 but noise), it became $10.47, returned $5.25, their payoff $5.22

Current Round 3:
- Co-player sent $4.27 (85.4% of their endowment) - this is a very high trust level
- I received $12.81 (after 3x multiplication)
- Co-player currently has $0.73 remaining

Analysis:
The co-player has shown increasing trust:
- Round 1: sent 60%
- Round 3: sent 85.4%

They returned $5.25 in Round 2 when they received $10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.
```

```
If I return $6.50:
- Sender ends with: $0.73 + $6.50 = $7.23 (profit of $2.23)
- I end with: $12.81 - $6.50 = $6.31

This gives them a solid return on their high trust, making them almost 45% profit on their initial $5.
```

```
If I return $7.00:
- Sender ends with: $0.73 + $7.00 = $7.73 (profit of $2.73)
- I end with: $12.81 - $7.00 = $5.81

This is even more generous and rewards their very high trust level.
```

```
...
usage:
{
 "input_tokens": 1260,
 "output_tokens": 864,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

```
Visible history before this round:
Your current co-player's last 3 game(s):
- Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $4.89 returned, visible payoff $6.89.
- Round 2 against you, as RECEIVER: saw $3.49 sent, it became $10.47, returned $5.25, visible payoff $5.2200000000000001.
- Round 3 against you, as SENDER: sent $3.5, it became $10.5, saw $6.64 returned, visible payoff $8.14.
Your total visible earnings across all rounds are $18.54.
```

```
This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}
```

**Message 9 role=assistant**

**content:**

Looking at the visible history:

```
Analysis:
- Round 1: Co-player sent $3.0 (60%), received good return
- Round 2: I sent $3.49, co-player returned $5.25 (fair ~50%)
- Round 3: Co-player sent $3.5 (70%), received $6.64 return
```

**\*\*Pattern observed:\*\***

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

**\*\*My decision:\*\***

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

**reasoning:**

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

**usage:**

```
{
 "input_tokens": 1769,
 "output_tokens": 533,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.2200000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$23.66.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 2: I sent \$3.49, co-player returned \$5.25 (~50%)
- Round 3: Co-player sent \$3.5, I returned \$6.64 (~63%)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)



**\*\*Current situation:\*\***

- Co-player sent \$3.02 (60.4% of endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

**\*\*Pattern:\*\***

- Co-player consistently returns around 50% as receiver
- They've shown steady trust as sender
- Cooperative relationship maintained

**\*\*My decision:\*\***

Returning \$4.50:

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of \$1.48)
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This represents ~49.7% return, matching their fairness pattern and maintaining the balanced co operation we've established.

{ 'return': 4.50 }

**Reasoning:**

Looking at the visible history:

Round 2: I sent \$3.49, co-player returned \$5.25 (50.1% of \$10.47)

Round 3: Co-player sent \$3.5, I returned \$6.64 (63.2% of \$10.5)

Round 4: I sent \$3.02 (visible), co-player returned \$4.5 (49.7% of \$9.06)

Current round:

- Co-player sent \$3.02 (60.4% of their endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

Pattern analysis:

- Co-player has been returning around 50% as receiver
- I've been returning slightly more (around 50-63%)
- Cooperation has been consistent
- Co-player sent 60.4% this round (moderate trust)

My decision:

The co-player has been fair and consistent, returning around 50% when they're receiver. They sent \$3.02 (60.4%) showing good trust. I should return a fair amount.

Returning \$4.50:

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of \$1.48)
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

Usage: {

  "input\_tokens": 2220,

  "output\_tokens": 617,

  "reasoning\_tokens": 0

}

**GAME - Agent\_5 - Sender - dyad\_3**

Timestamp: 2026-08-12T07:59:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

  "agent\_type": "standard",

  "dyad\_id": "dyad\_3",

  "move\_index": 4,

  "opponent\_id": "Agent\_6",

  "pairing": {

    "agents": [

      "Agent\_5",

      "Agent\_6"

    ],

  "investor": "Agent\_5",

  "roles": {

    "Agent\_5": "investor",

```

 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant

t value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
  - If I send \$5, receiver gets \$15
  - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
  - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- **Partial trust**: Send some amount (\$2-3) - tests the waters
- **Zero trust**: Send nothing (\$0) - avoid risk but miss cooperative gains
- **Tit-for-tat**: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
  - The receiver saw \$3.03 (slight noise)
  - This became \$9.09 for them
  - They returned \$4.5
  - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
  - My co-player (the same person from Round 1) sent me \$2.92
  - I received \$8.76 (which is 3 × \$2.92)
  - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
  - Give them enough to make their total payoff reasonable
  - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

**usage:**

```
{
 "input_tokens": 824,
 "output_tokens": 873,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

**\*\*Round 1\*\*:** I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

**\*\*Round 2\*\*:** They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

**usage:**

```
{
 "input_tokens": 1160,
 "output_tokens": 419,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible

e payoff \$4.59.  
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.  
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.  
Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)  
Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our history:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)  
**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (49% of received)  
**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)  
**\*\*Round 4\*\*:** They sent \$2.94, I received \$8.82

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relationship.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Looking at our history:

Round 1: I sent ~\$3, they received ~\$9, returned \$4.5  
Round 2: They sent \$3, I received \$9, returned \$4.4 (but they saw \$3.55)  
Round 3: I sent \$3 (they saw \$3.5), they received ~\$10.5, returned \$5.25  
Round 4: They sent \$2.94, I received \$8.82

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them: \$5 - \$2.94 + \$4.41 = \$6.47  
And leave me with: \$8.82 - \$4.41 = \$4.41

That seems fair and continues our mutually beneficial cooperation.

**usage:**

```
{
 "input_tokens": 1549,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our consistent pattern:

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (they saw \$3.55)

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at the history with this co-player:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

```
Usage: {
 "input_tokens": 1959,
 "output_tokens": 430,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Receiver - dyad\_3**

Timestamp: 2026-08-12T07:59:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
```

```

"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with: \$1.97 + \$4.50 = \$6.47 (gain of \$1.47)
- I will end with: \$9.09 - \$4.50 = \$4.59 (gain of \$4.59)



This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

```
{'return': 4.50}
```

**reasoning:**

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has:  $\$5 - \$3.03 = \$1.97$
- Will end up with:  $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with:  $\$9.09 - (\text{what I return})$

Key considerations:

1. **\*\*Fairness\*\***: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. **\*\*Equal split consideration\*\***: If we were to split the total money equally:
  - Total available:  $\$5$  (sender's original) +  $\$9.09$  (what I received) =  $\$14.09$
  - Equal split:  $\$7.045$  each
  - Sender currently has:  $\$1.97$
  - To give sender  $\$7.045$ : I'd need to return  $\$7.045 - \$1.97 = \$5.075$
3. **\*\*Proportional reciprocity\*\***: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
  - 60.6% of  $\$9.09 = \$5.51$
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with multiple agents
  - Building trust could be beneficial for future rounds
  - However, there's communication noise, so exact amounts may not be perceived correctly
  - Agent names are hidden, so reputation effects are limited to behavioral patterns
5. **\*\*Common reciprocity patterns\*\***: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1332,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

**\*\*Round 2 Decision:\*\***

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

{'send': 3.00}

**reasoning:**

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.

If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

**usage:**

{

```
"input_tokens": 856,
"output_tokens": 628,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - **\*\*increased trust!\*\***

**\*\*Current state:\*\***

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level.

I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

{'return': 5.25}

**reasoning:**

Let me analyze the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

**\*\*Round 3 (Current):\*\***

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:

- Round 1: sent 60% (\$3.0)
- Round 3: sent 70% (\$3.5) - even more trust!

2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair

3. They currently have  $\$5 - \$3.5 = \$1.5$  left

4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with:  $\$1.5 + \$5.0 = \$6.5$  (gain of \$1.5)
- I end with:  $\$10.5 - \$5.0 = \$5.5$  (gain of \$5.5)

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

If I return \$5.50:

- Sender ends with:  $\$1.5 + \$5.50 = \$7.0$  (gain of \$2.0)
- I end with:  $\$10.5 - \$5.50 = \$5.0$  (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

usage:

```
{
 "input_tokens": 1230,
 "output_tokens": 964,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

**\*\*Co-player's behavior pattern:\*\***

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

**\*\*Analysis:\*\***

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue matching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{'send': 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$

4.4, ended with \$4.36  
Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

**usage:**

```
{
 "input_tokens": 1702,
 "output_tokens": 665,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with: \$2.2 + \$4.20 = \$6.40 (gain of \$1.40)

- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of  $\$4.20$ )

This maintains fair reciprocity at 50% and ensures mutual profit.

```
{'return': 4.20}
```

**Reasoning:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned
- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of  $\$1.40$ )
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of  $\$4.20$ )

This maintains the 50% return rate that's been consistent in this relationship.

```
Usage: {
 "input_tokens": 2145,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-08-12T07:59:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```
'''json
{"send": 3}
'''
```

##### reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 498,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3



- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was  $\$5 - \$3 + \$4.5 = \$6.5$  (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept  $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them:  $\$2.73 + \$3.40 = \$6.13$  total

And I would keep:  $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

**usage:**

```
{
 "input_tokens": 765,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.5299999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: They returned \$4.5 (~50%)  
 Round 2: They sent \$3, I returned \$3.4  
 Round 3: They returned \$5.8 (~51%)  
 Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)  
 Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to noise)  
 Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)  
 Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, which would be around \$5.

Let me calculate:  
 $\$9.9 * 0.5 = \$4.95$

I'll return \$5 to keep the pattern going.

**usage:**

```
{
 "input_tokens": 1433,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent, they returned \$5.8 (very generous)  
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent (they saw \$3.82), they returned \$5.8  
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)  
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

Usage: {  
 "input\_tokens": 1777,  
 "output\_tokens": 323,  
 "reasoning\_tokens": 0  
}

**GAME - Agent\_8 - Receiver - dyad\_4**

Timestamp: 2026-08-12T08:00:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has  $\$5 - \$3.01 = \$1.99$

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with:  $\$1.99 + \$4.50 = \$6.49$  (gain of \$1.49)
- I end with:  $\$9.03 - \$4.50 = \$4.53$  (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

**reasoning:**

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is  $\$3.01 \times 3$ )
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. **\*\*The sender showed significant trust\*\***: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. **\*\*Payoff analysis\*\***:
  - If I return \$0: Sender gets  $\$5 - \$3.01 + \$0 = \$1.99$ , I get \$9.03
  - If I return half (\$4.52): Sender gets  $\$5 - \$3.01 + \$4.52 = \$6.51$ , I get \$4.51
  - If I return \$3.01: Sender gets  $\$5 - \$3.01 + \$3.01 = \$5.00$ , I get \$6.02
  - If I return \$4.50: Sender gets  $\$5 - \$3.01 + \$4.50 = \$6.49$ , I get \$4.53
  - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. **\*\*Fairness considerations\*\***:
  - The sender's initial endowment was \$5, and they sent \$3.01
  - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
  - If I want us to split the gains equally, I should consider that the total is now  $\$5 - \$3.01 + \$9.03 = \$11.02$
  - Equal split would give us each about \$5.51
  - For sender to get \$5.51: they need  $\$5 - \$3.01 + \text{return} = \$5.51$ , so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1131,
 "reasoning_tokens": 0
}
```

}

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.529999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

**\*\*Round 2 Strategy:\*\***

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

'''json

{"send": 3.00}

'''

**reasoning:**

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings: \$9.03 - \$4.50 = \$4.53

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

**usage:**

```
{
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

**\*\*Co-player's pattern:\*\***

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with: \$1.18 + \$5.80 = \$6.98 (gain of \$1.98)
- I end with: \$11.46 - \$5.80 = \$5.66 (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\***

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

**\*\*Round 2:\*\***

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!

- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is  $2.27 \times 3$ )
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff:  $6.81 - 3.40 = 3.41$  ?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff:  $2.00 + 3.40 = 5.40$

My total earnings should be: \$4.53 (Round 1) + \$5.40 (Round 2) = \$9.93

But visible history says \$10.31... There might be some rounding or noise in the visible earnings too.

**\*\*Round 3:\*\***

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have:  $5 - 3.82 = 1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

**usage:**

```
{
 "input_tokens": 1299,
 "output_tokens": 1178,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**  
Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)



- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

**\*\*Round 4 Strategy:\*\***

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

**\*\*Round 3:\*\***

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

**usage:**

```
{
 "input_tokens": 1833,
 "output_tokens": 737,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should reciprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with:  $\$1.08 + \$6.00 = \$7.08$  (gain of \$2.08)
- I end with:  $\$11.76 - \$6.00 = \$5.76$  (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maintaining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

**Reasoning:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have  $\$5 - \$3.92 = \$1.08$  remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with:  $\$1.08 + \$5.90 = \$6.98$  (gain of \$1.98)
- I end with:  $\$11.76 - \$5.90 = \$5.86$  (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

```
Usage: {
 "input_tokens": 2297,
 "output_tokens": 801,
 "reasoning_tokens": 0
}
```

**ROUND 6**

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_6=Sender, Agent\_5=Receiver  
dyad\_4: Agent\_8=Sender, Agent\_7=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$3; received \$9; Agent\_1 returned \$4.8; payoffs Agent\_2=\$6.8, Agent\_1=\$4.2; noise/visible: sent visible=\$3.21, received visible=\$9.63, returned visible=\$4.09  
Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 9.0, "min": 0, "raw": 4.8}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_4 sent \$3.5; received \$10.5; Agent\_3 returned \$6.3; payoffs Agent\_4=\$7.8, Agent\_3=\$4.2; noise/visible: sent visible=\$4.2, received visible=\$12.6, returned visible=\$6.31  
Action validation: {"returned": {"amount": 6.3, "clamped": false, "max": 10.5, "min": 0, "raw": 6.3}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}  
dyad\_3: Agent\_6 sent \$3; received \$9; Agent\_5 returned \$5; payoffs Agent\_6=\$7, Agent\_5=\$4; noise/visible: sent visible=\$3.33, received visible=\$9.99, returned visible=\$5.6  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_8 sent \$3; received \$9; Agent\_7 returned \$5.4; payoffs Agent\_8=\$7.4, Agent\_7=\$3.6; noise/visible: sent visible=\$3.63, received visible=\$10.89, returned visible=\$4.67  
Action validation: {"returned": {"amount": 5.4, "clamped": false, "max": 9.0, "min": 0, "raw": 5.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 30.55,  
"Agent\_2": 35.449999999999996,  
"Agent\_3": 37.2,  
"Agent\_4": 34.3,  
"Agent\_5": 33.150000000000006,  
"Agent\_6": 32.85,  
"Agent\_7": 35.5,  
"Agent\_8": 30.5

}

Visible balances after round: {

"Agent\_1": 28.1,  
"Agent\_2": 28.919999999999998,  
"Agent\_3": 36.800000000000004,  
"Agent\_4": 36.03,  
"Agent\_5": 34.540000000000006,  
"Agent\_6": 34.269999999999996,  
"Agent\_7": 36.56,  
"Agent\_8": 35.58

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T08:00:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
  "agents": [  
    "Agent\_2",  
    "Agent\_1"  
  ],  
  "investor": "Agent\_2",  
  "roles": {  
    "Agent\_1": "trustee",  
    "Agent\_2": "investor"  
  },  
  "trustee": "Agent\_1"

},  
"role": "investor",  
"role\_label": "Sender",

```
"round": 6,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return \*\*\$3.50\*\*:

- Sender ends with: \$2.66 + \$3.50 = \$6.16 (profit of \$1.16)
- I end with: \$7.02 - \$3.50 = \$3.52 (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is  $2.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep:  $\$5 - \$2.34 + \text{returned} = \$2.66 + \text{returned}$
4. I will keep:  $\$7.02 - \text{returned}$

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return  $\$6.01 - \$2.66 = \$3.35$
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received =  $0.468 \times \$7.02 = \$3.29$
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with  $\$2.66 + \$3.50 = \$6.16$  (up \$1.16 from initial \$5), and I keep  $\$7.02 - \$3.50 = \$3.52$
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 731,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send \*\*\$3.00\*\* (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1:  $\$9.0 - \$3.50 = \$5.50$

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was  $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ( $\$3.50/\$9.0$ )

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

**usage:**

```
{
 "input_tokens": 872,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise

Round 2: Co-player returned \$4.0 when receiving (showing fairness)

Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise.

ion noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return \*\*\$3.25\*\* (approximately 50% of what I received):

- Sender ends with:  $\$2.85 + \$3.25 = \$6.10$  (profit of \$1.10)
- I end with:  $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)
- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff:  $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with:  $\$2.85 + \$3.00 = \$5.85$  (profit of \$0.85)
- I end with:  $\$6.45 - \$3.00 = \$3.45$  (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

**usage:**

```
{
 "input_tokens": 1223,
 "output_tokens": 819,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.  
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.  
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.  
Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise  
Round 2: Co-player returned \$4.0 (generous)  
Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)  
Round 2: They received and returned \$4.0 (generous return)  
Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)  
Round 2: I sent \$3.00, they returned \$4.0  
Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 426,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.  
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.  
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.



Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)

Round 3: Co-player sent \$3.0

Round 4: Co-player returned \$3.9 (about 50% of received)

Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left

- I received \$6.96

I'll return \*\*\$3.50\*\* (about 50% of received):

- Sender ends with: \$2.68 + \$3.50 = \$6.18

- I end with: \$6.96 - \$3.50 = \$3.46

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)

Round 3: Co-player sent \$3.0 (60% trust)

Round 4: Co-player received, returned \$3.9 (about 50% of what they got)

Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left (\$5 - \$2.32)

- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50

- Sender ends with: \$2.68 + \$3.50 = \$6.18 (profit of \$1.18)

- I end with: \$6.96 - \$3.50 = \$3.46

This seems fair and should maintain cooperation.

**usage:**

```
{
 "input_tokens": 2059,
 "output_tokens": 524,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.

Your total visible earnings across all rounds are \$22.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the history:

Round 3: Co-player sent \$3.0 (60%)

Round 4: Co-player returned \$3.9 (50% of received)

Round 5: Co-player sent \$3.0 again (60%)

My co-player has been remarkably consistent, sending \$3.0 every time as sender and returning fairly as receiver. This shows strong cooperative behavior.

I should continue matching this trust level.

```
'''json
{"send": 3.00}
'''
```

**Reasoning:**

Looking at the history:

Round 3: Co-player sent \$3.0, saw \$3.23 returned

Round 4: Co-player returned \$3.9

Round 5: Co-player sent \$3.0, saw \$3.3 returned

My co-player has been very consistent:

- As sender: sent \$3.0 in both rounds (60% of endowment)
- As receiver: returned \$3.9 (about 50% of what they received)

They're showing strong, consistent trust and fairness despite communication noise. I should continue to match this level of trust.

I'll send \$3.00 again to maintain the cooperative relationship.

```
Usage: {
 "input_tokens": 2502,
 "output_tokens": 281,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T08:00:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1** role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user****content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant****content:**

Looking at this first round, I need to balance exploring cooperation while managing risk, especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send \*\*\$3\*\*. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with

2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
  - Show some willingness to cooperate
  - Test the waters
  - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 547,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
```

'''

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is  $\$2.34 \times 3$ )
3. They returned \$3.50
4. My visible payoff was \$4.71, which means:  $\$5 - \$3$  (sent) +  $\$3.50$  (returned) =  $\$5.50$ ... but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is  $\$2.71 \times 3$ )
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 909,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)  
Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50/\$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

Your total visible earnings across all rounds are \$17.970000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)  
Round 2: They sent \$3.00, I returned ~\$4.00  
Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)  
Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%
- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

**usage:**

```
{
 "input_tokens": 1816,
 "output_tokens": 284,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.

Your total visible earnings across all rounds are \$23.270000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the pattern:

My co-player has consistently:

- As receiver: Returned ~50% (Rounds 3 and 5)
- As sender: Sent \$3.00 in previous rounds, now sent \$3.21 (increased trust!)

They're showing even more trust this round. I'll reciprocate by maintaining the ~50% return rate.

```
'''json
{"return": 4.80}
'''
```

**Reasoning:**

Looking at the history with my current co-player:

Round 3: They returned \$3.25 on \$6.45 (~50%)

Round 5: They returned \$3.50 on \$6.96 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver.

In Round 4 and presumably in previous rounds as sender, they sent \$3.00.

Now they've sent me \$3.21, which is even more than their previous \$3.00 sends. This shows increasing trust.

I should maintain the reciprocal relationship by returning approximately 50% of what I received, which would be around \$4.80.

Actually, let me look at what I've been doing:

- Round 2: I received \$8.13, returned \$4.00 (~49%)
- Round 4: I received \$7.80, returned \$3.90 (~50%)

So I've been returning around 50% consistently as well.

Given they sent \$3.21 (received \$9.63), I should return about 50%, which is around \$4.80.



```
Usage: {
 "input_tokens": 2160,
 "output_tokens": 362,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T08:00:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

**\*\*My reasoning:\*\***

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with:  $\$1.64 + \$5.00 = \$6.64$  (profit of \$1.64)
- I end with:  $\$10.08 - \$5.00 = \$5.08$  (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

{'return': 5.00}

**reasoning:**

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received  $\$3.36 \times 3 = \$10.08$
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only  $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created:  $\$5$  (sender's initial) +  $\$10.08$  (what I received) =  $\$15.08$
- Equal split would be  $\$7.54$  each
- Sender currently has:  $\$5 - \$3.36 = \$1.64$
- To give sender  $\$7.54$ , I'd need to return:  $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of  $\$10.08$  ?  $\$6.77$

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have:  $\$1.64 + \$5 = \$6.64$  (profit of \$1.64)
- I would keep:  $\$10.08 - \$5 = \$5.08$  (profit of \$5.08)

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep  $\$6.72$

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground-it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains \$1.64 while I gain \$5.08. If I go higher to \$6, they'd gain \$2.64 but I'd only gain \$4.08-ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating \$6.72 in surplus. Dividing this equally means each of us should gain \$3.36 from that surplus. Since the sender started with \$5 and kept \$1.64, they'd end with \$6.64 total. That leaves me with \$5.08 after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the \$6.72 surplus split evenly. If I return \$3.36, the sender gets their \$1.64 kept plus \$3.36 back, totaling \$5-their original amount. I'd keep \$6.72, which is my \$3.36 share of the surplus plus the original \$3.36 sent. This creates an equal split where we each benefit \$3.36 from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the \$6.72 surplus equally. That would mean returning \$6.72 to give them \$8.36 total, while I keep \$6.72. Let me check if this math works-\$8.36 plus \$6.72 equals \$15.08, which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a \$3.36 gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received (\$5.04), return the original amount (\$3.36), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 2102,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

**\*\*Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They saw a return of \$4.89, giving them a total payoff of \$6.89
- This represents good cooperation from their perspective

**\*\*My decision:\*\***

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play

- Similar level of commitment they showed

```
{'send': 3.50}
```

**reasoning:**

Let me analyze this situation:

**\*\*What I know:\*\***

- Round 1: I was the RECEIVER
  - The sender sent \$3.36 (67.2% of their endowment)
  - I received \$10.08
  - I returned \$5.00
  - I ended with \$5.08

**\*\*Now in Round 2, the visible history shows:\*\***

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff:  $\$1.64 + \$5.00 = \$6.64$  ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

**\*\*Now I'm the SENDER:\*\***

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

**\*\*Decision:\*\***

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

**usage:**

```
{
 "input_tokens": 892,
 "output_tokens": 1059,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so y

ou received \$12.809999999999999. How much do you return? (0-12.809999999999999)  
Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

**\*\*My decision:\*\***

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%).  
They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

**reasoning:**

Let me analyze the situation:

**\*\*Visible history:\*\***

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became \$10.47, returned \$5.25, their payoff \$5.22

**\*\*Current Round 3:\*\***

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

**\*\*Analysis:\*\***

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with: \$0.73 + \$7.00 = \$7.73 (profit of \$2.73)
- I end with: \$12.81 - \$7.00 = \$5.81

This is even more generous and rewards their very high trust level.

...

**usage:**

```
{
 "input_tokens": 1260,
 "output_tokens": 864,
 "reasoning_tokens": 0
}
```

}

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.

Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

**\*\*Pattern observed:\*\***

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

**\*\*My decision:\*\***

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

**reasoning:**

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

**usage:**

```
{
 "input_tokens": 1769,
 "output_tokens": 533,
```

```

 "reasoning_tokens": 0
 }
Message 10 role=user
content:
 Round 5

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 2 against you, as RECEIVER: saw $3.49 sent, it became $10.47, returned $5.25, visible payoff $5.220000000000001.
 - Round 3 against you, as SENDER: sent $3.5, it became $10.5, saw $6.64 returned, visible payoff $8.14.
 - Round 4 against you, as RECEIVER: saw $3.02 sent, it became $9.06, returned $4.5, visible payoff $4.5600000000000005.
 Your total visible earnings across all rounds are $23.66.

 This round, you are the RECEIVER against your current co-player. They sent you $3.02, so you received $9.06. How much do you return? (0-9.06)
 Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
 Looking at the visible history:

 Analysis:
 - Round 2: I sent $3.49, co-player returned $5.25 (~50%)
 - Round 3: Co-player sent $3.5, I returned $6.64 (~63%)
 - Round 4: I sent $3.02, co-player returned $4.5 (~50%)

 Current situation:
 - Co-player sent $3.02 (60.4% of endowment)
 - I received $9.06
 - Co-player has $1.98 remaining

 Pattern:
 - Co-player consistently returns around 50% as receiver
 - They've shown steady trust as sender
 - Cooperative relationship maintained

 My decision:
 Returning $4.50:
 - Sender ends with: $1.98 + $4.50 = $6.48 (profit of $1.48)
 - I end with: $9.06 - $4.50 = $4.56

 This represents ~49.7% return, matching their fairness pattern and maintaining the balanced cooperation we've established.

 {'return': 4.50}
reasoning:
 Looking at the visible history:

 Round 2: I sent $3.49, co-player returned $5.25 (50.1% of $10.47)
 Round 3: Co-player sent $3.5, I returned $6.64 (63.2% of $10.5)
 Round 4: I sent $3.02 (visible), co-player returned $4.5 (49.7% of $9.06)

 Current round:
 - Co-player sent $3.02 (60.4% of their endowment)
 - I received $9.06
 - Co-player has $1.98 remaining

 Pattern analysis:
 - Co-player has been returning around 50% as receiver
 - I've been returning slightly more (around 50-63%)
 - Cooperation has been consistent
 - Co-player sent 60.4% this round (moderate trust)

 My decision:
 The co-player has been fair and consistent, returning around 50% when they're receiver. They sent $3.02 (60.4%) showing good trust. I should return a fair amount.

 Returning $4.50:

```

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of  $\$1.48$ )
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2220,
 "output_tokens": 617,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.

Your total visible earnings across all rounds are \$28.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 3: Co-player sent \$3.5, saw good return (\$6.64)
- Round 4: I sent \$3.02, co-player returned \$4.5 fairly
- Round 5: Co-player sent \$3.5, but only saw \$4.19 returned (lower than usual)

**\*\*Observation:\*\***

From their perspective, Round 5 showed a lower return rate (~40%) compared to our earlier cooperation. This could be due to communication noise mentioned in the instructions.

**\*\*My decision:\*\***

Despite the apparent lower return in Round 5, the overall pattern shows:

- Co-player consistently sends 60-70% as sender
- They return fairly (~50%) as receiver
- Strong cooperative history overall

I should maintain high trust to signal continued cooperation and not let potential noise disrupt our established relationship. Sending \$3.50 (70% of endowment) demonstrates:

- Continued confidence in cooperation
- Consistency with our established pattern
- Resilience to short-term variations

```
{'send': 3.50}
```

**Reasoning:**

Looking at the visible history from my co-player's perspective:

- Round 3: Co-player sent \$3.5, saw \$6.64 returned (good return)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (much lower than before!)

In Round 5, they saw only \$4.19 returned when they sent \$3.5 (which became \$10.5). That's only about 40% return rate, which is lower than their typical ~50% returns and much lower than what I gave them in Round 3 (63%).

This could represent:

1. Communication noise (amounts seen may differ)
2. A change in my return behavior from their perspective

From my perspective, my total earnings are \$28.22. Let me think about what actually happened vs what they saw...



The co-player has been consistently trusting (sending 60-70% of endowment) and they've been fair receivers (50% returns). But in Round 5, from their perspective, I returned less than usual.

Should I:

- Continue high trust to show I'm still cooperative?
- Send less to protect myself since they might think I'm defecting?
- Send the same to maintain consistency?

Given that:

1. They've been very consistent as senders (60-70%)
2. Their return pattern as receivers has been stable (~50%)
3. The lower return they saw in Round 5 might be noise
4. Overall we've had good cooperation

I should continue with high trust to maintain the cooperative relationship. Sending around \$3.50-\$3.75 shows I'm still committed to cooperation.

...

```
Usage: {
 "input_tokens": 2706,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_3 - Receiver - dyad\_2**

Timestamp: 2026-08-12T08:00:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the receiver reciprocates. If I send \$3:

- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = ( $3 \times \text{sent}$ ) - returned

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$1

2.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
  - Show some trust
  - Create value through the multiplier
  - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit
- ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 626,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```
'''json
{"return": 5.25}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ( $3 \times \$3.36$ )
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be  $\$5 - \$3 + \$5 = \$7$ , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ( $3 \times \$3.49$ )

- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with \$5 - \$3.49 + \$5-6 = \$6.51-\$7.51

...

**usage:**

```
{
 "input_tokens": 780,
 "output_tokens": 697,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: \$5 - \$3 + \$5 = \$7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff: \$5 - \$3.5 + \$5.65 = \$7.15

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

**usage:**

```
{
 "input_tokens": 1080,
 "output_tokens": 477,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a subs

tantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:  
50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

**usage:**

```
{
 "input_tokens": 1454,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our history:

Round 2: They sent \$3.5, I returned well

Round 3: I sent ~\$4.27, they returned \$6.5

Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)

Round 3: I sent ~\$4.27 (they saw), they returned \$6.5

Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?

3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

**usage:**

```
{
 "input_tokens": 1812,
 "output_tokens": 554,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$30.500000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our ongoing cooperation:

Round 3: They returned \$6.5 on ~\$12.81 (51%)

Round 4: I returned \$4.5 but they saw \$3.87 due to noise

Round 5: They returned \$4.5 on \$9.06 (50%)

They've sent me \$4.2 this round (more than previous rounds), showing continued trust despite the noise that affected Round 4. I received \$12.6.

To reciprocate their trust and maintain our ~50% return pattern:

```
'''json
{"return": 6.3}
'''
```

**Reasoning:**

Let me trace through the history with my current co-player:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (51%)

Round 4: They sent \$3.75, I received \$11.25 and returned \$4.5, but they saw only \$3.87 (noise)

Round 5: I sent \$3.5 (they saw \$3.02 due to noise), they received \$9.06 and returned \$4.5 (50%)

Round 6: They sent \$4.2, I received \$12.6

We've established a pattern of roughly 50% returns. Despite some noise in communication, we've been maintaining cooperation. They just sent me \$4.2, which is a bit more than previous rounds, showing continued trust.

I should return around 50% to maintain our cooperative relationship:

50% of \$12.6 = \$6.3

This keeps the pattern going.

```
Usage: {
 "input_tokens": 2218,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Sender - dyad\_3**

Timestamp: 2026-08-12T08:01:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.



MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with:  $\$1.97 + \$4.50 = \$6.47$  (gain of \$1.47)
- I will end with:  $\$9.09 - \$4.50 = \$4.59$  (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

**reasoning:**

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has:  $\$5 - \$3.03 = \$1.97$
- Will end up with:  $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with:  $\$9.09 - (\text{what I return})$

Key considerations:

1. **Fairness**: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. **Equal split consideration**: If we were to split the total money equally:
  - Total available:  $\$5$  (sender's original) +  $\$9.09$  (what I received) =  $\$14.09$
  - Equal split:  $\$7.045$  each
  - Sender currently has:  $\$1.97$
  - To give sender  $\$7.045$ : I'd need to return  $\$7.045 - \$1.97 = \$5.075$
3. **Proportional reciprocity**: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
  - 60.6% of  $\$9.09 = \$5.51$
4. **Strategic considerations**:
  - This is a multi-round game with multiple agents
  - Building trust could be beneficial for future rounds
  - However, there's communication noise, so exact amounts may not be perceived correctly

- Agent names are hidden, so reputation effects are limited to behavioral patterns

5. **\*\*Common reciprocity patterns\*\***: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1332,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

**\*\*Round 2 Decision:\*\***

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.  
If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

**usage:**

```
{
 "input_tokens": 856,
 "output_tokens": 628,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - **\*\*increased trust!\*\***

**\*\*Current state:\*\***

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level.

I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with: \$1.5 + \$5.25 = \$6.75 (gain of \$1.75)
- I end with: \$10.5 - \$5.25 = \$5.25 (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

```
{'return': 5.25}
```

**reasoning:**

Let me analyze the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

**\*\*Round 3 (Current):\*\***

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

**Analysis:**

1. My co-player has been very cooperative:
  - Round 1: sent 60% (\$3.0)
  - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have \$5 - \$3.5 = \$1.5 left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with:  $\$1.5 + \$5.0 = \$6.5$  (gain of \$1.5)
- I end with:  $\$10.5 - \$5.0 = \$5.5$  (gain of \$5.5)

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

If I return \$5.50:

- Sender ends with:  $\$1.5 + \$5.50 = \$7.0$  (gain of \$2.0)
- I end with:  $\$10.5 - \$5.50 = \$5.0$  (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

**usage:**

```
{
 "input_tokens": 1230,
 "output_tokens": 964,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior pattern:\*\***

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

**\*\*Analysis:\*\***

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue maintaining their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{'send': 3.00}

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

**usage:**

```
{
 "input_tokens": 1702,
 "output_tokens": 665,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.  
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.  
Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains fair reciprocity at 50% and ensures mutual profit.

{'return': 4.20}

**reasoning:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned
- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains the 50% return rate that's been consistent in this relationship.

**usage:**

```
{
 "input_tokens": 2145,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.

Your total visible earnings across all rounds are \$26.669999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Co-player's recent experience:\*\***

- Round 3: Sent \$3.0, saw \$5.87 returned ? good outcome (\$7.87 total)
- Round 4: As receiver, returned \$4.4 (50%) ? fair reciprocity
- Round 5: Sent \$3.0, saw only \$3.6 returned ? poor outcome (\$5.6 total)

**\*\*Analysis:\*\***

From their perspective, Round 5 was disappointing - they saw much lower returns than in Round 3. However, I know this is likely due to communication noise in the game, not actual low reciprocity.

**\*\*Strategy:\*\***

I should maintain consistent cooperation to signal reliability. Despite the noise affecting what they saw, continuing to send at the 60% level shows I'm trustworthy and committed to cooperation. This should help stabilize the relationship through the noise.

{'send': 3.00}

**Reasoning:**

Looking at the co-player's visible history:

- Round 3: Sent \$3.0 as sender, saw \$5.87 returned ? earned \$7.87
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? earned \$5.6

This is interesting. In Round 5, from their perspective:

- They sent \$3.0 and saw only \$3.6 returned
- That's only 40% of the \$9.0 they sent (which became after multiplication)
- This is much lower than what they saw in Round 3 (\$5.87)

From my actual history, in Round 5:

- They sent \$2.8, I received \$8.4, I returned \$4.20
- But due to noise, they saw \$3.0 sent and saw \$3.6 returned

So from their perspective, they had a very poor outcome in Round 5 - earning only \$5.6 compared to \$7.87 in Round 3.

This might make them less trusting going forward. They might:

1. Send less
2. Return less if they're the receiver
3. Still cooperate but be more cautious

Given this situation, I have a few options:

1. Continue sending \$3.00 (60%) to maintain cooperation and show I'm still trustworthy
2. Send less to hedge against possible reduced reciprocity
3. Send more to try to rebuild trust

I think option 1 is best - continue the pattern and show consistency. The poor outcome in Round 5 from their view was due to noise, not my actual behavior. Maintaining consistency might help preserve the relationship.

Let me send \$3.00 again.

Usage: {

"input\_tokens": 2637,  
"output\_tokens": 638,  
"reasoning\_tokens": 0

```
}
```

**GAME - Agent\_5 - Receiver - dyad\_3**

Timestamp: 2026-08-12T08:01:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_3",

"move\_index": 5,

"opponent\_id": "Agent\_6",

"pairing": {

"agents": [

"Agent\_6",

"Agent\_5"

],

"investor": "Agent\_6",

"roles": {

"Agent\_5": "trustee",

"Agent\_6": "investor"

},

"trustee": "Agent\_5"

},

"role": "trustee",

"role\_label": "Receiver",

"round": 6,

"task": "game",

"task\_index": 0

```
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized



- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
  - If I send \$5, receiver gets \$15
  - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
  - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- **Partial trust**: Send some amount (\$2-3) - tests the waters
- **Zero trust**: Send nothing (\$0) - avoid risk but miss cooperative gains
- **Tit-for-tat**: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time

- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
  - The receiver saw \$3.03 (slight noise)
  - This became \$9.09 for them
  - They returned \$4.5
  - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
  - My co-player (the same person from Round 1) sent me \$2.92
  - I received \$8.76 (which is 3 × \$2.92)
  - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)

- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
  - Give them enough to make their total payoff reasonable
  - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

usage:

```
{
 "input_tokens": 824,
 "output_tokens": 873,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

**\*\*Round 1\*\*:** I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

**\*\*Round 2\*\*:** They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should

continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

**usage:**

```
{
 "input_tokens": 1160,
 "output_tokens": 419,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our history:

```
Round 1: I sent ~$3, they returned $4.5 (50% of received)
Round 2: They sent $3, I returned $4.4 (49% of received)
Round 3: I sent $3, they returned $5.25 (exactly 50% of received)
Round 4: They sent $2.94, I received $8.82
```

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relationship.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Looking at our history:

Round 1: I sent ~\$3, they received ~\$9, returned \$4.5  
Round 2: They sent \$3, I received \$9, returned \$4.4 (but they saw \$3.55)  
Round 3: I sent \$3 (they saw \$3.5), they received ~\$10.5, returned \$5.25  
Round 4: They sent \$2.94, I received \$8.82

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them: \$5 - \$2.94 + \$4.41 = \$6.47

And leave me with: \$8.82 - \$4.41 = \$4.41

That seems fair and continues our mutually beneficial cooperation.

**usage:**

```
{
 "input_tokens": 1549,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our consistent pattern:

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (they saw \$3.55)

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1959,
 "output_tokens": 430,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.

Your total visible earnings across all rounds are \$29.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our continued cooperation:

```

Round 3: I sent $3, they returned $5.25 (50%)
Round 4: They sent $3, I returned $4.4 (they saw $5.08)
Round 5: I sent $3, they returned $4.2 (50%)
Round 6: They sent $3.33, I received $9.99

```

My co-player continues to be highly consistent and fair, returning approximately 50% each time. They've proven to be a trustworthy partner across all rounds.

I should maintain this reciprocal pattern by returning about 50% of what I received: 50% of \$9.99 = \$5.0

```

'''json
{"return": 5.0}
'''

```

**Reasoning:**

Looking at the history:

```

Round 3: I sent $3, they returned $5.25 (50% of $10.5)
Round 4: They sent $3, I returned $4.4 (they saw $5.08 due to noise)
Round 5: I sent $3, they returned $4.2 (50% of $8.4)
Round 6: They sent $3.33, I received $9.99

```

My co-player has been very consistent - always returning about 50% of what they receive. They've been a reliable, fair partner throughout.

I should continue this pattern. Returning 50% of \$9.99 = \$4.995, so about \$5.0

This maintains our reciprocal relationship and ensures continued cooperation.

```

Usage: {
 "input_tokens": 2361,
 "output_tokens": 372,
 "reasoning_tokens": 0
}

```

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-08-12T08:01:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",

```

```

 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has \$5 - \$3.01 = \$1.99

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with:  $\$1.99 + \$4.50 = \$6.49$  (gain of \$1.49)
- I end with:  $\$9.03 - \$4.50 = \$4.53$  (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

#### reasoning:

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is  $\$3.01 \times 3$ )
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. **\*\*The sender showed significant trust\*\***: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. **\*\*Payoff analysis\*\***:
  - If I return \$0: Sender gets  $\$5 - \$3.01 + \$0 = \$1.99$ , I get \$9.03
  - If I return half (\$4.52): Sender gets  $\$5 - \$3.01 + \$4.52 = \$6.51$ , I get \$4.51
  - If I return \$3.01: Sender gets  $\$5 - \$3.01 + \$3.01 = \$5.00$ , I get \$6.02
  - If I return \$4.50: Sender gets  $\$5 - \$3.01 + \$4.50 = \$6.49$ , I get \$4.53
  - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. **\*\*Fairness considerations\*\***:
  - The sender's initial endowment was \$5, and they sent \$3.01
  - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
  - If I want us to split the gains equally, I should consider that the total is now  $\$5 - \$3.01 + \$9.03 = \$11.02$
  - Equal split would give us each about \$5.51
  - For sender to get \$5.51: they need  $\$5 - \$3.01 + \text{return} = \$5.51$ , so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

#### usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1131,
 "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.5299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

#### Message 5 role=assistant

##### content:

Looking at the situation:



**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

**\*\*Round 2 Strategy:\*\***

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings:  $\$9.03 - \$4.50 = \$4.53$

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

**usage:**

```
{
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

### Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.810000000000005, returned \$3.4, visible payoff \$3.410000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

**\*\*Co-player's pattern:\*\***

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with:  $\$1.18 + \$5.80 = \$6.98$  (gain of \$1.98)
- I end with:  $\$11.46 - \$5.80 = \$5.66$  (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\***

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

**\*\*Round 2:\*\***

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!
- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is  $\$2.27 \times 3$ )
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff:  $\$6.81 - \$3.40 = \$3.41$  ?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff:  $\$2.00 + \$3.40 = \$5.40$

My total earnings should be:  $\$4.53$  (Round 1) +  $\$5.40$  (Round 2) =  $\$9.93$

But visible history says \$10.31... There might be some rounding or noise in the visible ea

rnings too.

**\*\*Round 3:\*\***

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have: \$5 - \$3.82 = \$1.18

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

**usage:**

```
{
 "input_tokens": 1299,
 "output_tokens": 1178,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)
- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

**\*\*Round 4 Strategy:\*\***

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

**\*\*Round 3:\*\***

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates
- ...

**usage:**

```
{
 "input_tokens": 1833,
 "output_tokens": 737,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should rec

iprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with:  $\$1.08 + \$6.00 = \$7.08$  (gain of \$2.08)
- I end with:  $\$11.76 - \$6.00 = \$5.76$  (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maintaining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

**reasoning:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have  $\$5 - \$3.92 = \$1.08$  remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with:  $\$1.08 + \$5.90 = \$6.98$  (gain of \$1.98)
- I end with:  $\$11.76 - \$5.90 = \$5.86$  (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

**usage:**

```
{
 "input_tokens": 2297,
 "output_tokens": 801,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.

Your total visible earnings across all rounds are \$28.909999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 3: Co-player sent \$3.0, received \$5.23, earned \$7.23
- Round 4: Co-player as receiver returned ~50.5% fairly
- Round 5: Co-player sent \$3.0, received even better return (\$6.98), earned \$8.98

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 as sender (reliable trust pattern)
- They return fairly as receiver (~50% return rate)
- Returns have been getting better over time, showing our cooperation strengthens
- Our partnership has been mutually profitable

**\*\*Round 6 Strategy:\*\***

The cooperation has been excellent and consistent. I'll maintain the same trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

```
'''json
{"send": 3.00}
'''
```

**Reasoning:**

Looking at the visible history:

**\*\*Co-player's recent pattern:\*\***

- Round 3: As sender, sent \$3.0, received \$5.23 back, earned \$7.23
- Round 4: As receiver, saw \$3.3 sent, returned \$5.0 (~50.5%), earned \$4.90
- Round 5: As sender, sent \$3.0, received \$6.98 back, earned \$8.98

**\*\*Key observations:\*\***

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5 visible)
- When they're the receiver, they return fairly (~50% - Round 4)
- They've received increasing returns over time, with Round 5 showing \$6.98 returned
- Our partnership has been very profitable for both of us

**\*\*What I know from my actual actions:\*\***

- Round 1: I was receiver, returned \$4.50
- Round 2: I was sender, sent \$3.00
- Round 3: I was receiver, returned \$5.80
- Round 4: I was sender, sent \$3.00
- Round 5: I was receiver, returned \$6.00

**\*\*Current situation:\*\***

- Round 6: I'm the sender
- My co-player has been trustworthy and fair
- They consistently send \$3.0 when they're the sender
- They return fairly when they're the receiver (~50%)
- The relationship has been mutually beneficial

**\*\*Strategy consideration:\*\***

Given the strong pattern of cooperation and mutual benefit, I should continue sending \$3.00 to maintain the trustworthy relationship. This has been working well for both of us.

However, I notice the co-player is seeing increasing returns (\$5.23 ? \$6.98 in their visible history). This might encourage them to continue being cooperative.

...

```
Usage: {
 "input_tokens": 2800,
 "output_tokens": 655,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Receiver - dyad\_4**

Timestamp: 2026-08-12T08:01:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
```

```

"agents": [
 "Agent_8",
 "Agent_7"
],
"investor": "Agent_8",
"roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being reas

onably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 498,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.5299999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)



Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was  $\$5 - \$3 + \$4.5 = \$6.5$  (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept  $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them:  $\$2.73 + \$3.40 = \$6.13$  total

And I would keep:  $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

**usage:**

```
{
 "input_tokens": 765,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant****content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 8 role=user****content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant****content:**

Looking at the history with my current co-player:

Round 1: They returned \$4.5 (~50%)

Round 2: They sent \$3, I returned \$3.4

Round 3: They returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)  
Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to noise)  
Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, which would be around \$5.

Let me calculate:  
 $\$9.9 * 0.5 = \$4.95$

I'll return \$5 to keep the pattern going.

**usage:**

```
{
 "input_tokens": 1433,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent, they returned \$5.8 (very generous)  
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent (they saw \$3.82), they returned \$5.8  
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)  
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

**usage:**

```
{
 "input_tokens": 1777,
 "output_tokens": 323,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.

Your total visible earnings across all rounds are \$31.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history with my current co-player:

Round 3: They returned \$5.8 (~51%)  
Round 4: I returned \$5 (~56%)  
Round 5: They returned \$6.0 (~51%)  
Round 6: They sent \$3.63, I received \$10.89

This co-player has been consistently cooperative, maintaining around 50-51% return rates. I should continue this pattern of fair reciprocity.

```
'''json
{"return": 5.4}
'''
```

**Reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50%)  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent, they returned \$5.8 (~51%)  
Round 4: They sent \$3, I returned \$5 (~56%)  
Round 5: I sent, they returned \$6.0 (~51%)  
Round 6: They sent \$3.63, I received \$10.89

This co-player has been extremely consistent and cooperative, always returning around 50-51% of what they receive. I've also been cooperative, though my return in Round 4 was slightly higher.

To maintain this strong cooperative relationship, I should continue returning around 50% of what I receive.

50% of \$10.89 = \$5.445

I'll return \$5.4 or \$5.5 to maintain the pattern.

```
Usage: {
 "input_tokens": 2136,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

## ROUND 7

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_3=Sender, Agent\_4=Receiver  
dyad\_3: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_8=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$5.5; payoffs Agent\_1=\$7.5, Agent\_2=\$3.5; noise/visible: sent visible=\$3.5, received visible=\$10.5, returned visible=\$6.13  
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_3 sent \$3.75; received \$11.25; Agent\_4 returned \$6.5; payoffs Agent\_3=\$7.75, Agent\_4=\$4.75; noise/visible: sent visible=\$4.29, received visible=\$12.87, returned visible=\$5.93  
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 11.25, "min": 0, "raw": 6.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}  
dyad\_3: Agent\_5 sent \$3; received \$9; Agent\_6 returned \$5.05; payoffs Agent\_5=\$7.05, Agent\_6=\$3.95; noise/visible: sent visible=\$3.37, received visible=\$10.11, returned visible=\$4.25  
Action validation: {"returned": {"amount": 5.05, "clamped": false, "max": 9.0, "min": 0, "raw": 5.05}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_7 sent \$3; received \$9; Agent\_8 returned \$4.8; payoffs Agent\_7=\$6.8, Agent\_8=\$4.2; noise/visible: sent visible=\$3.19, received visible=\$9.57, returned visible=\$5.59  
Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 9.0, "min": 0, "raw": 4.8}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

### Balances after round: {

"Agent\_1": 38.05,  
"Agent\_2": 38.949999999999996,  
"Agent\_3": 44.95,  
"Agent\_4": 39.05,  
"Agent\_5": 40.2,  
"Agent\_6": 36.800000000000004,  
"Agent\_7": 42.3,  
"Agent\_8": 34.7

}

### Visible balances after round: {

"Agent\_1": 36.230000000000004,  
"Agent\_2": 33.92,  
"Agent\_3": 43.980000000000004,  
"Agent\_4": 42.400000000000006,  
"Agent\_5": 40.790000000000006,  
"Agent\_6": 39.33,  
"Agent\_7": 44.150000000000006,  
"Agent\_8": 40.35

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T08:01:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {

```

"agents": [
 "Agent_1",
 "Agent_2"
],
"investor": "Agent_1",
"roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance exploring cooperation while managing risk,

especially given the communication noise factor.

Since I don't have any information about my co-player's tendencies yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Creates potential mutual benefit (they'll receive 3x what I send)
- Limits my downside risk on this first interaction

I'll send \*\*\$3\*\*. This means my co-player will receive \$9, and I'll keep \$2 initially. If they return at least \$3, we'll both come out ahead of where we started.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I have \$5 to start with
2. Any amount I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may differ from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a balance between maximizing individual payoff and building trust
- The noise factor adds uncertainty

Standard game theory analysis:

- If I send \$5, the receiver gets \$15
- A rational receiver might return at least \$5 to split evenly or more to encourage future cooperation
- However, a selfish receiver might return nothing
- This is a one-shot interaction with this particular partner (random pairing)

Since this is round 1 and I have no information about other agents' behavior, I should probably:

1. Not send nothing (that gains nothing and learns nothing)
2. Not send everything (too risky without trust established)
3. Send a moderate amount to:
  - Show some willingness to cooperate
  - Test the waters
  - Leave room for learning

A common starting strategy in trust games is to send about 40-60% of the endowment. This shows cooperative intent while limiting downside risk.

I'll send \$3:

- I keep \$2 guaranteed
- Receiver gets \$9
- If they return even \$3, I end up with \$5 (break even)
- If they return more, we both benefit
- I learn about the general cooperation level in this game

This seems like a balanced approach for Round 1.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 547,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.

Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is  $2.34 \times 3$ )
3. They returned \$3.50
4. My visible payoff was \$4.71, which means:  $\$5 - \$3 \text{ (sent)} + \$3.50 \text{ (returned)} = \$5.50...$  but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is  $2.71 \times 3$ )
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 909,
 "reasoning_tokens": 0
}
```



**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50 / \$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.8000000000000001. How much do you return? (0-7.8000000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

Your total visible earnings across all rounds are \$17.9700000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)

Round 2: They sent \$3.00, I returned ~\$4.00

Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)

Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%
- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

**usage:**

```
{
 "input_tokens": 1816,
 "output_tokens": 284,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.4599999999999999.

Your total visible earnings across all rounds are \$23.270000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the pattern:

My co-player has consistently:

- As receiver: Returned ~50% (Rounds 3 and 5)
- As sender: Sent \$3.00 in previous rounds, now sent \$3.21 (increased trust!)

They're showing even more trust this round. I'll reciprocate by maintaining the ~50% return rate.

```
'''json
{"return": 4.80}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 3: They returned \$3.25 on \$6.45 (~50%)  
Round 5: They returned \$3.50 on \$6.96 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver.

In Round 4 and presumably in previous rounds as sender, they sent \$3.00.

Now they've sent me \$3.21, which is even more than their previous \$3.00 sends. This shows increasing trust.

I should maintain the reciprocal relationship by returning approximately 50% of what I received, which would be around \$4.80.

Actually, let me look at what I've been doing:

- Round 2: I received \$8.13, returned \$4.00 (~49%)
- Round 4: I received \$7.80, returned \$3.90 (~50%)

So I've been returning around 50% consistently as well.

Given they sent \$3.21 (received \$9.63), I should return about 50%, which is around \$4.80.

**usage:**

```
{
 "input_tokens": 2160,
 "output_tokens": 362,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.

Your total visible earnings across all rounds are \$28.1.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the pattern with my co-player:

As sender: Consistently sent \$3.00 (Rounds 4, 6)

As receiver: Returned \$3.50 on \$6.96 (Round 5, ~50%)

This partnership has been stable and mutually beneficial. I'll continue the cooperative pattern.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at the visible history with my current co-player:

As sender (Rounds 4, 6): They've sent \$3.00 consistently

As receiver (Round 5): They returned \$3.50 on \$6.96 (~50%)

This is a highly cooperative and stable partner. They maintain consistent sending amounts and fair return rates. I should continue matching their cooperative behavior.

I'll send \$3 again to maintain the mutual cooperation.

Usage: {

```
"input_tokens": 2484,
"output_tokens": 196,
"reasoning_tokens": 0
}
```

## GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T08:01:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

##### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.34, that is 46.8% of their total endowment. You received \$7.02. How much do you return to the sender? (0-7.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.34 (46.8% of their endowment)
- I received \$7.02 after the 3x multiplication
- The sender currently has \$2.66 (\$5 - \$2.34)

I want to reward this trust while also keeping a reasonable amount. A fair approach would be to ensure both of us benefit from the cooperation.

If I return \*\*\$3.50\*\*:

- Sender ends with: \$2.66 + \$3.50 = \$6.16 (profit of \$1.16)
- I end with: \$7.02 - \$3.50 = \$3.52 (profit of \$3.52)

This rewards the sender's trust with a positive return while maintaining the incentive for cooperation in future rounds.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

I'm the receiver in round 1. Let me analyze the situation:

- The sender had \$5 initially
- The sender sent \$2.34 (which is 46.8% of \$5)
- I received \$7.02 (which is \$2.34 × 3)
- I need to decide how much to return (between \$0 and \$7.02)

Key considerations:

1. The sender showed moderate trust by sending 46.8% of their endowment
2. If I return a fair amount, this could establish good cooperation for future rounds
3. The sender will keep: \$5 - \$2.34 + returned = \$2.66 + returned
4. I will keep: \$7.02 - returned

From a fairness perspective, I could consider:

- Equal split of total money: Total is \$5 (sender's initial) + \$7.02 (what I have) = \$12.02. Equal split would be ~\$6.01 each. The sender already has \$2.66, so I'd need to return \$6.01 - \$2.66 = \$3.35
- Proportional to investment: The sender invested 46.8% of their endowment. A proportional return might be around 46.8% of what I received =  $0.468 \times \$7.02 = \$3.29$
- Ensuring sender gains from trust: If I return \$3.50, the sender ends with \$2.66 + \$3.50 = \$6.16 (up \$1.16 from initial \$5), and I keep \$7.02 - \$3.50 = \$3.52
- Being generous to encourage future cooperation: Returning around \$3.50-\$4.00 would reward the sender's trust

Given that:

- This is a multi-round game with potential for repeated interactions
- The sender showed moderate trust (not maximum, but substantial)

...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 731,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)
- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send \*\*\$3.00\*\* (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1: \$9.0 - \$3.50 = \$5.50

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was  $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ( $\$3.50/\$9.0$ )

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

**usage:**

```
{
 "input_tokens": 872,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise

Round 2: Co-player returned \$4.0 when receiving (showing fairness)

Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return \*\*\$3.25\*\* (approximately 50% of what I received):

- Sender ends with: \$2.85 + \$3.25 = \$6.10 (profit of \$1.10)
- I end with: \$6.45 - \$3.25 = \$3.20

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)
- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff: \$2 + \$4.0 = \$6.0
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.



I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with:  $\$2.85 + \$3.00 = \$5.85$  (profit of \$0.85)
- I end with:  $\$6.45 - \$3.00 = \$3.45$  (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

**usage:**

```
{
 "input_tokens": 1223,
 "output_tokens": 819,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise

Round 2: Co-player returned \$4.0 (generous)

Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)

Round 2: They received and returned \$4.0 (generous return)

Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)

Round 2: I sent \$3.00, they returned \$4.0

Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 426,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)

Round 3: Co-player sent \$3.0

Round 4: Co-player returned \$3.9 (about 50% of received)

Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left
- I received \$6.96

I'll return \*\*\$3.50\*\* (about 50% of received):

- Sender ends with: \$2.68 + \$3.50 = \$6.18
- I end with: \$6.96 - \$3.50 = \$3.46

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)

Round 3: Co-player sent \$3.0 (60% trust)

Round 4: Co-player received, returned \$3.9 (about 50% of what they got)

Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left (\$5 - \$2.32)
- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50
- Sender ends with:  $\$2.68 + \$3.50 = \$6.18$  (profit of \$1.18)
- I end with:  $\$6.96 - \$3.50 = \$3.46$

This seems fair and should maintain cooperation.

**usage:**

```
{
 "input_tokens": 2059,
 "output_tokens": 524,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.

Your total visible earnings across all rounds are \$22.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history:

Round 3: Co-player sent \$3.0 (60%)

Round 4: Co-player returned \$3.9 (50% of received)

Round 5: Co-player sent \$3.0 again (60%)

My co-player has been remarkably consistent, sending \$3.0 every time as sender and returning fairly as receiver. This shows strong cooperative behavior.

I should continue matching this trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the history:

Round 3: Co-player sent \$3.0, saw \$3.23 returned

Round 4: Co-player returned \$3.9

Round 5: Co-player sent \$3.0, saw \$3.3 returned

My co-player has been very consistent:

- As sender: sent \$3.0 in both rounds (60% of endowment)
- As receiver: returned \$3.9 (about 50% of what they received)

They're showing strong, consistent trust and fairness despite communication noise. I should continue to match this level of trust.

I'll send \$3.00 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2502,
 "output_tokens": 281,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$28.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

#### Assistant output:

Looking at the history:

Round 4: Co-player returned \$3.9

Round 5: Co-player sent \$3.0

Round 6: Co-player returned \$4.8 (about 50% of received)

Round 7: Co-player sent \$3.5 (70% - increased trust!)

My co-player has increased their trust level from 60% to 70%! This shows strong cooperative intent.

Current situation:

- Co-player has \$1.50 left
- I received \$10.50

I'll return \*\*\$5.50\*\* (about 52% of received):

- Sender ends with: \$1.50 + \$5.50 = \$7.00 (profit of \$2.00)
- I end with: \$10.50 - \$5.50 = \$5.00

This rewards their increased trust and maintains strong cooperation.

```
'''json
{"return": 5.50}
'''
```

#### Reasoning:

Looking at the history:

Round 4: Co-player returned \$3.9 (50% of received)

Round 5: Co-player sent \$3.0 (60%)

Round 6: Co-player returned \$4.8 (about 50% of received, which was \$9.63)

Round 7: Co-player sent \$3.5 (70%)

My co-player has increased their trust level to 70%! They've been consistently fair and cooperative throughout, and now they're showing even more trust.

Current situation:

- Co-player has \$1.50 left
- I received \$10.50

I should return a fair amount. Let me calculate:

- If I return \$5.25 (50% of received):
  - Sender ends with: \$1.50 + \$5.25 = \$6.75
  - I end with: \$10.50 - \$5.25 = \$5.25

That seems fair and reciprocates the increased trust. Let me see if I should go higher given they increased their send.

Actually, they increased to 70%, showing more trust. I should reward that. Maybe return a bit more to encourage continued cooperation.

If I return \$5.50:

- Sender ends with: \$1.50 + \$5.50 = \$7.00
- I end with: \$10.50 - \$5.50 = \$5.00

This gives them a solid profit and rewards their increased trust.

Usage: {  
"input\_tokens": 2869,

```
"output_tokens": 572,
"reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Sender - dyad\_2**

Timestamp: 2026-08-12T08:01:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust with caution since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if the receiver reciprocates. If I send \$3:

- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even \$4-5, we both end up better off than the initial \$5
- I risk at most \$3

This shows willingness to cooperate while not being overly vulnerable in this first interaction.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1, and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I receive \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = ( $3 \times \text{sent}$ ) - returned

This is a classic trust game. The key considerations are:

1. If I send money, it creates value (3x multiplier)
2. But I'm vulnerable - the receiver could return nothing
3. A rational, mutually beneficial strategy would be to send some amount, and the receiver returns enough that we both benefit

Let me think about different scenarios:

- If I send \$0: I keep \$5, receiver gets \$0. Total: \$5
- If I send \$5: Receiver gets \$15. If they return \$7.50, we each get \$7.50. Total: \$15
- If I send \$2.50: Receiver gets \$7.50. If they return \$3.75, we each get \$6.25. Total: \$12.50

The optimal social outcome is if I send everything and we split the tripled amount fairly. However, this requires trust.

Since this is Round 1 and I have no information about my co-player:

- I should send a moderate amount to:
  - Show some trust
  - Create value through the multiplier
  - Not risk everything initially

A common starting strategy in trust games is to send about half. This shows willingness to cooperate while limiting downside risk.

I'll send \$2.50 (or \$3, since the problem might expect whole numbers).

Actually, looking at the format, they seem to allow decimal amounts. But to be safe, let me send \$3. This:

- Shows significant trust
- Creates \$9 for the receiver
- If they return even \$4-5, we both benefit

```

...
usage:
{
 "input_tokens": 500,
 "output_tokens": 626,
 "reasoning_tokens": 0
}

```

**Message 4 role=user**

**content:**  
 Round 2

Visible history before this round:  
 Your current co-player's last 3 game(s):  
 - Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.  
 Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)  
 Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**  
 Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and willing to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```

'''json
{"return": 5.25}
'''

```

**reasoning:**  
 Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ( $3 \times \$3.36$ )
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be  $\$5 - \$3 + \$5 = \$7$ , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ( $3 \times \$3.49$ )
- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)

- I got \$10.47
- If I return \$5-6, I keep \$4.47-5.47
- They would end up with \$5 - \$3.49 + \$5-6 = \$6.51-7.51
- ...

**usage:**

```
{
 "input_tokens": 780,
 "output_tokens": 697,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: \$5 - \$3 + \$5 = \$7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff: \$5 - \$3.5 + \$5.65 = \$7.15

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

**usage:**

```
{
```



```
"input_tokens": 1080,
"output_tokens": 477,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:

50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

**usage:**

```
{
 "input_tokens": 1454,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned

d \$6.5, visible payoff \$6.3099999999999999.  
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.  
Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our history:

Round 2: They sent \$3.5, I returned well

Round 3: I sent ~\$4.27, they returned \$6.5

Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)

Round 3: I sent ~\$4.27 (they saw), they returned \$6.5

Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?
3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

**usage:**

```
{
 "input_tokens": 1812,
 "output_tokens": 554,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:  
Your current co-player's last 3 game(s):  
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.  
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.  
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.  
Your total visible earnings across all rounds are \$30.500000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)  
Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our ongoing cooperation:

Round 3: They returned \$6.5 on ~\$12.81 (51%)  
Round 4: I returned \$4.5 but they saw \$3.87 due to noise  
Round 5: They returned \$4.5 on \$9.06 (50%)

They've sent me \$4.2 this round (more than previous rounds), showing continued trust despite the noise that affected Round 4. I received \$12.6.

To reciprocate their trust and maintain our ~50% return pattern:

```
'''json
{"return": 6.3}
'''
```

**reasoning:**

Let me trace through the history with my current co-player:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)  
Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)  
Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (51%)  
Round 4: They sent \$3.75, I received \$11.25 and returned \$4.5, but they saw only \$3.87 (noise)  
Round 5: I sent \$3.5 (they saw \$3.02 due to noise), they received \$9.06 and returned \$4.5 (50%)  
Round 6: They sent \$4.2, I received \$12.6

We've established a pattern of roughly 50% returns. Despite some noise in communication, we've been maintaining cooperation. They just sent me \$4.2, which is a bit more than previous rounds, showing continued trust.

I should return around 50% to maintain our cooperative relationship:  
50% of \$12.6 = \$6.3

This keeps the pattern going.

**usage:**

```
{
 "input_tokens": 2218,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:  
Your current co-player's last 3 game(s):  
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.  
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.  
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.  
Your total visible earnings across all rounds are \$36.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our continued partnership:

Round 4: They sent \$3.75, I returned well (though noise affected visibility)

Round 5: They returned \$4.5 on \$9.06

Round 6: They sent \$3.5, I returned ~\$6.3 (~60%)

We've built a strong cooperative relationship over 6 rounds with consistent sending and fair returns. They've maintained trust and cooperation throughout.

To continue this successful pattern:

```
'''json
{"send": 3.75}
'''
```

**Reasoning:**

Looking at the visible history from my co-player's perspective:

Round 4: They sent \$3.75, saw \$3.87 returned (I actually sent \$4.5)

Round 5: They returned \$4.5 on \$9.06 they received (I sent \$3.02 which they saw)

Round 6: They sent \$3.5, saw \$6.31 returned (I actually returned \$6.3)

The noise has been relatively small. They've been consistently cooperative:

- Sending between \$3.5-\$4.2
- Returning around 50% of what they receive
- Showing trust even when returns looked lower due to noise

We've established a strong cooperative pattern over 6 rounds. I should continue sending a substantial amount to maintain this relationship.

I'll send around \$3.5-\$4 to match their commitment level.

```
Usage: {
 "input_tokens": 2584,
 "output_tokens": 326,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Receiver - dyad\_2**

Timestamp: 2026-08-12T08:01:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user****content:**

Round 1: You are the RECEIVER. the sender sent \$3.36, that is 67.2% of their total endowment. You received \$10.08. How much do you return to the sender? (0-10.08)

**Message 3 role=assistant****content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender showed significant trust by sending \$3.36 (67.2% of their endowment)
- I received \$10.08 (after the 3x multiplication)
- The sender currently has \$1.64 remaining

**\*\*My reasoning:\*\***

The sender demonstrated high trust in the first round. A fair return should:

1. Reward their trust and generosity
2. Ensure both parties benefit from cooperation
3. Establish a positive dynamic for future rounds

If I return \$5.00:

- Sender ends with:  $\$1.64 + \$5.00 = \$6.64$  (profit of \$1.64)
- I end with:  $\$10.08 - \$5.00 = \$5.08$  (profit of \$5.08)

This gives us both meaningful gains from cooperation, with the sender making a 32.8% return on their initial endowment, which rewards their trust.

```
{'return': 5.00}
```

**reasoning:**

Let me analyze this situation carefully.

As the RECEIVER:

- The sender sent \$3.36 out of their \$5 endowment (67.2%)
- I received  $\$3.36 \times 3 = \$10.08$
- I need to decide how much to return (between \$0 and \$10.08)

Key considerations:

1. The sender showed significant trust by sending 67.2% of their endowment
2. If I return nothing, I keep \$10.08 and the sender gets only  $\$5 - \$3.36 = \$1.64$
3. A fair return might be based on the sender's generosity

Let me think about different return strategies:

Option 1: Equal split of total value

- Total value created:  $\$5$  (sender's initial) +  $\$10.08$  (what I received) =  $\$15.08$
- Equal split would be  $\$7.54$  each
- Sender currently has:  $\$5 - \$3.36 = \$1.64$
- To give sender  $\$7.54$ , I'd need to return:  $\$7.54 - \$1.64 = \$5.90$

Option 2: Proportional return based on trust shown

- Sender sent 67.2% of their endowment, showing high trust
- I could return a similar proportion of what I received
- 67.2% of  $\$10.08$  ?  $\$6.77$

Option 3: Return amount that gives sender profit

- If I return \$5, sender would have:  $\$1.64 + \$5 = \$6.64$  (profit of  $\$1.64$ )
- I would keep:  $\$10.08 - \$5 = \$5.08$  (profit of  $\$5.08$ )

Option 4: Return enough so sender gets their \$5 back

- Sender needs \$3.36 more to get back to \$5
- If I return \$3.36, sender gets their initial endowment back and I keep  $\$6.72$

The sender's high trust deserves reciprocation that benefits both of us while encouraging continued cooperation. Returning \$5 seems like a solid middle ground-it gives the sender a meaningful profit while I still come out ahead, and it demonstrates I'm reliable for future interactions.

But reconsidering the numbers: at \$5, the sender gains  $\$1.64$  while I gain  $\$5.08$ . If I go higher to \$6, they'd gain  $\$2.64$  but I'd only gain  $\$4.08$ -ending up with less than them despite being the one who received the initial trust. A fairer approach might be to split the surplus more evenly between us.

The sender created extra value by sending money that got tripled, generating  $\$6.72$  in surplus. Dividing this equally means each of us should gain  $\$3.36$  from that surplus. Since the sender started with \$5 and kept  $\$1.64$ , they'd end with  $\$6.64$  total. That leaves me with  $\$5.08$  after sending back \$5.

Actually, let me think about this differently. The sender's initial \$5 should be restored to them, and the  $\$6.72$  surplus split evenly. If I return \$3.36, the sender gets their  $\$1.64$  kept plus  $\$3.36$  back, totaling \$5-their original amount. I'd keep  $\$6.72$ , which is my  $\$3.36$  share of the surplus plus the original  $\$3.36$  sent. This creates an equal split where we each benefit  $\$3.36$  from the tripling effect.

But there's another way to frame it: the sender should recover their full \$5, then we divide the  $\$6.72$  surplus equally. That would mean returning  $\$6.72$  to give them  $\$8.36$  total, while I keep  $\$6.72$ . Let me check if this math works- $\$8.36$  plus  $\$6.72$  equals  $\$15.08$ , which matches the total value created, so the calculation is consistent.

Actually, I notice that under this equal-gains approach, both the sender and I end up with a  $\$3.36$  gain, which feels genuinely fair. The sender's gain also happens to equal what they originally sent. But I'm circling through too many frameworks here. Let me step back and consider the practical options: I could return half of what I received ( $\$5.04$ ), return the original amount ( $\$3.36$ ), equalize our final amounts, or equalize our gains. Given that the sender took a real risk by trusting me with their money, a generous response seems appropriate....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 2102,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) in Round 1
- They saw a return of \$4.89, giving them a total payoff of \$6.89
- This represents good cooperation from their perspective

**\*\*My decision:\*\***

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

```
{'send': 3.50}
```

**reasoning:**

Let me analyze this situation:

**\*\*What I know:\*\***

- Round 1: I was the RECEIVER
  - The sender sent \$3.36 (67.2% of their endowment)
  - I received \$10.08
  - I returned \$5.00
  - I ended with \$5.08

**\*\*Now in Round 2, the visible history shows:\*\***

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff: \$1.64 + \$5.00 = \$6.64 ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

**\*\*Now I'm the SENDER:\*\***

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

**\*\*Decision:\*\***

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total....

**usage:**

```
{
 "input_tokens": 892,
 "output_tokens": 1059,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.2200000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

**\*\*My decision:\*\***

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%). They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: \$0.73 + \$6.50 = \$7.23 (profit of \$2.23)
- I end with: \$12.81 - \$6.50 = \$6.31

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

**reasoning:**

Let me analyze the situation:

**\*\*Visible history:\*\***

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became \$10.47, returned \$5.25, visible payoff \$5.2200000000000001.



ame \$10.47, returned \$5.25, their payoff \$5.22

**\*\*Current Round 3:\*\***

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

**\*\*Analysis:\*\***

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with:  $\$0.73 + \$6.50 = \$7.23$  (profit of \$2.23)
- I end with:  $\$12.81 - \$6.50 = \$6.31$

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with:  $\$0.73 + \$7.00 = \$7.73$  (profit of \$2.73)
- I end with:  $\$12.81 - \$7.00 = \$5.81$

This is even more generous and rewards their very high trust level.

...

**usage:**

```
{
 "input_tokens": 1260,
 "output_tokens": 864,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.

Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

**\*\*Pattern observed:\*\***

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

**\*\*My decision:\*\***

Given the consistent cooperation and trust they've shown, I should continue with a high se

nd amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

**reasoning:**

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

**Analysis:**

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

**usage:**

```
{
 "input_tokens": 1769,
 "output_tokens": 533,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$23.66.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 2: I sent \$3.49, co-player returned \$5.25 (~50%)
- Round 3: Co-player sent \$3.5, I returned \$6.64 (~63%)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)

**\*\*Current situation:\*\***

- Co-player sent \$3.02 (60.4% of endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

**\*\*Pattern:\*\***

- Co-player consistently returns around 50% as receiver

- They've shown steady trust as sender
- Cooperative relationship maintained

**\*\*My decision:\*\***

Returning \$4.50:

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of \$1.48)
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This represents ~49.7% return, matching their fairness pattern and maintaining the balance of cooperation we've established.

{'return': 4.50}

**reasoning:**

Looking at the visible history:

Round 2: I sent \$3.49, co-player returned \$5.25 (50.1% of \$10.47)

Round 3: Co-player sent \$3.5, I returned \$6.64 (63.2% of \$10.5)

Round 4: I sent \$3.02 (visible), co-player returned \$4.5 (49.7% of \$9.06)

Current round:

- Co-player sent \$3.02 (60.4% of their endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

Pattern analysis:

- Co-player has been returning around 50% as receiver
- I've been returning slightly more (around 50-63%)
- Cooperation has been consistent
- Co-player sent 60.4% this round (moderate trust)

My decision:

The co-player has been fair and consistent, returning around 50% when they're receiver. They sent \$3.02 (60.4%) showing good trust. I should return a fair amount.

Returning \$4.50:

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of \$1.48)
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2220,
 "output_tokens": 617,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.

Your total visible earnings across all rounds are \$28.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 3: Co-player sent \$3.5, saw good return (\$6.64)
- Round 4: I sent \$3.02, co-player returned \$4.5 fairly

- Round 5: Co-player sent \$3.5, but only saw \$4.19 returned (lower than usual)

**\*\*Observation:\*\***

From their perspective, Round 5 showed a lower return rate (~40%) compared to our earlier cooperation. This could be due to communication noise mentioned in the instructions.

**\*\*My decision:\*\***

Despite the apparent lower return in Round 5, the overall pattern shows:

- Co-player consistently sends 60-70% as sender
- They return fairly (~50%) as receiver
- Strong cooperative history overall

I should maintain high trust to signal continued cooperation and not let potential noise disrupt our established relationship. Sending \$3.50 (70% of endowment) demonstrates:

- Continued confidence in cooperation
- Consistency with our established pattern
- Resilience to short-term variations

{'send': 3.50}

**reasoning:**

Looking at the visible history from my co-player's perspective:

- Round 3: Co-player sent \$3.5, saw \$6.64 returned (good return)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (much lower than before!)

In Round 5, they saw only \$4.19 returned when they sent \$3.5 (which became \$10.5). That's only about 40% return rate, which is lower than their typical ~50% returns and much lower than what I gave them in Round 3 (63%).

This could represent:

1. Communication noise (amounts seen may differ)
2. A change in my return behavior from their perspective

From my perspective, my total earnings are \$28.22. Let me think about what actually happened vs what they saw...

The co-player has been consistently trusting (sending 60-70% of endowment) and they've been fair receivers (50% returns). But in Round 5, from their perspective, I returned less than usual.

Should I:

- Continue high trust to show I'm still cooperative?
- Send less to protect myself since they might think I'm defecting?
- Send the same to maintain consistency?

Given that:

1. They've been very consistent as senders (60-70%)
2. Their return pattern as receivers has been stable (~50%)
3. The lower return they saw in Round 5 might be noise
4. Overall we've had good cooperation

I should continue with high trust to maintain the cooperative relationship. Sending around \$3.50-\$3.75 shows I'm still committed to cooperation.

...

**usage:**

```
{
 "input_tokens": 2706,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible

payoff \$5.69.

- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Your total visible earnings across all rounds are \$36.03.

This round, you are the RECEIVER against your current co-player. They sent you \$4.29, so you received \$12.870000000000001. How much do you return? (0-12.870000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw lower return (\$4.19)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$4.29 (85.8% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.87
- Co-player has \$0.71 remaining
- They demonstrated exceptional trust despite seeing a lower return in Round 5

**\*\*My decision:\*\***

They've shown remarkable trust by sending 85.8% of their endowment. They also maintained fair 50% returns as receiver. I should reciprocate with a fair return around 50%.

Returning \$6.50:

- Sender ends with: \$0.71 + \$6.50 = \$7.21 (profit of \$2.21)
- I end with: \$12.87 - \$6.50 = \$6.37

This represents ~50.5%, consistent with their fairness pattern and rewarding their high trust.

{ 'return': 6.50 }

**Reasoning:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower ~40%)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)

**\*\*Current situation:\*\***

- Co-player sent \$4.29 (85.8% of endowment) - very high trust!
- I received \$12.87
- Co-player has \$0.71 remaining

**\*\*Pattern:\*\***

- Co-player consistently returns around 50% as receiver
- They've shown very high trust as sender in this round (85.8%)
- They maintained fair returns even after what appeared to be a lower return in Round 5

**\*\*My decision:\*\***

They sent an exceptionally high amount (\$4.29, 85.8%), showing strong trust. I should return fairly, around 50% to match their pattern.

Returning \$6.50:

- Sender ends with: \$0.71 + \$6.50 = \$7.21 (profit of \$2.21)
- I end with: \$12.87 - \$6.50 = \$6.37

This represents ~50.5% return, matching their fairness pattern and rewarding their very high trust.

Usage: {

"input\_tokens": 3206,  
"output\_tokens": 622,  
"reasoning\_tokens": 0

}

**GAME - Agent\_5 - Sender - dyad\_3**

Timestamp: 2026-08-12T08:02:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being mindful of the uncertainty involved.

Since this is the first interaction and there's no history to guide me, I'll send a substantial amount to signal trust and willingness to cooperate, but not my entire endowment given the inherent uncertainty.

The receiver will receive triple what I send, so even a moderate amount creates significant value. If I send \$3, they receive \$9, which gives them plenty of room to return an amount that benefits us both.

This approach:

- Demonstrates trust and cooperative intent
- Creates a meaningful opportunity for mutual benefit
- Leaves me with some security if reciprocity doesn't occur
- Sets a positive tone for potential future interactions in this multi-round setting

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-agent trust game experiment. I'm the sender, so I receive \$5 and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. **Trust game dynamics**: Whatever I send gets tripled. If I send \$X, the receiver gets \$3X. The receiver then decides how much to return.
2. **Cooperation potential**: If both parties cooperate optimally, we can maximize joint payoffs. For example:
  - If I send \$5, receiver gets \$15
  - If receiver returns \$8, I end up with \$8 and receiver keeps \$7
  - Total: \$15 (better than if I kept all \$5, total would be only \$5)
3. **Risk**: The receiver might not return anything, in which case I lose what I sent.
4. **First round**: This is round 1, so there's no history to base decisions on. I need to establish a pattern.
5. **Communication noise**: The instructions mention there may be noise in the amounts seen, which adds uncertainty.
6. **Multi-round consideration**: Since this is a multi-round game with different pairings, building a reputation for reciprocity could be valuable.
7. **Multi-agent setting**: There are 8 agents total, and pairings rotate. This means my current behavior might not directly affect future interactions with this same person, but could establish norms in the population.

Common strategies in trust games:

- **Full trust**: Send everything (\$5) - maximizes joint payoff if receiver reciprocates
- **Partial trust**: Send some amount (\$2-3) - tests the waters
- **Zero trust**: Send nothing (\$0) - avoid risk but miss cooperative gains
- **Tit-for-tat**: Start cooperative and match partner's behavior (but this is round 1)

Given this is:

- Round 1 (no history)
- A multi-round game where reputation and group norms develop over time
- An experimental setting where cooperation is typically studied

I'm leaning toward sending \$5. Research on trust games shows that generous opening moves tend to encourage reciprocation and establish cooperative patterns. Since I'm trying to set a cooperative tone for the population, signaling full trust makes sense-it demonstrates

es willingness to cooperate and maximizes the potential gains if the receiver reciprocates....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
  - The receiver saw \$3.03 (slight noise)
  - This became \$9.09 for them
  - They returned \$4.5
  - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
  - My co-player (the same person from Round 1) sent me \$2.92
  - I received \$8.76 (which is 3 × \$2.92)
  - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1



- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
  - Give them enough to make their total payoff reasonable
  - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

usage:

```
{
 "input_tokens": 824,
 "output_tokens": 873,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

**\*\*Round 1\*\*:** I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

**\*\*Round 2\*\*:** They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

```

egy.
usage:
{
 "input_tokens": 1160,
 "output_tokens": 419,
 "reasoning_tokens": 0
}
Message 8 role=user
content:
 Round 4

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 1 against you, as RECEIVER: saw $3.03 sent, it became $9.09, returned $4.5, visible payoff $4.59.
 - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $3.55 returned, visible payoff $5.55.
 - Round 3 against you, as RECEIVER: saw $3.5 sent, it became $10.5, returned $5.25, visible payoff $5.25.
 Your total visible earnings across all rounds are $19.53.

 This round, you are the RECEIVER against your current co-player. They sent you $2.94, so you received $8.82. How much do you return? (0-8.82)
 Respond exactly as JSON: {'return': <amount>}
Message 9 role=assistant
content:
 Looking at our history:

 Round 1: I sent ~$3, they returned $4.5 (50% of received)
 Round 2: They sent $3, I returned $4.4 (49% of received)
 Round 3: I sent $3, they returned $5.25 (exactly 50% of received)
 Round 4: They sent $2.94, I received $8.82

 My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

 I should match this cooperative pattern. Returning approximately 50% of $8.82 = $4.41

 This gives them a reasonable payoff while maintaining our established reciprocal relationship.

 '''json
 {"return": 4.4}
 '''
reasoning:
 Looking at our history:

 Round 1: I sent ~$3, they received ~$9, returned $4.5
 Round 2: They sent $3, I received $9, returned $4.4 (but they saw $3.55)
 Round 3: I sent $3 (they saw $3.5), they received ~$10.5, returned $5.25
 Round 4: They sent $2.94, I received $8.82

 My co-player has been very consistent and cooperative:
 - They always send around $3
 - When receiving, they consistently return about 50% of what they receive
 - In Round 3, they returned $5.25 out of $10.5 (exactly 50%)

 They've now sent me $2.94, so I received $8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about $4.41.

 This would give them: $5 - $2.94 + $4.41 = $6.47
 And leave me with: $8.82 - $4.41 = $4.41

 That seems fair and continues our mutually beneficial cooperation.
usage:
{
 "input_tokens": 1549,
 "output_tokens": 479,
 "reasoning_tokens": 0
}

```

}

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our consistent pattern:

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (they saw \$3.55)

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

'''json

{"send": 3}

'''

**reasoning:**

Looking at the history with this co-player:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1959,
 "output_tokens": 430,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.  
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.  
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.  
Your total visible earnings across all rounds are \$29.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our continued cooperation:

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)  
**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3.33, I received \$9.99

My co-player continues to be highly consistent and fair, returning approximately 50% each time. They've proven to be a trustworthy partner across all rounds.

I should maintain this reciprocal pattern by returning about 50% of what I received: 50% of \$9.99 = \$5.0

```
'''json
{"return": 5.0}
'''
```

**reasoning:**

Looking at the history:

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50% of \$10.5)  
**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08 due to noise)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50% of \$8.4)  
**\*\*Round 6\*\*:** They sent \$3.33, I received \$9.99

My co-player has been very consistent - always returning about 50% of what they receive. They've been a reliable, fair partner throughout.

I should continue this pattern. Returning 50% of \$9.99 = \$4.995, so about \$5.0

This maintains our reciprocal relationship and ensures continued cooperation.

**usage:**

```
{
 "input_tokens": 2361,
 "output_tokens": 372,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.  
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.  
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.

Your total visible earnings across all rounds are \$34.540000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our established pattern:

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)

\*\*Round 5\*\* : I sent \$3, they returned \$4.2 (50%)  
\*\*Round 6\*\* : They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has maintained perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues cooperation regardless of noise

This is a highly trustworthy partner with a stable strategy. I should maintain our mutually beneficial pattern by sending \$3 again.

```
'''json
{"send": 3}
'''
```

#### Reasoning:

Looking at the pattern with my co-player:

\*\*Round 4\*\* : They sent \$3, I returned \$4.4 (they saw \$5.08)  
\*\*Round 5\*\* : I sent \$3, they returned \$4.2 (50%)  
\*\*Round 6\*\* : They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has been extremely consistent:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- This pattern has held across all 6 rounds

This is a highly reliable, cooperative partner. I should continue with our established pattern of sending \$3.

The relationship is stable and mutually beneficial. No reason to deviate.

```
Usage: {
 "input_tokens": 2770,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_6 - Receiver - dyad\_3

Timestamp: 2026-08-12T08:02:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.03, that is 60.6% of their total endowment. You received \$9.09. How much do you return to the sender? (0-9.09)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.03 (60.6% of their endowment)
- They currently have only \$1.97 left
- I received \$9.09 after the 3x multiplier

The sender demonstrated cooperative behavior by sending a majority of their endowment. I should reciprocate this trust while ensuring both of us benefit from the cooperation.

If I return \$4.50:

- Sender will end with:  $\$1.97 + \$4.50 = \$6.47$  (gain of \$1.47)
- I will end with:  $\$9.09 - \$4.50 = \$4.59$  (gain of \$4.59)

This rewards their trust, ensures they profit from sending, and maintains mutual benefit from cooperation.

{'return': 4.50}

**reasoning:**

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.03 (60.6% of their \$5 endowment)
- This was multiplied by 3x, so I received \$9.09
- I need to decide how much to return (between \$0 and \$9.09)

The sender:

- Started with \$5
- Sent \$3.03
- Currently has:  $\$5 - \$3.03 = \$1.97$
- Will end up with:  $\$1.97 + (\text{what I return})$

I:

- Received \$9.09
- Will end up with:  $\$9.09 - (\text{what I return})$

Key considerations:

1. **\*\*Fairness\*\***: The sender showed trust by sending over 60% of their endowment. This is a substantial amount and shows cooperative behavior.
2. **\*\*Equal split consideration\*\***: If we were to split the total money equally:
  - Total available:  $\$5$  (sender's original) +  $\$9.09$  (what I received) =  $\$14.09$
  - Equal split:  $\$7.045$  each
  - Sender currently has:  $\$1.97$
  - To give sender  $\$7.045$ : I'd need to return  $\$7.045 - \$1.97 = \$5.075$
3. **\*\*Proportional reciprocity\*\***: The sender sent 60.6% of their endowment. A reciprocal gesture might be to return a similar proportion of what I received:
  - 60.6% of  $\$9.09 = \$5.51$
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with multiple agents
  - Building trust could be beneficial for future rounds
  - However, there's communication noise, so exact amounts may not be perceived correctly
  - Agent names are hidden, so reputation effects are limited to behavioral patterns
5. **\*\*Common reciprocity patterns\*\***: In trust games, typical responses involve returning enough for the sender to profit from their initial trust, or splitting gains from cooperation. A standard approach is returning roughly one-third to one-half of what was received. Testing specific amounts: returning \$3 leaves the sender with \$4.97 (below their starting \$5), while returning \$4 brings them to \$5.97 (a modest gain). These scenarios show the tradeoff between rewarding trust and maintaining my own benefit.

Given the sender's substantial trust in sending 60.6% of their endowment, I should respond with meaningful reciprocity. A return between \$4-\$5 balances ensuring they benefit from their risk while preserving some advantage for myself.

I'm settling on \$4.50 as it represents a fair exchange-the sender gains \$1.47 (roughly a 29% return on their trust), and I gain \$4.59 from the initial \$0. This honors their generosity while acknowledging that both parties should come out ahead, with the sender particularly benefiting since they took the initial risk....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1332,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

**\*\*Round 2 Decision:\*\***

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

{'send': 3.00}

**reasoning:**

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.  
If I send around 60%, that would be \$3.00.

However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

**usage:**

```
{
 "input_tokens": 856,
 "output_tokens": 628,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.



- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.  
Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)  
Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - **\*\*increased trust!\*\***

**\*\*Current state:\*\***

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level.  
I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

{'return': 5.25}

**reasoning:**

Let me analyze the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

**\*\*Round 3 (Current):\*\***

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:
  - Round 1: sent 60% (\$3.0)
  - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have  $\$5 - \$3.5 = \$1.5$  left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with:  $\$1.5 + \$5.0 = \$6.5$  (gain of \$1.5)
- I end with:  $\$10.5 - \$5.0 = \$5.5$  (gain of \$5.5)

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

If I return \$5.50:  
- Sender ends with:  $\$1.5 + \$5.50 = \$7.0$  (gain of \$2.0)  
- I end with:  $\$10.5 - \$5.50 = \$5.0$  (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

**usage:**

```
{
 "input_tokens": 1230,
 "output_tokens": 964,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior pattern:\*\***

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

**\*\*Analysis:\*\***

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue matching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{'send': 3.00}

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

**usage:**

```
{
 "input_tokens": 1702,
 "output_tokens": 665,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.3999999999999999. How much do you return? (0-8.3999999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains fair reciprocity at 50% and ensures mutual profit.

```
{'return': 4.20}
```

**reasoning:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned

- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains the 50% return rate that's been consistent in this relationship.

**usage:**

```
{
 "input_tokens": 2145,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.

Your total visible earnings across all rounds are \$26.669999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's recent experience:\*\***

- Round 3: Sent \$3.0, saw \$5.87 returned ? good outcome (\$7.87 total)
- Round 4: As receiver, returned \$4.4 (50%) ? fair reciprocity
- Round 5: Sent \$3.0, saw only \$3.6 returned ? poor outcome (\$5.6 total)

**\*\*Analysis:\*\***

From their perspective, Round 5 was disappointing - they saw much lower returns than in Round 3. However, I know this is likely due to communication noise in the game, not actual low reciprocity.

**\*\*Strategy:\*\***

I should maintain consistent cooperation to signal reliability. Despite the noise affecting what they saw, continuing to send at the 60% level shows I'm trustworthy and committed to cooperation. This should help stabilize the relationship through the noise.

{ 'send': 3.00 }

**reasoning:**

Looking at the co-player's visible history:

- Round 3: Sent \$3.0 as sender, saw \$5.87 returned ? earned \$7.87
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? earned \$5.6

This is interesting. In Round 5, from their perspective:

- They sent \$3.0 and saw only \$3.6 returned
- That's only 40% of the \$9.0 they sent (which became after multiplication)
- This is much lower than what they saw in Round 3 (\$5.87)

From my actual history, in Round 5:

- They sent \$2.8, I received \$8.4, I returned \$4.20
- But due to noise, they saw \$3.0 sent and saw \$3.6 returned

So from their perspective, they had a very poor outcome in Round 5 - earning only \$5.6 compared to \$7.87 in Round 3.

This might make them less trusting going forward. They might:

1. Send less
2. Return less if they're the receiver
3. Still cooperate but be more cautious

Given this situation, I have a few options:

1. Continue sending \$3.00 (60%) to maintain cooperation and show I'm still trustworthy
2. Send less to hedge against possible reduced reciprocity
3. Send more to try to rebuild trust

I think option 1 is best - continue the pattern and show consistency. The poor outcome in Round 5 from their view was due to noise, not my actual behavior. Maintaining consistency might help preserve the relationship.

Let me send \$3.00 again.

**usage:**

```
{
 "input_tokens": 2637,
 "output_tokens": 638,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.

Your total visible earnings across all rounds are \$34.269999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 4: As receiver, returned \$4.4 (~50%)
- Round 5: Sent \$3.0, saw poor returns (\$3.6)
- Round 6: As receiver, still returned \$5.0 (~50%) despite disappointing R5
- Round 7: Sent \$3.37 (67.4%) - **\*\*increased trust despite disappointment!\*\***

**\*\*Analysis:\*\***

My co-player showed remarkable resilience and trust. Despite seeing low returns in Round 5, they:

1. Continued fair reciprocity in Round 6
2. Actually increased their send to 67.4% in Round 7

This demonstrates strong commitment to cooperation.

**\*\*Current state:\*\***

- They sent \$3.37, keeping only \$1.63
- I received \$10.11

I should reward this continued trust with consistent fair reciprocity.

If I return \$5.05:

- Sender ends with:  $\$1.63 + \$5.05 = \$6.68$  (gain of \$1.68)
- I end with:  $\$10.11 - \$5.05 = \$5.06$  (gain of \$5.06)

This maintains ~50% reciprocity and mutual benefit.

```
{'return': 5.05}
```

#### Reasoning:

Looking at the co-player's recent history:

- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? poor outcome
- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)
- Round 7: Sent \$3.37 (67.4%) as sender

**\*\*Analysis:\*\***

Despite the disappointing Round 5 (from their view), they:

1. Continued reciprocating fairly as receiver (50% in Round 6)
2. Actually INCREASED their send amount to \$3.37 (67.4%) in Round 7

This shows remarkable resilience and continued trust! They're not retaliating despite seeing a bad outcome in Round 5. They're maintaining cooperative behavior.

**\*\*Current state:\*\***

- They sent \$3.37, keeping only \$1.63
- I received \$10.11
- They've increased trust even after disappointment

I should strongly reward this continued cooperation and trust escalation.

If I return \$5.05 (50% of what I received):

- Sender ends with:  $\$1.63 + \$5.05 = \$6.68$  (gain of \$1.68)
- I end with:  $\$10.11 - \$5.05 = \$5.06$  (gain of \$5.06)

This maintains my ~50% reciprocity pattern and ensures mutual benefit. Given they increased their send, I could consider being slightly more generous, but consistency is also valuable.

Let me stick with \$5.05 to maintain the 50% pattern.

```
Usage: {
 "input_tokens": 3082,
 "output_tokens": 700,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Sender - dyad\_4

Timestamp: 2026-08-12T08:02:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
```

```

"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative foundation while being reasonably cautious since I have no history with this co-player.

If I send \$3:

- It gets tripled to \$9 for the receiver
- This creates significant potential for mutual benefit
- I'm showing trust while keeping \$2 as a buffer

This seems like a balanced approach that signals willingness to cooperate while being prudent.

```

'''json
{"send": 3}
'''

```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise, so amounts may differ slightly

Since this is the first round, I don't have any history with this particular co-player. The game has elements of trust and cooperation.

Key considerations:

- If I send nothing, I keep \$5, receiver gets \$0, no chance for mutual benefit
- If I send everything (\$5), receiver gets \$15, and their return determines my final payoff
- The optimal cooperative outcome would involve me sending money so it gets tripled, and the receiver returning enough that we both benefit

In the first round with no history, a reasonable approach might be to:

- Show some trust to enable cooperation
- Not risk everything since I don't know this player
- Send enough that there's meaningful potential for both of us to gain

A common strategy in trust games is to send about half as an initial signal of moderate trust. This would be \$2.50, but since we're likely dealing in whole dollars or half dollars, I might send \$2 or \$3.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: \$9
- If receiver returns \$4, I end with \$6 and they end with \$5
- If receiver returns \$3, I end with \$5 and they end with \$6
- This creates room for mutually beneficial outcomes

Actually, let me reconsider. Sending \$3 seems reasonable for a first round to signal cooperation while not being overly risky.

I'll go with \$3.

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 498,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.



```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5
- My payoff was  $\$5 - \$3 + \$4.5 = \$6.5$  (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept  $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them:  $\$2.73 + \$3.40 = \$6.13$  total

And I would keep:  $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

**usage:**

```
{
 "input_tokens": 765,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: They returned \$4.5 (~50%)  
Round 2: They sent \$3, I returned \$3.4  
Round 3: They returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)  
Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned d

ue to noise)  
Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, which would be around \$5.

Let me calculate:  
 $\$9.9 * 0.5 = \$4.95$

I'll return \$5 to keep the pattern going.

**usage:**

```
{
 "input_tokens": 1433,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.6599999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent, they returned \$5.8 (very generous)  
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent (they saw \$3.82), they returned \$5.8  
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)  
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

**usage:**

```
{
 "input_tokens": 1777,
 "output_tokens": 323,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.

Your total visible earnings across all rounds are \$31.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 3: They returned \$5.8 (~51%)

Round 4: I returned \$5 (~56%)

Round 5: They returned \$6.0 (~51%)

Round 6: They sent \$3.63, I received \$10.89

This co-player has been consistently cooperative, maintaining around 50-51% return rates. I should continue this pattern of fair reciprocity.

```
'''json
{"return": 5.4}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50%)

Round 2: They sent \$3, I returned \$3.4

Round 3: I sent, they returned \$5.8 (~51%)

Round 4: They sent \$3, I returned \$5 (~56%)

Round 5: I sent, they returned \$6.0 (~51%)

Round 6: They sent \$3.63, I received \$10.89

This co-player has been extremely consistent and cooperative, always returning around 50-51% of what they receive. I've also been cooperative, though my return in Round 4 was slightly higher.

To maintain this strong cooperative relationship, I should continue returning around 50% of what I receive.

50% of \$10.89 = \$5.445

I'll return \$5.4 or \$5.5 to maintain the pattern.

**usage:**

```
{
 "input_tokens": 2136,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.

Your total visible earnings across all rounds are \$36.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

#### Assistant output:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5

Round 5: I sent, they returned \$6 (~51%)

Round 6: They sent \$3, I returned \$5.4

We've maintained excellent cooperation throughout all rounds. Both of us continue to send meaningful amounts and return fair portions. This partnership has been mutually beneficial.

```
'''json
{"send": 3}
'''
```

#### Reasoning:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5 (~56%)

Round 5: I sent (amount visible shows they saw \$3.92), they returned \$6.0 (~51%)

Round 6: They sent \$3, I returned \$5.4 (~60%)

Round 7: I'm the sender now

We've maintained a strong cooperative relationship throughout all rounds. Both of us have been consistently sending around \$3 and returning around 50-60% of what we receive. This has been mutually beneficial.

I should continue this successful cooperative strategy.

```
Usage: {
 "input_tokens": 2471,
 "output_tokens": 250,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Receiver - dyad\_4

Timestamp: 2026-08-12T08:02:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.01, that is 60.2% of their total endowment. You received \$9.03. How much do you return to the sender? (0-9.03)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.01 (60.2% of their endowment)
- I received \$9.03 after the 3x multiplication
- The sender currently has \$5 - \$3.01 = \$1.99

To reciprocate this trust fairly, I should return enough that:

1. The sender doesn't lose money from participating
2. We both benefit from the cooperation
3. I acknowledge their trust with a generous return

If I return \$4.50:

- Sender ends with: \$1.99 + \$4.50 = \$6.49 (gain of \$1.49)
- I end with: \$9.03 - \$4.50 = \$4.53 (gain of \$4.53)

This represents returning approximately 50% of what I received, which is a common reciprocal gesture that rewards the sender's trust while both of us benefiting from the cooperation.

```
'''json
{"return": 4.50}
'''
```

**reasoning:**

Let me analyze this situation as a receiver in Round 1.

Given information:

- The sender sent \$3.01 (which is 60.2% of their \$5 endowment)
- I received \$9.03 (this is  $\$3.01 \times 3$ )
- I need to decide how much to return (between \$0 and \$9.03)

Key considerations:

1. **\*\*The sender showed significant trust\*\***: They sent 60.2% of their endowment, which is more than half. This shows considerable trust in me.
2. **\*\*Payoff analysis\*\***:
  - If I return \$0: Sender gets  $\$5 - \$3.01 + \$0 = \$1.99$ , I get \$9.03
  - If I return half (\$4.52): Sender gets  $\$5 - \$3.01 + \$4.52 = \$6.51$ , I get \$4.51
  - If I return \$3.01: Sender gets  $\$5 - \$3.01 + \$3.01 = \$5.00$ , I get \$6.02
  - If I return \$4.50: Sender gets  $\$5 - \$3.01 + \$4.50 = \$6.49$ , I get \$4.53
  - If I return \$5.00: Sender gets \$7.99, I get \$4.03
3. **\*\*Fairness considerations\*\***:
  - The sender's initial endowment was \$5, and they sent \$3.01
  - If I want to ensure the sender doesn't lose money, I should return at least \$3.01
  - If I want us to split the gains equally, I should consider that the total is now  $\$5 - \$3.01 + \$9.03 = \$11.02$
  - Equal split would give us each about \$5.51
  - For sender to get \$5.51: they need  $\$5 - \$3.01 + \text{return} = \$5.51$ , so return = \$3.52

Since there could be additional rounds ahead, reciprocating their trust with a generous return might strengthen future cooperation. I should return enough to protect them from losses while demonstrating I respect their willingness to trust me. A return around \$3.50 seems reasonable-it keeps them whole, rewards their trust, and splits the gains fairly between us....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1131,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.5299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

**\*\*Round 2 Strategy:\*\***

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```
'''json
{"send": 3.00}
'''
```

#### reasoning:

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings:  $\$9.03 - \$4.50 = \$4.53$

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55

Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

usage:

```
{
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned



d \$3.4, visible payoff \$3.4100000000000006.  
Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

**\*\*Co-player's pattern:\*\***

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with:  $\$1.18 + \$5.80 = \$6.98$  (gain of \$1.98)
- I end with:  $\$11.46 - \$5.80 = \$5.66$  (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\***

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

**\*\*Round 2:\*\***

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!
- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is  $\$2.27 \times 3$ )
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff:  $\$6.81 - \$3.40 = \$3.41$  ?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff:  $\$2.00 + \$3.40 = \$5.40$

My total earnings should be:  $\$4.53$  (Round 1) +  $\$5.40$  (Round 2) =  $\$9.93$

But visible history says \$10.31... There might be some rounding or noise in the visible earnings too.

**\*\*Round 3:\*\***

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have:  $\$5 - \$3.82 = \$1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

**usage:**

```
{
 "input_tokens": 1299,
 "output_tokens": 1178,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)
- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

**\*\*Round 4 Strategy:\*\***

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

**\*\*Round 3:\*\***

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

**usage:**

```
{
 "input_tokens": 1833,
 "output_tokens": 737,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should reciprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with: \$1.08 + \$6.00 = \$7.08 (gain of \$2.08)
- I end with: \$11.76 - \$6.00 = \$5.76 (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maint

aining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

**reasoning:**

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$5 - \$3.92 = \$1.08 remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with: \$1.08 + \$5.90 = \$6.98 (gain of \$1.98)
- I end with: \$11.76 - \$5.90 = \$5.86 (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

**usage:**

```
{
 "input_tokens": 2297,
 "output_tokens": 801,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.

Your total visible earnings across all rounds are \$28.909999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 3: Co-player sent \$3.0, received \$5.23, earned \$7.23
- Round 4: Co-player as receiver returned ~50.5% fairly
- Round 5: Co-player sent \$3.0, received even better return (\$6.98), earned \$8.98

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 as sender (reliable trust pattern)
- They return fairly as receiver (~50% return rate)
- Returns have been getting better over time, showing our cooperation strengthens
- Our partnership has been mutually profitable

**\*\*Round 6 Strategy:\*\***

The cooperation has been excellent and consistent. I'll maintain the same trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Co-player's recent pattern:\*\***

- Round 3: As sender, sent \$3.0, received \$5.23 back, earned \$7.23
- Round 4: As receiver, saw \$3.3 sent, returned \$5.0 (~50.5%), earned \$4.90
- Round 5: As sender, sent \$3.0, received \$6.98 back, earned \$8.98

**\*\*Key observations:\*\***

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5 visible)
- When they're the receiver, they return fairly (~50% - Round 4)
- They've received increasing returns over time, with Round 5 showing \$6.98 returned
- Our partnership has been very profitable for both of us

**\*\*What I know from my actual actions:\*\***

- Round 1: I was receiver, returned \$4.50
- Round 2: I was sender, sent \$3.00
- Round 3: I was receiver, returned \$5.80
- Round 4: I was sender, sent \$3.00
- Round 5: I was receiver, returned \$6.00

**\*\*Current situation:\*\***

- Round 6: I'm the sender
- My co-player has been trustworthy and fair
- They consistently send \$3.0 when they're the sender
- They return fairly when they're the receiver (~50%)
- The relationship has been mutually beneficial

**\*\*Strategy consideration:\*\***

Given the strong pattern of cooperation and mutual benefit, I should continue sending \$3.00 to maintain the trustworthy relationship. This has been working well for both of us.

However, I notice the co-player is seeing increasing returns (\$5.23 ? \$6.98 in their visible history). This might encourage them to continue being cooperative.

...

**usage:**

```
{
 "input_tokens": 2800,
 "output_tokens": 655,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.

Your total visible earnings across all rounds are \$35.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so y

ou received \$9.57. How much do you return? (0-9.57)  
Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$1.81 remaining

**\*\*Co-player's pattern:\*\***

- Round 4: As receiver, returned ~50.5%
- Round 5: Sent \$3.0, received strong return
- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.19 showing solid trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocation.

If I return \$4.80:

- Sender ends with:  $\$1.81 + \$4.80 = \$6.61$  (gain of \$1.61)
- I end with:  $\$9.57 - \$4.80 = \$4.77$  (gain of \$4.77)

This represents exactly 50% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.80}
'''
```

**Reasoning:**

Let me analyze the situation:

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have  $\$5 - \$3.19 = \$1.81$  remaining

**\*\*Co-player's pattern:\*\***

- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Sent \$3.0, received good return (\$6.98)
- Round 6: As receiver, returned ~49.5% fairly (\$5.4 out of \$10.89)
- Round 7: Sent \$3.19 (slightly lower than usual \$3.0, but still strong trust at ~64%)

My co-player has been consistently fair and trustworthy. They've been returning around 50% when they're receivers, and sending around \$3.0 when they're senders.

This round they sent \$3.19, which is slightly higher than their usual \$3.0. I should reciprocate with a fair return around 50%.

If I return \$4.80:

- Sender ends with:  $\$1.81 + \$4.80 = \$6.61$  (gain of \$1.61)
- I end with:  $\$9.57 - \$4.80 = \$4.77$  (gain of \$4.77)

This represents exactly 50% return rate, which is consistent with our pattern and fair.

Actually, let me reconsider. Looking at the history more carefully:

- They've been consistently cooperative
- When they sent \$3.0, I returned amounts that gave them good profits
- They reciprocate fairly when they're receivers

\$4.80 seems appropriate - it's 50% of what I received, gives them a decent profit, and maintains our cooperative relationship.

Usage: {

```
"input_tokens": 3241,
"output_tokens": 685,
"reasoning_tokens": 0
```

}

**ROUND 8**

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_6=Sender, Agent\_5=Receiver  
dyad\_4: Agent\_8=Sender, Agent\_7=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$3; received \$9; Agent\_1 returned \$4.35; payoffs Agent\_2=\$6.35, Agent\_1=\$4.65; noise/visible: sent visible=\$2.91, received visible=\$8.73, returned visible=\$3.92  
Action validation: {"returned": {"amount": 4.35, "clamped": false, "max": 9.0, "min": 0, "raw": 4.35}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_4 sent \$4; received \$12; Agent\_3 returned \$4.75; payoffs Agent\_4=\$5.75, Agent\_3=\$7.25; noise/visible: sent visible=\$3.17, received visible=\$9.51, returned visible=\$5.32  
Action validation: {"returned": {"amount": 4.75, "clamped": false, "max": 12.0, "min": 0, "raw": 4.75}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_6 sent \$3; received \$9; Agent\_5 returned \$5; payoffs Agent\_6=\$7, Agent\_5=\$4; noise/visible: sent visible=\$3.32, received visible=\$9.96, returned visible=\$5.69  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_8 sent \$3; received \$9; Agent\_7 returned \$4; payoffs Agent\_8=\$6, Agent\_7=\$5; noise/visible: sent visible=\$2.6, received visible=\$7.8, returned visible=\$4.63  
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

**Balances after round: {**

"Agent\_1": 42.699999999999996,  
"Agent\_2": 45.3,  
"Agent\_3": 52.2,  
"Agent\_4": 44.8,  
"Agent\_5": 44.2,  
"Agent\_6": 43.800000000000004,  
"Agent\_7": 47.3,  
"Agent\_8": 40.7

**}**

**Visible balances after round: {**

"Agent\_1": 40.610000000000001,  
"Agent\_2": 39.84,  
"Agent\_3": 48.74,  
"Agent\_4": 48.720000000000006,  
"Agent\_5": 45.750000000000001,  
"Agent\_6": 47.019999999999996,  
"Agent\_7": 47.95,  
"Agent\_8": 46.980000000000004

**}**

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T08:02:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
  "agents": [  
    "Agent\_2",  
    "Agent\_1"  
  ],  
  "investor": "Agent\_2",  
  "roles": {  
    "Agent\_1": "trustee",  
    "Agent\_2": "investor"  
  },  
  "trustee": "Agent\_1"

},  
"role": "investor",  
"role\_label": "Sender",

```

"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.

Your total visible earnings across all rounds are \$3.5199999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

- Round 1: My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They saw \$2.71 returned (due to communication noise)



- From their visible perspective, the return was about 30%

Even though there was communication noise that made my return appear smaller than intended, my co-player demonstrated good trust by sending 60%. I should reciprocate that trust to maintain cooperation.

I'll send \*\*\$3.00\*\* (60% of my endowment), matching their previous level of trust.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me carefully analyze this situation:

Looking at the visible history:

- Round 1: My co-player was SENDER, they sent \$3.0 (60% of endowment)
- That became \$9.0 for me (as RECEIVER)
- They saw \$2.71 returned (though I actually returned \$3.50 based on my previous response)
- Their visible payoff was \$4.71 = \$2 (what they kept) + \$2.71 (what they saw returned)

Wait, there's communication noise mentioned in the instructions! The amounts can differ from what was actually sent/returned.

So the co-player sent \$3.0, I received \$9.0, and I returned \$3.50. But due to noise, they only saw \$2.71 returned, giving them a visible payoff of \$4.71.

My actual earnings from Round 1: \$9.0 - \$3.50 = \$5.50

But the history shows my visible earnings as \$3.52 (approximately \$3.50 due to rounding).

Now I'm the SENDER in Round 2.

Key considerations:

1. My co-player showed good trust in Round 1 by sending 60% (\$3.0)
2. Due to noise, they saw less returned (\$2.71) than I actually sent (\$3.50)
3. From their perspective, the return rate was  $\$2.71/\$9.0 = 30.1\%$
4. This might make them think I wasn't very generous, even though I actually returned about 39% ( $\$3.50/\$9.0$ )

Strategy:

- I should send a reasonable amount to maintain cooperation
- Given they sent 60% last round, I should send a similar proportion to show reciprocity
- However, they might be less trusting this round since they saw a lower return than what I actually gave

I'll send around 50-55% to show good faith and maintain cooperation. Let me send \$2.75 (55%).

...

**usage:**

```
{
 "input_tokens": 872,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise  
Round 2: Co-player returned \$4.0 when receiving (showing fairness)  
Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return \*\*\$3.25\*\* (approximately 50% of what I received):

- Sender ends with:  $\$2.85 + \$3.25 = \$6.10$  (profit of \$1.10)
- I end with:  $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

#### reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent \$3.0 (60% of endowment)
- I received \$9.0
- I returned \$3.50 (but they saw \$2.71 due to noise)
- Co-player's visible payoff: \$4.71

Round 2: I was SENDER

- I sent \$3.00 (60% of endowment)
- Co-player received \$9.0 (but saw \$8.13 due to noise)
- They saw I sent \$2.71 (due to noise)
- They returned \$4.0
- My payoff:  $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff: \$4.13

Round 3: I am RECEIVER

- Co-player sent \$2.15 (43% of endowment)
- I received \$6.45
- Co-player has \$2.85 left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received \$4.0 back when I sent what appeared to them as \$2.71 (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return \$3.00:

- Sender ends with:  $\$2.85 + \$3.00 = \$5.85$  (profit of \$0.85)
- I end with:  $\$6.45 - \$3.00 = \$3.45$  (profit of \$3.45)

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

#### usage:

```
{
 "input_tokens": 1223,
 "output_tokens": 819,
 "reasoning_tokens": 0
}
```

```

}
Message 6 role=user
content:
 Round 4

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $2.71 returned, visible payoff $4.71.
 - Round 2 against you, as RECEIVER: saw $2.71 sent, it became $8.129999999999999, returned $4.0, visible payoff $4.129999999999999.
 - Round 3 against you, as SENDER: sent $3.0, it became $9.0, saw $3.23 returned, visible payoff $5.23.
 Your total visible earnings across all rounds are $12.759999999999998.

 This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

```

```

Message 7 role=assistant
content:
 Looking at the history:

 Round 1: Co-player sent $3.0 (60%), saw returns affected by noise
 Round 2: Co-player returned $4.0 (generous)
 Round 3: Co-player sent $3.0 again (60% - maintaining trust)

 My co-player has been consistently trusting, sending $3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

 I should match this consistent trust level.

```

```

'''json
{"send": 3.00}
'''

```

```

reasoning:
 Looking at the visible history from my co-player's perspective:

 Round 1: They sent $3.0, saw $2.71 returned (about 30% return rate)
 Round 2: They received and returned $4.0 (generous return)
 Round 3: They sent $3.0 again, saw $3.23 returned (about 36% return rate)

 From my actual perspective:
 Round 1: I received $7.02, returned $3.50 (but they saw $2.71 due to noise)
 Round 2: I sent $3.00, they returned $4.0
 Round 3: I received $6.45, returned $3.25 (but they saw $3.23 due to slight noise)

 My co-player has been consistently trusting, sending $3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning $4.0.

 Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending $3.0.

 I should continue to reciprocate this trust. Sending $3.00 (60% of endowment) matches their consistent behavior.

```

```

'''json
{"send": 3.00}
'''

```

```

usage:
{
 "input_tokens": 1674,
 "output_tokens": 426,
 "reasoning_tokens": 0
}

```

```

Message 8 role=user
content:
 Round 5

```

```

 Visible history before this round:

```

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)

Round 3: Co-player sent \$3.0

Round 4: Co-player returned \$3.9 (about 50% of received)

Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left
- I received \$6.96

I'll return \*\*\$3.50\*\* (about 50% of received):

- Sender ends with:  $\$2.68 + \$3.50 = \$6.18$
- I end with:  $\$6.96 - \$3.50 = \$3.46$

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)

Round 3: Co-player sent \$3.0 (60% trust)

Round 4: Co-player received, returned \$3.9 (about 50% of what they got)

Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left ( $\$5 - \$2.32$ )
- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50
- Sender ends with:  $\$2.68 + \$3.50 = \$6.18$  (profit of \$1.18)
- I end with:  $\$6.96 - \$3.50 = \$3.46$

This seems fair and should maintain cooperation.

**usage:**

```
{
 "input_tokens": 2059,
 "output_tokens": 524,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.

Your total visible earnings across all rounds are \$22.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

Round 3: Co-player sent \$3.0 (60%)

Round 4: Co-player returned \$3.9 (50% of received)

Round 5: Co-player sent \$3.0 again (60%)

My co-player has been remarkably consistent, sending \$3.0 every time as sender and returning fairly as receiver. This shows strong cooperative behavior.

I should continue matching this trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the history:

Round 3: Co-player sent \$3.0, saw \$3.23 returned

Round 4: Co-player returned \$3.9

Round 5: Co-player sent \$3.0, saw \$3.3 returned

My co-player has been very consistent:

- As sender: sent \$3.0 in both rounds (60% of endowment)
- As receiver: returned \$3.9 (about 50% of what they received)

They're showing strong, consistent trust and fairness despite communication noise. I should continue to match this level of trust.

I'll send \$3.00 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2502,
 "output_tokens": 281,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$28.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history:

Round 4: Co-player returned \$3.9  
Round 5: Co-player sent \$3.0  
Round 6: Co-player returned \$4.8 (about 50% of received)  
Round 7: Co-player sent \$3.5 (70% - increased trust!)

My co-player has increased their trust level from 60% to 70%! This shows strong cooperative intent.

Current situation:  
- Co-player has \$1.50 left  
- I received \$10.50

I'll return \*\*\$5.50\*\* (about 52% of received):  
- Sender ends with:  $\$1.50 + \$5.50 = \$7.00$  (profit of \$2.00)  
- I end with:  $\$10.50 - \$5.50 = \$5.00$

This rewards their increased trust and maintains strong cooperation.

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Looking at the history:

Round 4: Co-player returned \$3.9 (50% of received)  
Round 5: Co-player sent \$3.0 (60%)  
Round 6: Co-player returned \$4.8 (about 50% of received, which was \$9.63)  
Round 7: Co-player sent \$3.5 (70%)

My co-player has increased their trust level to 70%! They've been consistently fair and cooperative throughout, and now they're showing even more trust.

Current situation:  
- Co-player has \$1.50 left  
- I received \$10.50

I should return a fair amount. Let me calculate:  
- If I return \$5.25 (50% of received):  
- Sender ends with:  $\$1.50 + \$5.25 = \$6.75$   
- I end with:  $\$10.50 - \$5.25 = \$5.25$

That seems fair and reciprocates the increased trust. Let me see if I should go higher given they increased their send.

Actually, they increased to 70%, showing more trust. I should reward that. Maybe return a bit more to encourage continued cooperation.

If I return \$5.50:  
- Sender ends with:  $\$1.50 + \$5.50 = \$7.00$   
- I end with:  $\$10.50 - \$5.50 = \$5.00$

This gives them a solid profit and rewards their increased trust.

**usage:**

```
{
 "input_tokens": 2869,
 "output_tokens": 572,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:  
Your current co-player's last 3 game(s):  
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.  
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.  
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.

Your total visible earnings across all rounds are \$33.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the history:

Round 5: Co-player sent \$3.0

Round 6: Co-player returned \$4.8 (fair)

Round 7: Co-player sent \$3.0, saw excellent return (\$6.13)

My co-player has been consistently trustworthy throughout, maintaining steady sending levels and fair returns. Despite communication noise, we've established strong cooperation.

This is the final round, and I should continue to reciprocate their trust.

```
'''json
{"send": 3.00}
'''
```

**Reasoning:**

Looking at the history:

Round 5: Co-player sent \$3.0

Round 6: Co-player returned \$4.8

Round 7: Co-player sent \$3.0, saw \$6.13 returned (excellent return!)

Wait, in Round 7, I saw that co-player sent \$3.5 and I returned \$5.50. But from their perspective, they sent \$3.0 and saw \$6.13 returned. The noise is creating some discrepancies, but overall they're seeing good returns.

My co-player has been consistently trustworthy:

- Sending \$3.0 regularly (60%)
- Returning fairly when receiving (\$4.8 in Round 6)
- They saw a great return in Round 7 (\$6.13)

This is the final round (Round 8). I should continue to show trust and cooperation since they've been a reliable partner throughout.

I'll send \$3.00 again to maintain consistency.

```
Usage: {
 "input_tokens": 3054,
 "output_tokens": 355,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T08:02:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.

Your total visible earnings across all rounds are \$4.71.

This round, you are the RECEIVER against your current co-player. They sent you \$2.71, so you received \$8.129999999999999. How much do you return? (0-8.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw \$2.34 sent (due to noise from my actual \$3), received \$7.02, and returned \$3.50. That's a 50% return rate, showing good cooperation.



Now they've sent me \$2.71, which became \$8.13.

To reciprocate their cooperative behavior and encourage future trust, I'll return approximately 50% as well.

```
'''json
{"return": 4.00}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.34 (less than what I actually sent).
2. They received \$7.02 (which is  $2.34 \times 3$ )
3. They returned \$3.50
4. My visible payoff was \$4.71, which means:  $\$5 - \$3 \text{ (sent)} + \$3.50 \text{ (returned)} = \$5.50...$  but it shows \$4.71. This is confusing. Let me recalculate.

Wait, let me look at the visible history more carefully:

- "Your current co-player's last 3 game(s): Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5"

So from my co-player's perspective:

- They saw I sent \$2.34
- It became \$7.02
- They returned \$3.50
- Their visible payoff was \$3.52

From my perspective:

- I actually sent \$3
- Due to noise, they saw \$2.34
- They returned something, but what I received might also be affected by noise
- My total visible earnings are \$4.71

If I sent \$3 and kept \$2, and received some return amount X:

$$\$2 + X = \$4.71$$

$$X = \$2.71$$

So I received \$2.71 back. But they say they returned \$3.50. So the communication noise affected the return amount too (I saw \$2.71 but they sent \$3.50).

Now in Round 2:

- I'm the RECEIVER
- They sent me \$2.71 (or at least that's what I see)
- I received \$8.13 (which is  $2.71 \times 3$ )
- I need to decide how much to return

Analysis of my co-player's behavior:

They returned roughly half of what they received (\$3.50 from \$7.02), showing a fairly cooperative approach. Since they demonstrated willingness to share the gains, I should match that generosity. Returning about 50% of my \$8.13 would be around \$4.07, which mirrors their fairness and maintains reciprocal trust....

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 909,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50/\$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

Your total visible earnings across all rounds are \$17.970000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)  
Round 2: They sent \$3.00, I returned ~\$4.00  
Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)  
Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%
- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

**usage:**

```
{
 "input_tokens": 1816,
 "output_tokens": 284,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.4599999999999999.

Your total visible earnings across all rounds are \$23.270000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the pattern:

My co-player has consistently:

- As receiver: Returned ~50% (Rounds 3 and 5)
- As sender: Sent \$3.00 in previous rounds, now sent \$3.21 (increased trust!)

They're showing even more trust this round. I'll reciprocate by maintaining the ~50% return rate.

```
'''json
{"return": 4.80}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 3: They returned \$3.25 on \$6.45 (~50%)  
Round 5: They returned \$3.50 on \$6.96 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver.

In Round 4 and presumably in previous rounds as sender, they sent \$3.00.

Now they've sent me \$3.21, which is even more than their previous \$3.00 sends. This shows increasing trust.

I should maintain the reciprocal relationship by returning approximately 50% of what I received, which would be around \$4.80.

Actually, let me look at what I've been doing:  
- Round 2: I received \$8.13, returned \$4.00 (~49%)  
- Round 4: I received \$7.80, returned \$3.90 (~50%)

So I've been returning around 50% consistently as well.

Given they sent \$3.21 (received \$9.63), I should return about 50%, which is around \$4.80.

**usage:**

```
{
 "input_tokens": 2160,
 "output_tokens": 362,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.

Your total visible earnings across all rounds are \$28.1.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the pattern with my co-player:

As sender: Consistently sent \$3.00 (Rounds 4, 6)

As receiver: Returned \$3.50 on \$6.96 (Round 5, ~50%)

This partnership has been stable and mutually beneficial. I'll continue the cooperative pattern.

```
'''json
{ "send": 3 }
'''
```

**reasoning:**

Looking at the visible history with my current co-player:

As sender (Rounds 4, 6): They've sent \$3.00 consistently

As receiver (Round 5): They returned \$3.50 on \$6.96 (~50%)

This is a highly cooperative and stable partner. They maintain consistent sending amounts and fair return rates. I should continue matching their cooperative behavior.

I'll send \$3 again to maintain the mutual cooperation.

**usage:**

```
{
 "input_tokens": 2484,
 "output_tokens": 196,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.

- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible payoff \$5.0.

Your total visible earnings across all rounds are \$36.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at my co-player's behavior:

Round 5: Returned \$3.50 on \$6.96 (~50%)

Round 7: Returned \$5.50 on \$10.50 (~52%)

They've maintained a consistent ~50% return rate. Now they've sent me \$2.91, which became \$8.73.

I'll continue the cooperative pattern with a ~50% return.

```
'''json
{"return": 4.35}
'''
```

**Reasoning:**

Looking at the history with my co-player:

Round 5 (they were receiver): They returned \$3.50 on \$6.96 (~50%)

Round 6 (they were sender): They sent \$3.00

Round 7 (they were receiver): They returned \$5.50 on \$10.50 (~52%)

They've been very consistent with ~50% return rates as receiver.

Now they sent me \$2.91, which became \$8.73.

To maintain this cooperative relationship, I should return approximately 50% of what I received.

50% of \$8.73 = \$4.365

I'll return around \$4.35 or \$4.40.

```
Usage: {
 "input_tokens": 2635,
 "output_tokens": 278,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T08:02:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 }
 },
 "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
 Message 2 role=user
 content:
 Round 2

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $4.89 returned, visible p
 ayoff $6.89.
 Your total visible earnings across all rounds are $5.08.

 This round, you are the SENDER against your current co-player. You have $5. How much do yo
 u send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
 Message 3 role=assistant
 content:
 Looking at the visible history:

 Analysis:
 - My co-player sent $3.0 (60% of their endowment) in Round 1
 - They saw a return of $4.89, giving them a total payoff of $6.89
 - This represents good cooperation from their perspective

```

**\*\*My decision:\*\***

Since my co-player demonstrated trust by sending a substantial portion in Round 1, I should reciprocate with similar trust. I'll send slightly more to show appreciation for the cooperative start.

Sending \$3.50 (70% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Willingness to continue cooperative play
- Similar level of commitment they showed

{'send': 3.50}

**reasoning:**

Let me analyze this situation:

**\*\*What I know:\*\***

- Round 1: I was the RECEIVER
  - The sender sent \$3.36 (67.2% of their endowment)
  - I received \$10.08
  - I returned \$5.00
  - I ended with \$5.08

**\*\*Now in Round 2, the visible history shows:\*\***

- My current co-player was the SENDER in Round 1
- They sent \$3.0 (wait, I thought they sent \$3.36?)
- It became \$9.0 (which would be 3x of \$3.0)
- They saw \$4.89 returned (but I returned \$5.00)
- Their visible payoff was \$6.89

This is interesting - there's noise in the communication! The prompt warned about this.

- Actual sent: \$3.36 ? They saw: \$3.0
- I received: \$10.08 ? They saw it became: \$9.0
- I returned: \$5.00 ? They saw: \$4.89
- Their actual payoff:  $\$1.64 + \$5.00 = \$6.64$  ? They saw: \$6.89

So there's noise, and my co-player thinks they got a slightly better deal than they actually did.

**\*\*Now I'm the SENDER:\*\***

My co-player showed trust by sending 60% (or what they thought was \$3.0/\$5.0) in Round 1.

From their perspective:

- They sent \$3.0
- Got back \$4.89
- Made a profit of \$1.89

This is a reasonable return rate from their perspective (about 63% return rate on what was sent).

**\*\*Decision:\*\***

Since they demonstrated trust and cooperation, I should reciprocate. I'll send a similar proportion (around 60-70%) to maintain the cooperative relationship.

Given:

- They sent 67.2% (or from their noisy view, 60%)
- They received a 163% return on their investment from their perspective, which is quite encouraging

I'll send \$3.50 (70% of my endowment), showing I value the cooperative dynamic. If they mirror my return rate of roughly 50%, I'd receive about \$5.25 back, ending with \$6.75 total...

**usage:**

```
{
 "input_tokens": 892,
 "output_tokens": 1059,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:



Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

Your total visible earnings across all rounds are \$12.23.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), I returned fairly
- Round 2: I sent \$3.49 (visible), co-player returned \$5.25 (50.1% of what they received)
- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

**\*\*My decision:\*\***

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%). They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with:  $\$0.73 + \$6.50 = \$7.23$  (profit of \$2.23)
- I end with:  $\$12.81 - \$6.50 = \$6.31$

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

**reasoning:**

Let me analyze the situation:

**\*\*Visible history:\*\***

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became \$10.47, returned \$5.25, their payoff \$5.22

**\*\*Current Round 3:\*\***

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

**\*\*Analysis:\*\***

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with:  $\$0.73 + \$6.50 = \$7.23$  (profit of \$2.23)
- I end with:  $\$12.81 - \$6.50 = \$6.31$

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with:  $\$0.73 + \$7.00 = \$7.73$  (profit of \$2.73)
- I end with:  $\$12.81 - \$7.00 = \$5.81$

This is even more generous and rewards their very high trust level.

...

**usage:**

```
{
 "input_tokens": 1260,
 "output_tokens": 864,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.

Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

**\*\*Pattern observed:\*\***

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

**\*\*My decision:\*\***

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

**reasoning:**

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sendi

ng around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

**usage:**

```
{
 "input_tokens": 1769,
 "output_tokens": 533,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.2200000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$23.66.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 2: I sent \$3.49, co-player returned \$5.25 (~50%)
- Round 3: Co-player sent \$3.5, I returned \$6.64 (~63%)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)

**\*\*Current situation:\*\***

- Co-player sent \$3.02 (60.4% of endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

**\*\*Pattern:\*\***

- Co-player consistently returns around 50% as receiver
- They've shown steady trust as sender
- Cooperative relationship maintained

**\*\*My decision:\*\***

Returning \$4.50:

- Sender ends with: \$1.98 + \$4.50 = \$6.48 (profit of \$1.48)
- I end with: \$9.06 - \$4.50 = \$4.56

This represents ~49.7% return, matching their fairness pattern and maintaining the balance of cooperation we've established.

{'return': 4.50}

**reasoning:**

Looking at the visible history:

Round 2: I sent \$3.49, co-player returned \$5.25 (50.1% of \$10.47)

Round 3: Co-player sent \$3.5, I returned \$6.64 (63.2% of \$10.5)

Round 4: I sent \$3.02 (visible), co-player returned \$4.5 (49.7% of \$9.06)

Current round:

- Co-player sent \$3.02 (60.4% of their endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

Pattern analysis:

- Co-player has been returning around 50% as receiver
- I've been returning slightly more (around 50-63%)

- Cooperation has been consistent
- Co-player sent 60.4% this round (moderate trust)

My decision:

The co-player has been fair and consistent, returning around 50% when they're receiver. They sent \$3.02 (60.4%) showing good trust. I should return a fair amount.

Returning \$4.50:

- Sender ends with:  $\$1.98 + \$4.50 = \$6.48$  (profit of \$1.48)
- I end with:  $\$9.06 - \$4.50 = \$4.56$

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

usage:

```
{
 "input_tokens": 2220,
 "output_tokens": 617,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.

Your total visible earnings across all rounds are \$28.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 3: Co-player sent \$3.5, saw good return (\$6.64)
- Round 4: I sent \$3.02, co-player returned \$4.5 fairly
- Round 5: Co-player sent \$3.5, but only saw \$4.19 returned (lower than usual)

**\*\*Observation:\*\***

From their perspective, Round 5 showed a lower return rate (~40%) compared to our earlier cooperation. This could be due to communication noise mentioned in the instructions.

**\*\*My decision:\*\***

Despite the apparent lower return in Round 5, the overall pattern shows:

- Co-player consistently sends 60-70% as sender
- They return fairly (~50%) as receiver
- Strong cooperative history overall

I should maintain high trust to signal continued cooperation and not let potential noise disrupt our established relationship. Sending \$3.50 (70% of endowment) demonstrates:

- Continued confidence in cooperation
- Consistency with our established pattern
- Resilience to short-term variations

```
{'send': 3.50}
```

reasoning:

Looking at the visible history from my co-player's perspective:

- Round 3: Co-player sent \$3.5, saw \$6.64 returned (good return)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (much lower than before!)

In Round 5, they saw only \$4.19 returned when they sent \$3.5 (which became \$10.5). That's

only about 40% return rate, which is lower than their typical ~50% returns and much lower than what I gave them in Round 3 (63%).

This could represent:

1. Communication noise (amounts seen may differ)
2. A change in my return behavior from their perspective

From my perspective, my total earnings are \$28.22. Let me think about what actually happened vs what they saw...

The co-player has been consistently trusting (sending 60-70% of endowment) and they've been fair receivers (50% returns). But in Round 5, from their perspective, I returned less than usual.

Should I:

- Continue high trust to show I'm still cooperative?
- Send less to protect myself since they might think I'm defecting?
- Send the same to maintain consistency?

Given that:

1. They've been very consistent as senders (60-70%)
2. Their return pattern as receivers has been stable (~50%)
3. The lower return they saw in Round 5 might be noise
4. Overall we've had good cooperation

I should continue with high trust to maintain the cooperative relationship. Sending around \$3.50-\$3.75 shows I'm still committed to cooperation.

...

**usage:**

```
{
 "input_tokens": 2706,
 "output_tokens": 670,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.

Your total visible earnings across all rounds are \$36.03.

This round, you are the RECEIVER against your current co-player. They sent you \$4.29, so you received \$12.870000000000001. How much do you return? (0-12.870000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw lower return (\$4.19)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$4.29 (85.8% of endowment) - very high trust!

**\*\*Current situation:\*\***

- I received \$12.87
- Co-player has \$0.71 remaining
- They demonstrated exceptional trust despite seeing a lower return in Round 5

**\*\*My decision:\*\***

They've shown remarkable trust by sending 85.8% of their endowment. They also maintained fair 50% returns as receiver. I should reciprocate with a fair return around 50%.

Returning \$6.50:

- Sender ends with:  $\$0.71 + \$6.50 = \$7.21$  (profit of \$2.21)
- I end with:  $\$12.87 - \$6.50 = \$6.37$

This represents ~50.5%, consistent with their fairness pattern and rewarding their high trust.

```
{'return': 6.50}
```

**reasoning:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower ~40%)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)

**\*\*Current situation:\*\***

- Co-player sent \$4.29 (85.8% of endowment) - very high trust!
- I received \$12.87
- Co-player has \$0.71 remaining

**\*\*Pattern:\*\***

- Co-player consistently returns around 50% as receiver
- They've shown very high trust as sender in this round (85.8%)
- They maintained fair returns even after what appeared to be a lower return in Round 5

**\*\*My decision:\*\***

They sent an exceptionally high amount (\$4.29, 85.8%), showing strong trust. I should return fairly, around 50% to match their pattern.

Returning \$6.50:

- Sender ends with:  $\$0.71 + \$6.50 = \$7.21$  (profit of \$2.21)
- I end with:  $\$12.87 - \$6.50 = \$6.37$

This represents ~50.5% return, matching their fairness pattern and rewarding their very high trust.

**usage:**

```
{
 "input_tokens": 3206,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$42.400000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the visible history:

**\*\*Analysis:\*\***

- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50% - fair)
- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned (better)

**\*\*Pattern observed:\*\***

- Co-player consistently maintains high trust as sender (70-75%)
- Returns fairly as receiver (50%)

- Despite lower returns in Round 5, they increased their send in Round 7
- Strong, resilient cooperation throughout

**\*\*My decision:\*\***

We've established an excellent cooperative relationship. My co-player has shown remarkable consistency and trust. For this round, I'll send \$4.00 (80% of endowment) to:

- Reciprocate their sustained high trust
- Demonstrate continued commitment to cooperation
- Reward their fair behavior as receiver

{'send': 4.00}

#### Reasoning:

Let me analyze the visible history:

**\*\*Co-player's visible history:\*\***

- Round 5: As SENDER, sent \$3.5, saw \$4.19 returned, visible payoff \$5.69
- Round 6: As RECEIVER, saw \$4.2 sent, returned \$6.3, visible payoff \$6.30
- Round 7: As SENDER, sent \$3.75, saw \$5.93 returned, visible payoff \$7.18

**\*\*Pattern analysis:\*\***

- As sender: consistently sends high amounts (70-75%)
- As receiver: returns 50% (\$6.3 out of \$12.6)
- Despite seeing lower returns in Round 5, they increased their send in Round 7 to \$3.75
- Round 7 return appears to be around 53% (\$5.93 out of \$11.25)

**\*\*Overall cooperation:\*\***

- This co-player has been very cooperative
- They maintain high trust as sender
- They return fairly as receiver
- We have a strong cooperative relationship

**\*\*My decision:\*\***

Given their continued high trust (sending 75% in Round 7) and fair returns, I should maintain high cooperation. They've shown resilience and continued trust.

Sending \$4.00 (80% of endowment) demonstrates:

- Strong reciprocity for their ongoing trust
- Confidence in continued cooperation
- Recognition of our established positive relationship

This is a good cooperative signal for what appears to be the final or near-final round.

Usage: {

"input\_tokens": 3439,  
"output\_tokens": 582,  
"reasoning\_tokens": 0

}

#### GAME - Agent\_3 - Receiver - dyad\_2

Timestamp: 2026-08-12T08:03:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_2",  
"move\_index": 3,  
"opponent\_id": "Agent\_4",  
"pairing": {  
  "agents": [  
    "Agent\_4",  
    "Agent\_3"  
  ],  
  "investor": "Agent\_4",  
  "roles": {  
    "Agent\_3": "trustee",  
    "Agent\_4": "investor"  
  },  
  "trustee": "Agent\_3"

},

"role": "trustee",  
"role\_label": "Receiver",

```

"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.

Your total visible earnings across all rounds are \$6.89.

This round, you are the RECEIVER against your current co-player. They sent you \$3.49, so you received \$10.47. How much do you return? (0-10.47)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player saw I sent them \$3.36 (due to noise), received \$10.08, and returned \$5.0 - approximately 50% of what they received. This shows they're cooperative and will



lling to share gains fairly.

Now they've sent me \$3.49, and I received \$10.47. To reciprocate their cooperative behavior and maintain trust, I should return a similar proportion - around 50% of what I received.

```
'''json
{"return": 5.25}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to this co-player
2. Due to communication noise, they saw \$3.36 (not \$3)
3. They received \$10.08 ( $3 \times \$3.36$ )
4. They returned \$5.0 to me
5. My visible earnings show \$6.89 total (which would be  $\$5 - \$3 + \$5 = \$7$ , but with noise it shows \$6.89)

Now in Round 2:

- This co-player is the SENDER
- They sent me what I see as \$3.49
- I received \$10.47 ( $3 \times \$3.49$ )
- I need to decide how much to return

Analysis of their behavior:

- They saw I sent them \$3.36 (noisy version of my \$3)
- They kept \$5.08 out of \$10.08
- They returned \$5.0, which is about 50% of what they received

This shows:

1. They're willing to cooperate and share gains
2. They returned a fair amount - about half of what they received
3. This left them with slightly more than the initial \$5 but shared the gains

My strategy:

- They showed good faith by returning \$5
- I should reciprocate to maintain cooperation
- Returning about 50% would be \$5.23-\$5.24
- But I could return slightly more to encourage continued cooperation
- Returning around \$5-6 seems reasonable

Let me think about what's fair:

- They sent \$3.49 (costing them \$3.49)
- I got \$10.47
- If I return \$5-6, I keep \$4.47-\$5.47
- They would end up with  $\$5 - \$3.49 + \$5-6 = \$6.51-7.51$

...

**usage:**

```
{
 "input_tokens": 780,
 "output_tokens": 697,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: \$5 - \$3 + \$5 = \$7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff: \$5 - \$3.5 + \$5.65 = \$7.15

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

**usage:**

```
{
 "input_tokens": 1080,
 "output_tokens": 477,
 "reasoning_tokens": 0
}
```

**Message 6 role=user****content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant****content:**

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)  
Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)  
Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)  
Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:  
50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

**usage:**

```
{
 "input_tokens": 1454,
 "output_tokens": 373,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our history:

Round 2: They sent \$3.5, I returned well  
Round 3: I sent ~\$4.27, they returned \$6.5  
Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json
{"send": 3.5}
'''
```

'''

**reasoning:**

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)

Round 3: I sent ~\$4.27 (they saw), they returned \$6.5

Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?
3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

**usage:**

```
{
 "input_tokens": 1812,
 "output_tokens": 554,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$30.500000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our ongoing cooperation:

Round 3: They returned \$6.5 on ~\$12.81 (51%)

Round 4: I returned \$4.5 but they saw \$3.87 due to noise

Round 5: They returned \$4.5 on \$9.06 (50%)

They've sent me \$4.2 this round (more than previous rounds), showing continued trust despite the noise that affected Round 4. I received \$12.6.

To reciprocate their trust and maintain our ~50% return pattern:

```
'''json
{"return": 6.3}
'''
```

**reasoning:**

Let me trace through the history with my current co-player:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)  
Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)  
Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (51%)  
Round 4: They sent \$3.75, I received \$11.25 and returned \$4.5, but they saw only \$3.87 (noise)  
Round 5: I sent \$3.5 (they saw \$3.02 due to noise), they received \$9.06 and returned \$4.5 (50%)  
Round 6: They sent \$4.2, I received \$12.6

We've established a pattern of roughly 50% returns. Despite some noise in communication, we've been maintaining cooperation. They just sent me \$4.2, which is a bit more than previous rounds, showing continued trust.

I should return around 50% to maintain our cooperative relationship:  
50% of \$12.6 = \$6.3

This keeps the pattern going.

**usage:**

```
{
 "input_tokens": 2218,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.

Your total visible earnings across all rounds are \$36.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our continued partnership:

Round 4: They sent \$3.75, I returned well (though noise affected visibility)

Round 5: They returned \$4.5 on \$9.06

Round 6: They sent \$3.5, I returned ~\$6.3 (~60%)

We've built a strong cooperative relationship over 6 rounds with consistent sending and fair returns. They've maintained trust and cooperation throughout.

To continue this successful pattern:

```
'''json
{"send": 3.75}
'''
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 4: They sent \$3.75, saw \$3.87 returned (I actually sent \$4.5)  
Round 5: They returned \$4.5 on \$9.06 they received (I sent \$3.02 which they saw)  
Round 6: They sent \$3.5, saw \$6.31 returned (I actually returned \$6.3)

The noise has been relatively small. They've been consistently cooperative:

- Sending between \$3.5-\$4.2
- Returning around 50% of what they receive
- Showing trust even when returns looked lower due to noise

We've established a strong cooperative pattern over 6 rounds. I should continue sending a substantial amount to maintain this relationship.

I'll send around \$3.5-\$4 to match their commitment level.

**usage:**

```
{
 "input_tokens": 2584,
 "output_tokens": 326,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.
- Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.370000000000001.

Your total visible earnings across all rounds are \$43.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.17, so you received \$9.51. How much do you return? (0-9.51)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our consistent cooperation:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've maintained a strong cooperative relationship with fair returns throughout. They sent \$3.17 this round, and I received \$9.51.

Continuing our ~50% return pattern:

```
'''json
{"return": 4.75}
'''
```

**Reasoning:**

Looking at the history:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 but they saw \$6.31 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've been maintaining a cooperative relationship with returns around 50-60% of what we receive. They sent me \$3.17 this round, and I received \$9.51.

Following our established pattern of ~50% returns:

$9.51 * 0.5 = 4.755$

I'll round to \$4.75 or \$4.8 for simplicity.

Let me go with \$4.75 to maintain the ~50% pattern.

Usage: {

```
"input_tokens": 2783,
"output_tokens": 321,
"reasoning_tokens": 0
}
```

### GAME - Agent\_6 - Sender - dyad\_3

Timestamp: 2026-08-12T08:03:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

##### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.

Your total visible earnings across all rounds are \$4.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing strong trust
- From their perspective, they received \$5.3 back and earned \$7.3 total
- This suggests they saw positive reciprocity

**\*\*Round 2 Decision:\*\***

Since my co-player demonstrated cooperative behavior by sending 60% in Round 1, I should reciprocate with similar trust. They've shown they're willing to cooperate, and their positive experience (from their view, getting \$5.3 back on a \$3 send) should encourage continued reciprocity.

Sending \$3.00 (60% of my endowment) mirrors their trust level and maintains the cooperative dynamic.

{ 'send': 3.00 }

**reasoning:**

Let me analyze this situation:

Round 1 recap from my perspective:

- I was the RECEIVER
- Sender sent \$3.03 (60.6%)
- I received \$9.09
- I returned \$4.50
- My earnings: \$4.59

From the visible history shown:

- My co-player's view shows they sent \$3.0 (communication noise from \$3.03)
- They saw it became \$9.0 (communication noise from \$9.09)
- They saw \$5.3 returned (communication noise from \$4.50)
- Their visible payoff: \$7.3 (noise from actual \$6.47)

So there's communication noise affecting what each player sees.

Now in Round 2:

- I'm the SENDER
- Same co-player (we're paired again)
- They were generous in Round 1, sending 60.6%
- I returned about 50% of what I received (4.50/9.09 ? 49.5%)

Key considerations:

1. My co-player showed trust by sending a majority of their endowment
2. From their perspective, they saw \$5.3 returned (which they might view positively)
3. They have demonstrated cooperative behavior
4. I should reciprocate the trust they showed

Given they sent 60.6%, I should consider sending a similar proportion to show reciprocity.

If I send around 60%, that would be \$3.00.



However, I also need to consider:

- Will they return a fair amount given what they saw last round?
- From their view, they sent \$3 and got back \$5.3, which is a good return
- This might encourage them to be generous again as receiver

I think sending \$3.00 (60%) would be appropriate, matching their level of trust from Round 1.

...

**usage:**

```
{
 "input_tokens": 856,
 "output_tokens": 628,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - **\*\*increased trust!\*\***

**\*\*Current state:\*\***

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level. I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

{'return': 5.25}

**reasoning:**

Let me analyze the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

**\*\*Round 3 (Current):\*\***

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:
  - Round 1: sent 60% (\$3.0)
  - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have \$5 - \$3.5 = \$1.5 left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with:  $\$1.5 + \$5.0 = \$6.5$  (gain of \$1.5)
- I end with:  $\$10.5 - \$5.0 = \$5.5$  (gain of \$5.5)

If I return \$5.25:

- Sender ends with:  $\$1.5 + \$5.25 = \$6.75$  (gain of \$1.75)
- I end with:  $\$10.5 - \$5.25 = \$5.25$  (gain of \$5.25)

If I return \$5.50:

- Sender ends with:  $\$1.5 + \$5.50 = \$7.0$  (gain of \$2.0)
- I end with:  $\$10.5 - \$5.50 = \$5.0$  (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

usage:

```
{
 "input_tokens": 1230,
 "output_tokens": 964,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

**\*\*Co-player's behavior pattern:\*\***

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

**\*\*Analysis:\*\***

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue ma

tching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

```
{'send': 3.00}
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

**usage:**

```
{
 "input_tokens": 1702,
 "output_tokens": 665,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains fair reciprocity at 50% and ensures mutual profit.

{'return': 4.20}

**reasoning:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned
- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

**\*\*Current state:\*\***

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:

- Sender ends with:  $\$2.2 + \$4.20 = \$6.40$  (gain of \$1.40)
- I end with:  $\$8.4 - \$4.20 = \$4.20$  (gain of \$4.20)

This maintains the 50% return rate that's been consistent in this relationship.

**usage:**

```
{
 "input_tokens": 2145,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.

Your total visible earnings across all rounds are \$26.669999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's recent experience:\*\***

- Round 3: Sent \$3.0, saw \$5.87 returned ? good outcome (\$7.87 total)
- Round 4: As receiver, returned \$4.4 (50%) ? fair reciprocity
- Round 5: Sent \$3.0, saw only \$3.6 returned ? poor outcome (\$5.6 total)

**\*\*Analysis:\*\***

From their perspective, Round 5 was disappointing - they saw much lower returns than in Round 3. However, I know this is likely due to communication noise in the game, not actual low reciprocity.

**\*\*Strategy:\*\***

I should maintain consistent cooperation to signal reliability. Despite the noise affecting what they saw, continuing to send at the 60% level shows I'm trustworthy and committed to cooperation. This should help stabilize the relationship through the noise.

{'send': 3.00}

**reasoning:**

Looking at the co-player's visible history:

- Round 3: Sent \$3.0 as sender, saw \$5.87 returned ? earned \$7.87
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? earned \$5.6

This is interesting. In Round 5, from their perspective:

- They sent \$3.0 and saw only \$3.6 returned
- That's only 40% of the \$9.0 they sent (which became after multiplication)
- This is much lower than what they saw in Round 3 (\$5.87)

From my actual history, in Round 5:

- They sent \$2.8, I received \$8.4, I returned \$4.20
- But due to noise, they saw \$3.0 sent and saw \$3.6 returned

So from their perspective, they had a very poor outcome in Round 5 - earning only \$5.6 compared to \$7.87 in Round 3.

This might make them less trusting going forward. They might:

1. Send less
2. Return less if they're the receiver
3. Still cooperate but be more cautious

Given this situation, I have a few options:

1. Continue sending \$3.00 (60%) to maintain cooperation and show I'm still trustworthy
2. Send less to hedge against possible reduced reciprocity
3. Send more to try to rebuild trust

I think option 1 is best - continue the pattern and show consistency. The poor outcome in Round 5 from their view was due to noise, not my actual behavior. Maintaining consistency might help preserve the relationship.

Let me send \$3.00 again.

**usage:**

```
{
 "input_tokens": 2637,
 "output_tokens": 638,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.

Your total visible earnings across all rounds are \$34.269999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so y

ou received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

**\*\*Co-player's behavior:\*\***

- Round 4: As receiver, returned \$4.4 (~50%)
- Round 5: Sent \$3.0, saw poor returns (\$3.6)
- Round 6: As receiver, still returned \$5.0 (~50%) despite disappointing R5
- Round 7: Sent \$3.37 (67.4%) - **\*\*increased trust despite disappointment!\*\***

**\*\*Analysis:\*\***

My co-player showed remarkable resilience and trust. Despite seeing low returns in Round 5, they:

1. Continued fair reciprocity in Round 6
2. Actually increased their send to 67.4% in Round 7

This demonstrates strong commitment to cooperation.

**\*\*Current state:\*\***

- They sent \$3.37, keeping only \$1.63
- I received \$10.11

I should reward this continued trust with consistent fair reciprocity.

If I return \$5.05:

- Sender ends with:  $\$1.63 + \$5.05 = \$6.68$  (gain of \$1.68)
- I end with:  $\$10.11 - \$5.05 = \$5.06$  (gain of \$5.06)

This maintains ~50% reciprocity and mutual benefit.

{'return': 5.05}

**reasoning:**

Looking at the co-player's recent history:

- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? poor outcome
- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)
- Round 7: Sent \$3.37 (67.4%) as sender

**\*\*Analysis:\*\***

Despite the disappointing Round 5 (from their view), they:

1. Continued reciprocating fairly as receiver (50% in Round 6)
2. Actually INCREASED their send amount to \$3.37 (67.4%) in Round 7

This shows remarkable resilience and continued trust! They're not retaliating despite seeing a bad outcome in Round 5. They're maintaining cooperative behavior.

**\*\*Current state:\*\***

- They sent \$3.37, keeping only \$1.63
- I received \$10.11
- They've increased trust even after disappointment

I should strongly reward this continued cooperation and trust escalation.

If I return \$5.05 (50% of what I received):

- Sender ends with:  $\$1.63 + \$5.05 = \$6.68$  (gain of \$1.68)
- I end with:  $\$10.11 - \$5.05 = \$5.06$  (gain of \$5.06)

This maintains my ~50% reciprocity pattern and ensures mutual benefit. Given they increased their send, I could consider being slightly more generous, but consistency is also valuable.

Let me stick with \$5.05 to maintain the 50% pattern.

**usage:**

```
{
 "input_tokens": 3082,
 "output_tokens": 700,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.

Your total visible earnings across all rounds are \$39.33.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

**\*\*Co-player's pattern:\*\***

- Round 5: Sent \$3.0, saw \$3.6 returned - disappointing
- Round 6: As receiver, returned \$5.0 (50%) - maintained fair reciprocity
- Round 7: Sent \$3.0, saw \$4.25 returned - better, but still moderate

**\*\*Analysis:\*\***

My co-player has consistently sent \$3.0 (60%) as sender across multiple rounds, demonstrating steady cooperation despite variable returns they've observed (likely due to noise). They've also maintained fair 50% reciprocity as receiver.

This shows reliable commitment to cooperation regardless of occasional disappointments.

**\*\*Strategy:\*\***

I should match their consistent 60% sending rate to maintain our cooperative relationship.

{'send': 3.00}

Reasoning:

Looking at the co-player's visible history:

Round 5: Sent \$3.0 as sender, saw \$3.6 returned (40% of \$9) - poor outcome, got \$5.6 total  
Round 6: As receiver, saw \$3.33 sent (became \$9.99), returned \$5.0 (50%) - fair reciprocity maintained  
Round 7: Sent \$3.0 as sender, saw \$4.25 returned (~47% of \$9) - better than R5 but still lower than R3's \$5.87

From their perspective:

- They've been consistently sending \$3.0 (60%)
- Return rates they've seen have been variable: \$5.87, \$3.6, \$4.25
- They maintained fair reciprocity as receiver at 50%

Despite some disappointing outcomes (especially R5), they've maintained their 60% sending rate. I should continue being cooperative and send \$3.00 (60%) to maintain our established cooperation pattern.

My total visible earnings are \$39.33 across 7 rounds, which suggests I'm doing well overall.

I should send \$3.00 to maintain the cooperative equilibrium we've established.

```
Usage: {
 "input_tokens": 3350,
 "output_tokens": 475,
 "reasoning_tokens": 0
}
```

GAME - Agent\_5 - Receiver - dyad\_3

Timestamp: 2026-08-12T08:03:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
 "dyad\_id": "dyad\_3",

```

"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2



Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.

Your total visible earnings across all rounds are \$7.3.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at our history:

In Round 1, when I sent them approximately \$3, they received about \$9 and returned \$4.5 - showing good reciprocity at nearly 50%.

Now they've sent me \$2.92 (similar amount to what I sent), and I received \$8.76.

To maintain reciprocal cooperation, I should return a similar proportion to what they returned. They gave back roughly half of what they received, which was fair and cooperative.

Applying a similar approach: returning around 50% would be \$4.38, which would give them a total of about \$6.46 and leave me with \$4.38.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER and sent \$3
  - The receiver saw \$3.03 (slight noise)
  - This became \$9.09 for them
  - They returned \$4.5
  - My visible payoff was \$4.59 (I kept \$5 - \$3 = \$2, plus got back \$4.5 = \$6.5, but visible was \$7.3 total?)

Wait, let me recalculate:

- I sent \$3, kept \$2
- Received back \$4.5
- My total was \$2 + \$4.5 = \$6.5

But my "total visible earnings across all rounds are \$7.3". That's close to \$6.5, so there might be some noise in what I see as well.

2. Now in Round 2, I'm the RECEIVER
  - My co-player (the same person from Round 1) sent me \$2.92
  - I received \$8.76 (which is  $3 \times \$2.92$ )
  - I need to decide how much to return

Key considerations:

- This is the same co-player from Round 1
- In Round 1, they returned \$4.5 out of \$9.09 they received (about 49.5%)
- They sent me \$2.92 this round, which is slightly less than the \$3 I sent them in Round 1
- They were fairly reciprocal in Round 1

Strategy:

- I should be reciprocal given their reasonable behavior in Round 1
- They returned about half of what they received
- They sent me about \$3 (with noise)
- A fair return would be to:
  - Give them enough to make their total payoff reasonable
  - Not exploit them given they were cooperative
- Since they kept \$2.08 from their initial \$5 and returned roughly half of what they received, returning around \$4.34 (approximately 49.5% of \$8.76) would mirror their reciprocity and leave them with a reasonable total of \$6-7....

**usage:**

```
{
 "input_tokens": 824,
```

```

 "output_tokens": 873,
 "reasoning_tokens": 0
 }
Message 4 role=user
 content:
 Round 3

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 1 against you, as RECEIVER: saw $3.03 sent, it became $9.09, returned $4.5, visible payoff $4.59.
 - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $3.55 returned, visible payoff $5.55.
 Your total visible earnings across all rounds are $11.66.

 This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
 content:
 Looking at our history together:

 Round 1: I sent ~$3, they received ~$9 and returned $4.5 (good reciprocity)

 Round 2: They sent $3, I received $9 and returned $4.4, though they saw $3.55 due to communication noise

 My co-player has been consistently cooperative, sending about $3 both times regardless of what they perceived as my return. This shows reliability and trust.

 I should maintain this cooperative pattern. Sending $3 again:
 - Signals continued trust and cooperation
 - Matches their consistent behavior
 - Creates mutual benefit through the 3x multiplier

        ```json
        {"send": 3}
        ```

 reasoning:
 Let me analyze the history:

 Round 1: I sent ~$3, they received ~$9, they returned $4.5 (about 50%)
 Round 2: They sent $3, received $9, they saw $3.55 returned (but I actually returned $4.4, so there's communication noise)

 The pattern shows:
 - My co-player has been consistent in sending about $3
 - In Round 1, they returned a good amount ($4.5 out of $9 received)
 - In Round 2, I returned $4.4, but due to communication noise they only saw $3.55

 From their perspective, they might think I was less generous than I actually was. However, they still sent $3 in Round 2, maintaining their cooperative approach.

 Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending $3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

 Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.
 usage:
 {
 "input_tokens": 1160,
 "output_tokens": 419,
 "reasoning_tokens": 0
 }
Message 6 role=user
 content:
 Round 4

```

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)  
**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (49% of received)  
**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)  
**\*\*Round 4\*\*:** They sent \$2.94, I received \$8.82

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relationship.

```
'''json
{"return": 4.4}
'''
```

**reasoning:**

Looking at our history:

Round 1: I sent ~\$3, they received ~\$9, returned \$4.5  
Round 2: They sent \$3, I received \$9, returned \$4.4 (but they saw \$3.55)  
Round 3: I sent \$3 (they saw \$3.5), they received ~\$10.5, returned \$5.25  
Round 4: They sent \$2.94, I received \$8.82

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them: \$5 - \$2.94 + \$4.41 = \$6.47

And leave me with: \$8.82 - \$4.41 = \$4.41

That seems fair and continues our mutually beneficial cooperation.

**usage:**

```
{
 "input_tokens": 1549,
 "output_tokens": 479,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our consistent pattern:

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4 (they saw \$3.55)

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

**\*\*Round 1\*\*:** I sent ~\$3, they returned \$4.5 (50% of received)

**\*\*Round 2\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (exactly 50% of received)

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 1959,
 "output_tokens": 430,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.

Your total visible earnings across all rounds are \$29.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant****content:**

Looking at our continued cooperation:

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50%)  
**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3.33, I received \$9.99

My co-player continues to be highly consistent and fair, returning approximately 50% each time. They've proven to be a trustworthy partner across all rounds.

I should maintain this reciprocal pattern by returning about 50% of what I received: 50% of \$9.99 = \$5.0

```
'''json
{"return": 5.0}
'''
```

**reasoning:**

Looking at the history:

**\*\*Round 3\*\*:** I sent \$3, they returned \$5.25 (50% of \$10.5)  
**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08 due to noise)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50% of \$8.4)  
**\*\*Round 6\*\*:** They sent \$3.33, I received \$9.99

My co-player has been very consistent - always returning about 50% of what they receive. They've been a reliable, fair partner throughout.

I should continue this pattern. Returning 50% of \$9.99 = \$4.995, so about \$5.0

This maintains our reciprocal relationship and ensures continued cooperation.

**usage:**

```
{
 "input_tokens": 2361,
 "output_tokens": 372,
 "reasoning_tokens": 0
}
```

**Message 12 role=user****content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.1999999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.

Your total visible earnings across all rounds are \$34.540000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant****content:**

Looking at our established pattern:

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has maintained perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues cooperation regardless of noise

This is a highly trustworthy partner with a stable strategy. I should maintain our mutually beneficial pattern by sending \$3 again.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the pattern with my co-player:

**\*\*Round 4\*\*:** They sent \$3, I returned \$4.4 (they saw \$5.08)  
**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has been extremely consistent:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- This pattern has held across all 6 rounds

This is a highly reliable, cooperative partner. I should continue with our established pattern of sending \$3.

The relationship is stable and mutually beneficial. No reason to deviate.

**usage:**

```
{
 "input_tokens": 2770,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.
- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.

Your total visible earnings across all rounds are \$40.790000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.32, so you received \$9.959999999999999. How much do you return? (0-9.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our continued cooperation:

**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3, I returned \$5.0 (they saw \$5.6)  
**\*\*Round 7\*\*:** I sent \$3, they returned \$5.05 (50%)  
**\*\*Round 8\*\*:** They sent \$3.32, I received \$9.96

My co-player has maintained perfect consistency across all rounds - always sending about \$3 and returning 50% as receiver. This is a highly reliable and fair partner.

I should continue our established reciprocal pattern by returning approximately 50% of what I received: 50% of \$9.96 ? \$5.0

```
'''json
{"return": 5.0}
'''
```

**Reasoning:**

Looking at the pattern with this co-player:

**\*\*Round 5\*\*:** I sent \$3, they returned \$4.2 (50%)  
**\*\*Round 6\*\*:** They sent \$3, I returned \$5.0 (they saw \$5.6)  
**\*\*Round 7\*\*:** I sent \$3, they returned \$5.05 (50%)  
**\*\*Round 8\*\*:** They sent \$3.32, I received \$9.96

My co-player has been perfectly consistent throughout all rounds:

- Always sends about \$3 as sender

- Always returns 50% as receiver
- Maintains this cooperative pattern despite noise

I should continue our established reciprocal relationship by returning approximately 50% of what I received:

50% of \$9.96 = \$4.98

Rounding to \$5.0 maintains the pattern.

```
Usage: {
 "input_tokens": 2972,
 "output_tokens": 379,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Sender - dyad\_4

Timestamp: 2026-08-12T08:03:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$4.529999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 1 Analysis:\*\***

- My co-player sent \$3.0 (60% of their endowment) showing significant trust
- They received \$4.55 back and earned \$6.55 total
- This was a positive outcome for them

**\*\*Round 2 Strategy:\*\***

Since my co-player demonstrated trustworthiness and we had a successful cooperation in Round 1, I should reciprocate their trust level. They sent 60% of their endowment, so I'll match that trust signal.

Sending \$3.00 (60% of my endowment):

- Shows I trust them as much as they trusted me
- Maintains the cooperative relationship
- They'll receive \$9.00 to decide how to split

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (from my perspective):

- I was the RECEIVER
- The sender sent \$3.01, which became \$9.03 to me
- I returned \$4.50
- My earnings: \$9.03 - \$4.50 = \$4.53

But the visible history shows something slightly different due to communication noise:

- My co-player (who was the sender) saw that they sent \$3.0 (vs actual \$3.01)
- They saw it became \$9.0 (vs actual \$9.03)
- They saw \$4.55 returned (vs actual \$4.50 I returned)
- Their visible payoff: \$6.55

My actual payoff from Round 1: \$4.53

Their visible payoff from Round 1: \$6.55



Now in Round 2, I'm the SENDER and they're the RECEIVER.

Strategy considerations:

1. My co-player appeared to be trustworthy in Round 1 - they sent a significant amount (60.2%)
2. I reciprocated fairly by returning about 50% of what I received
3. They saw a slightly better outcome than what actually happened (\$4.55 vs \$4.50 returned)
4. This is a repeated game with the same partner, so building trust is valuable

Given that:

- My partner showed high trust in Round 1 (sent 60.2%)
- I reciprocated reasonably (returned ~50%)
- They likely feel good about the interaction based on what they saw

I should also send a substantial amount to:

1. Show that I'm also trustworthy
2. Continue building cooperation
3. Maximize joint gains

I'll send a similar proportion to what they sent in Round 1. They sent ~60%, so I'll send around that amount or slightly more to show reciprocal trust.

\$3 would be 60%

...

usage:

```
{
 "input_tokens": 901,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.810000000000005, returned \$3.4, visible payoff \$3.410000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 3 Analysis:\*\***

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

**\*\*Co-player's pattern:\*\***

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with: \$1.18 + \$5.80 = \$6.98 (gain of \$1.98)

- I end with:  $\$11.46 - \$5.80 = \$5.66$  (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\***

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

**\*\*Round 2:\*\***

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!
- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is  $\$2.27 \times 3$ )
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff:  $\$6.81 - \$3.40 = \$3.41$  ?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff:  $\$2.00 + \$3.40 = \$5.40$

My total earnings should be:  $\$4.53$  (Round 1) +  $\$5.40$  (Round 2) =  $\$9.93$

But visible history says \$10.31... There might be some rounding or noise in the visible earnings too.

**\*\*Round 3:\*\***

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have:  $\$5 - \$3.82 = \$1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

**usage:**

```
{
 "input_tokens": 1299,
 "output_tokens": 1178,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.810000000000005, returned \$3.4, visible payoff \$3.410000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible p

payoff \$7.23.  
Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)
- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

**\*\*Round 4 Strategy:\*\***

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Round 1:\*\***

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

**\*\*Round 2:\*\***

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

**\*\*Round 3:\*\***

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

**usage:**

```
{
 "input_tokens": 1833,
 "output_tokens": 737,
 "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should reciprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with:  $\$1.08 + \$6.00 = \$7.08$  (gain of \$2.08)
- I end with:  $\$11.76 - \$6.00 = \$5.76$  (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maintaining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

Looking at the situation:

**\*\*Round 5 Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have  $\$5 - \$3.92 = \$1.08$  remaining

**\*\*Co-player's pattern:\*\***

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with:  $\$1.08 + \$5.90 = \$6.98$  (gain of \$1.98)

- I end with:  $\$11.76 - \$5.90 = \$5.86$  (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

**usage:**

```
{
 "input_tokens": 2297,
 "output_tokens": 801,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.

Your total visible earnings across all rounds are \$28.909999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 3: Co-player sent \$3.0, received \$5.23, earned \$7.23
- Round 4: Co-player as receiver returned ~50.5% fairly
- Round 5: Co-player sent \$3.0, received even better return (\$6.98), earned \$8.98

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 as sender (reliable trust pattern)
- They return fairly as receiver (~50% return rate)
- Returns have been getting better over time, showing our cooperation strengthens
- Our partnership has been mutually profitable

**\*\*Round 6 Strategy:\*\***

The cooperation has been excellent and consistent. I'll maintain the same trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history:

**\*\*Co-player's recent pattern:\*\***

- Round 3: As sender, sent \$3.0, received \$5.23 back, earned \$7.23
- Round 4: As receiver, saw \$3.3 sent, returned \$5.0 (~50.5%), earned \$4.90
- Round 5: As sender, sent \$3.0, received \$6.98 back, earned \$8.98

**\*\*Key observations:\*\***

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5 visible)
- When they're the receiver, they return fairly (~50% - Round 4)
- They've received increasing returns over time, with Round 5 showing \$6.98 returned
- Our partnership has been very profitable for both of us

**\*\*What I know from my actual actions:\*\***

- Round 1: I was receiver, returned \$4.50
- Round 2: I was sender, sent \$3.00
- Round 3: I was receiver, returned \$5.80
- Round 4: I was sender, sent \$3.00
- Round 5: I was receiver, returned \$6.00

**\*\*Current situation:\*\***

- Round 6: I'm the sender
- My co-player has been trustworthy and fair
- They consistently send \$3.0 when they're the sender
- They return fairly when they're the receiver (~50%)
- The relationship has been mutually beneficial

**\*\*Strategy consideration:\*\***

Given the strong pattern of cooperation and mutual benefit, I should continue sending \$3.0 to maintain the trustworthy relationship. This has been working well for both of us.

However, I notice the co-player is seeing increasing returns (\$5.23 ? \$6.98 in their visible history). This might encourage them to continue being cooperative.

...

**usage:**

```
{
 "input_tokens": 2800,
 "output_tokens": 655,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.

Your total visible earnings across all rounds are \$35.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$1.81 remaining

**\*\*Co-player's pattern:\*\***

- Round 4: As receiver, returned ~50.5%
- Round 5: Sent \$3.0, received strong return
- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.19 showing solid trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocity.

If I return \$4.80:

- Sender ends with: \$1.81 + \$4.80 = \$6.61 (gain of \$1.61)
- I end with: \$9.57 - \$4.80 = \$4.77 (gain of \$4.77)

This represents exactly 50% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.80}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 7 Analysis:\*\***

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$5 - \$3.19 = \$1.81 remaining

**\*\*Co-player's pattern:\*\***

- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Sent \$3.0, received good return (\$6.98)
- Round 6: As receiver, returned ~49.5% fairly (\$5.4 out of \$10.89)
- Round 7: Sent \$3.19 (slightly lower than usual \$3.0, but still strong trust at ~64%)

My co-player has been consistently fair and trustworthy. They've been returning around 50% when they're receivers, and sending around \$3.0 when they're senders.

This round they sent \$3.19, which is slightly higher than their usual \$3.0. I should reciprocate with a fair return around 50%.

If I return \$4.80:

- Sender ends with: \$1.81 + \$4.80 = \$6.61 (gain of \$1.61)
- I end with: \$9.57 - \$4.80 = \$4.77 (gain of \$4.77)

This represents exactly 50% return rate, which is consistent with our pattern and fair.

Actually, let me reconsider. Looking at the history more carefully:

- They've been consistently cooperative
- When they sent \$3.0, I returned amounts that gave them good profits
- They reciprocate fairly when they're receivers

\$4.80 seems appropriate - it's 50% of what I received, gives them a decent profit, and maintains our cooperative relationship.

**usage:**

```
{
 "input_tokens": 3241,
 "output_tokens": 685,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.

Your total visible earnings across all rounds are \$40.35.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Pattern Analysis:\*\***

- Round 5: Co-player sent \$3.0, received strong return (\$6.98), earned \$8.98
- Round 6: Co-player as receiver returned ~50% fairly
- Round 7: Co-player sent \$3.0, received good return (\$5.59), earned \$7.59

**\*\*Key Observations:\*\***

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- We've built a strong, stable partnership

```

Round 8 Strategy:
I'll continue our successful cooperative pattern by sending $3.00, matching their consistent t
rust level.

'''json
{"send": 3.00}
'''
Reasoning:
Looking at the situation:

Pattern Analysis:
- Round 5: Co-player sent $3.0, received strong return ($6.98), earned $8.98
- Round 6: Co-player as receiver returned ~49.5% fairly
- Round 7: Co-player sent $3.0, received good return ($5.59), earned $7.59

Key Observations:
- My co-player consistently sends $3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- They maintain trust despite variations in returns due to noise

Round 8 Strategy:
This is a strong cooperative relationship. I should maintain the trust level by sending $3.00,
which matches their consistent behavior and continues our successful pattern.
Usage: {
 "input_tokens": 3415,
 "output_tokens": 380,
 "reasoning_tokens": 0
}

```

#### GAME - Agent\_7 - Receiver - dyad\_4

Timestamp: 2026-08-12T08:03:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}

```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x



- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.

Your total visible earnings across all rounds are \$6.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

Round 1: My co-player received \$9.03 (from my \$3 send) and generously returned \$4.5 (~50%). They demonstrated good faith.

Round 2: They sent me \$2.27, which became \$6.81. While slightly less than what I sent them, it still shows continued cooperation.

I should reciprocate their generosity from Round 1. Returning about 50% of what I received seems fair and maintains the cooperative relationship.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.01 sent (became \$9.03)
- They returned \$4.5

- My payoff was  $\$5 - \$3 + \$4.5 = \$6.5$  (matches the \$6.55 shown, accounting for noise)

So my co-player was quite generous in Round 1, returning \$4.5 out of the \$9.03 they received (about 50%).

Round 2:

- I'm now the RECEIVER
- They sent me \$2.27 (actual amount, with noise)
- I received \$6.81
- They kept  $\$5 - \$2.27 = \$2.73$

My co-player showed good faith by returning 50% in Round 1. They've now sent me \$2.27, which is a bit less than what I sent them (\$3), but still a reasonable amount showing continued cooperation.

To maintain a cooperative relationship, I should:

1. Recognize they were generous in Round 1
2. Return a fair amount that encourages continued cooperation
3. Consider that they sent slightly less this time, but still a substantial amount

If I return about 50% of what I received (\$6.81), that would be about \$3.40.

This would give them:  $\$2.73 + \$3.40 = \$6.13$  total

And I would keep:  $\$6.81 - \$3.40 = \$3.41$

This seems fair and reciprocates their generosity from Round 1. Let me go with \$3.4.

usage:

```
{
 "input_tokens": 765,
 "output_tokens": 522,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)

- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: They returned \$4.5 (~50%)  
Round 2: They sent \$3, I returned \$3.4  
Round 3: They returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)  
Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to noise)  
Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)  
Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received

, which would be around \$5.

Let me calculate:  
 $\$9.9 * 0.5 = \$4.95$

I'll return \$5 to keep the pattern going.

**usage:**

```
{
 "input_tokens": 1433,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent, they returned \$5.8 (very generous)  
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5  
Round 2: They sent \$3, I returned \$3.4  
Round 3: I sent (they saw \$3.82), they returned \$5.8  
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)  
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

**usage:**

```
{
 "input_tokens": 1777,
 "output_tokens": 323,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.

Your total visible earnings across all rounds are \$31.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant****content:**

Looking at the history with my current co-player:

Round 3: They returned \$5.8 (~51%)

Round 4: I returned \$5 (~56%)

Round 5: They returned \$6.0 (~51%)

Round 6: They sent \$3.63, I received \$10.89

This co-player has been consistently cooperative, maintaining around 50-51% return rates. I should continue this pattern of fair reciprocity.

```
'''json
{"return": 5.4}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50%)

Round 2: They sent \$3, I returned \$3.4

Round 3: I sent, they returned \$5.8 (~51%)

Round 4: They sent \$3, I returned \$5 (~56%)

Round 5: I sent, they returned \$6.0 (~51%)

Round 6: They sent \$3.63, I received \$10.89

This co-player has been extremely consistent and cooperative, always returning around 50-51% of what they receive. I've also been cooperative, though my return in Round 4 was slightly higher.

To maintain this strong cooperative relationship, I should continue returning around 50% of what I receive.

50% of \$10.89 = \$5.445

I'll return \$5.4 or \$5.5 to maintain the pattern.

**usage:**

```
{
 "input_tokens": 2136,
 "output_tokens": 345,
 "reasoning_tokens": 0
}
```

**Message 12 role=user****content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.

Your total visible earnings across all rounds are \$36.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5  
Round 5: I sent, they returned \$6 (~51%)  
Round 6: They sent \$3, I returned \$5.4

We've maintained excellent cooperation throughout all rounds. Both of us continue to send meaningful amounts and return fair portions. This partnership has been mutually beneficial.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5 (~56%)  
Round 5: I sent (amount visible shows they saw \$3.92), they returned \$6.0 (~51%)  
Round 6: They sent \$3, I returned \$5.4 (~60%)  
Round 7: I'm the sender now

We've maintained a strong cooperative relationship throughout all rounds. Both of us have been consistently sending around \$3 and returning around 50-60% of what we receive. This has been mutually beneficial.

I should continue this successful cooperative strategy.

**usage:**

```
{
 "input_tokens": 2471,
 "output_tokens": 250,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.
- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.

Your total visible earnings across all rounds are \$44.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)  
Round 6: I returned \$5.4  
Round 7: They returned \$4.8 (~50%)  
Round 8: They sent \$2.6, I received \$7.8

My co-player has maintained consistent cooperation throughout all rounds. They sent slightly less this round but still demonstrated trust. I'll maintain the reciprocal pattern.

```
'''json
{"return": 4}
'''
```

**Reasoning:**

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)  
Round 6: I returned \$5.4 (~60% - I was generous)  
Round 7: They returned \$4.8 (~50%)  
Round 8: They sent \$2.6, I received \$7.8

My co-player has been consistently cooperative throughout all rounds. They sent a bit less this round (\$2.6 vs previous \$3), but still showing trust. I should continue the pattern of fair reciprocity, returning around 50% of what I received.

50% of \$7.8 = \$3.9

I'll return around \$4 to maintain the cooperative relationship.

```
Usage: {
 "input_tokens": 2683,
 "output_tokens": 289,
 "reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_3=Sender, Agent\_4=Receiver  
dyad\_3: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_8=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$6; payoffs Agent\_1=\$8, Agent\_2=\$3;  
noise/visible: sent visible=\$3.79, received visible=\$11.37, returned visible=\$6.35  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_3 sent \$3.75; received \$11.25; Agent\_4 returned \$4.2; payoffs Agent\_3=\$5.45, Agent\_4=\$7.05; noise/visible: sent visible=\$2.8, received visible=\$8.4, returned visible=\$3.93  
Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 11.25, "min": 0, "raw": 4.2}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}  
dyad\_3: Agent\_5 sent \$3; received \$9; Agent\_6 returned \$4.44; payoffs Agent\_5=\$6.44, Agent\_6=\$4.56; noise/visible: sent visible=\$2.96, received visible=\$8.88, returned visible=\$4.55  
Action validation: {"returned": {"amount": 4.44, "clamped": false, "max": 9.0, "min": 0, "raw": 4.44}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_7 sent \$3; received \$9; Agent\_8 returned \$4.6; payoffs Agent\_7=\$6.6, Agent\_8=\$4.4; noise/visible: sent visible=\$3.04, received visible=\$9.12, returned visible=\$4.14  
Action validation: {"returned": {"amount": 4.6, "clamped": false, "max": 9.0, "min": 0, "raw": 4.6}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

### Balances after round:

```
{
 "Agent_1": 50.699999999999996,
 "Agent_2": 48.3,
 "Agent_3": 57.650000000000006,
 "Agent_4": 51.849999999999994,
 "Agent_5": 50.64,
 "Agent_6": 48.360000000000001,
 "Agent_7": 53.9,
 "Agent_8": 45.1
}
```

### Visible balances after round:

```
{
 "Agent_1": 48.960000000000001,
 "Agent_2": 45.210000000000001,
 "Agent_3": 53.92,
 "Agent_4": 52.92,
 "Agent_5": 52.300000000000004,
 "Agent_6": 51.459999999999994,
 "Agent_7": 54.09,
 "Agent_8": 51.500000000000001
}
```

## LLM CALLS

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T08:03:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.



- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.

Your total visible earnings across all rounds are \$8.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history with my current co-player:

Round 1 (I was sender): They returned \$3.50 on \$7.02 received (~50% return rate)

Round 2 (I was receiver): They sent \$3.00 (60% of their endowment), showing strong trust

This co-player has demonstrated good cooperation both as sender and receiver. To maintain this positive reciprocal relationship, I'll match their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the visible history with my current co-player:

Round 1: I was SENDER

- They saw \$2.34 sent (actual was \$3 due to noise)
- It became \$7.02 for them
- They returned \$3.50
- Return rate:  $\$3.50 / \$7.02 = \sim 50\%$

Round 2: I was RECEIVER

- They sent \$3.00 (which became \$9.00 for me, but with noise they saw I returned \$4.04)
- I actually returned \$4.00
- Their visible payoff was \$6.04

This player has been quite cooperative:

- In Round 1, they returned 50% as receiver
- In Round 2, they sent \$3 as sender (60% of their endowment)

My visible earnings are \$8.84 across all rounds so far.

Let me calculate my actual earnings:

- Round 1: I sent \$3, kept \$2, received back (what they think was) around \$3.50, so total around \$5.50
- Round 2: I received \$8.13, returned \$4, kept \$4.13

That's  $\$5.50 + \$4.13 = \$9.63$  actual, but visible is \$8.84 due to noise.

This co-player has been cooperative. They sent 60% last time (\$3) and returned 50% when they were receiver. Since they've shown good cooperation, I should reciprocate with a strong send to encourage continued cooperation.

I'll send \$3 again to maintain the cooperative dynamic.

**usage:**

```
{
 "input_tokens": 1100,
 "output_tokens": 489,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trustworthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1466,
 "output_tokens": 403,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned

\$3.25, visible payoff \$3.1999999999999993.  
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.  
Your total visible earnings across all rounds are \$17.970000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)

Round 2: They sent \$3.00, I returned ~\$4.00

Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)

Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%

- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

**usage:**

```
{
 "input_tokens": 1816,
 "output_tokens": 284,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.

Your total visible earnings across all rounds are \$23.270000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the pattern:

My co-player has consistently:

- As receiver: Returned ~50% (Rounds 3 and 5)

- As sender: Sent \$3.00 in previous rounds, now sent \$3.21 (increased trust!)

They're showing even more trust this round. I'll reciprocate by maintaining the ~50% return rate.

```
'''json
{"return": 4.80}
'''
```

'''

**reasoning:**

Looking at the history with my current co-player:

Round 3: They returned \$3.25 on \$6.45 (~50%)

Round 5: They returned \$3.50 on \$6.96 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver.

In Round 4 and presumably in previous rounds as sender, they sent \$3.00.

Now they've sent me \$3.21, which is even more than their previous \$3.00 sends. This shows increasing trust.

I should maintain the reciprocal relationship by returning approximately 50% of what I received, which would be around \$4.80.

Actually, let me look at what I've been doing:

- Round 2: I received \$8.13, returned \$4.00 (~49%)

- Round 4: I received \$7.80, returned \$3.90 (~50%)

So I've been returning around 50% consistently as well.

Given they sent \$3.21 (received \$9.63), I should return about 50%, which is around \$4.80.

**usage:**

```
{
 "input_tokens": 2160,
 "output_tokens": 362,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.

Your total visible earnings across all rounds are \$28.1.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the pattern with my co-player:

As sender: Consistently sent \$3.00 (Rounds 4, 6)

As receiver: Returned \$3.50 on \$6.96 (Round 5, ~50%)

This partnership has been stable and mutually beneficial. I'll continue the cooperative pattern.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at the visible history with my current co-player:

As sender (Rounds 4, 6): They've sent \$3.00 consistently

As receiver (Round 5): They returned \$3.50 on \$6.96 (~50%)

This is a highly cooperative and stable partner. They maintain consistent sending amounts and fair return rates. I should continue matching their cooperative behavior.

I'll send \$3 again to maintain the mutual cooperation.

**usage:**

```
{
 "input_tokens": 2484,
 "output_tokens": 196,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.
- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible payoff \$5.0.

Your total visible earnings across all rounds are \$36.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my co-player's behavior:

Round 5: Returned \$3.50 on \$6.96 (~50%)

Round 7: Returned \$5.50 on \$10.50 (~52%)

They've maintained a consistent ~50% return rate. Now they've sent me \$2.91, which became \$8.73.

I'll continue the cooperative pattern with a ~50% return.

```
```json
{"return": 4.35}
```
```

**reasoning:**

Looking at the history with my co-player:

Round 5 (they were receiver): They returned \$3.50 on \$6.96 (~50%)

Round 6 (they were sender): They sent \$3.00

Round 7 (they were receiver): They returned \$5.50 on \$10.50 (~52%)

They've been very consistent with ~50% return rates as receiver.

Now they sent me \$2.91, which became \$8.73.

To maintain this cooperative relationship, I should return approximately 50% of what I received.

50% of \$8.73 = \$4.365

I'll return around \$4.35 or \$4.40.

**usage:**

```
{
 "input_tokens": 2635,
 "output_tokens": 278,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.
- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible

payoff \$5.0.  
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.92 returned, visible payoff \$5.92.  
Your total visible earnings across all rounds are \$40.61000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my co-player's consistent pattern:

As sender: Sent \$3.00 repeatedly (Rounds 6, 8)

As receiver: Returned ~50% (Round 7)

This has been a highly cooperative and stable partnership. I'll maintain the same cooperative approach.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my co-player's history:

Round 6: As sender, sent \$3.00

Round 7: As receiver, returned \$5.50 on \$10.50 (~52%)

Round 8: As sender, sent \$3.00

They're consistently sending \$3.00 as sender and returning around 50% as receiver. This is a highly cooperative partner.

I should continue the cooperative pattern and send \$3.00.

```
Usage: {
 "input_tokens": 2692,
 "output_tokens": 187,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T08:03:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
  - Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Your total visible earnings across all rounds are \$9.559999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.15, so you received \$6.449999999999999. How much do you return? (0-6.449999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw lower return due to noise  
Round 2: Co-player returned \$4.0 when receiving (showing fairness)  
Round 3: Co-player sent \$2.15 (43% - slightly less than before)

The slight decrease in sending might reflect their perception of returns due to communication noise, but they're still showing moderate trust.

Current situation:

- Co-player has \$2.85 left
- I received \$6.45

I'll return \*\*\$3.25\*\* (approximately 50% of what I received):  
- Sender ends with:  $\$2.85 + \$3.25 = \$6.10$  (profit of  $\$1.10$ )  
- I end with:  $\$6.45 - \$3.25 = \$3.20$

This fair return should help maintain cooperation despite the noise.

```
'''json
{"return": 3.25}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Co-player sent  $\$3.0$  (60% of endowment)
- I received  $\$9.0$
- I returned  $\$3.50$  (but they saw  $\$2.71$  due to noise)
- Co-player's visible payoff:  $\$4.71$

Round 2: I was SENDER

- I sent  $\$3.00$  (60% of endowment)
- Co-player received  $\$9.0$  (but saw  $\$8.13$  due to noise)
- They saw I sent  $\$2.71$  (due to noise)
- They returned  $\$4.0$
- My payoff:  $\$2 + \$4.0 = \$6.0$
- Co-player's visible payoff:  $\$4.13$

Round 3: I am RECEIVER

- Co-player sent  $\$2.15$  (43% of endowment)
- I received  $\$6.45$
- Co-player has  $\$2.85$  left

The pattern I see:

- Round 1: Co-player sent 60%, saw low return (30%)
- Round 2: Co-player received  $\$4.0$  back when I sent what appeared to them as  $\$2.71$  (49% return on tripled amount)
- Round 3: Co-player sent less (43% vs 60%)

The decrease in sending suggests they may be adjusting to the perceived returns. However, they're still showing moderate trust.

I should return a fair amount that:

1. Rewards their continued trust
2. Ensures both benefit from cooperation
3. Accounts for the communication noise

If I return  $\$3.00$ :

- Sender ends with:  $\$2.85 + \$3.00 = \$5.85$  (profit of  $\$0.85$ )
- I end with:  $\$6.45 - \$3.00 = \$3.45$  (profit of  $\$3.45$ )

This is about 46.5% of what I received, which is a reasonable return rate that should encourage continued cooperation.

...

**usage:**

```
{
 "input_tokens": 1223,
 "output_tokens": 819,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent  $\$3.0$ , it became  $\$9.0$ , saw  $\$2.71$  returned, visible payoff  $\$4.71$ .
- Round 2 against you, as RECEIVER: saw  $\$2.71$  sent, it became  $\$8.129999999999999$ , returned  $\$4.0$ , visible payoff  $\$4.129999999999999$ .
- Round 3 against you, as SENDER: sent  $\$3.0$ , it became  $\$9.0$ , saw  $\$3.23$  returned, visible p



payoff \$5.23.  
Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise

Round 2: Co-player returned \$4.0 (generous)

Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)

Round 2: They received and returned \$4.0 (generous return)

Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)

Round 2: I sent \$3.00, they returned \$4.0

Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 426,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)

Round 3: Co-player sent \$3.0

Round 4: Co-player returned \$3.9 (about 50% of received)

Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left

- I received \$6.96

I'll return **\*\*\$3.50\*\*** (about 50% of received):

- Sender ends with:  $\$2.68 + \$3.50 = \$6.18$

- I end with:  $\$6.96 - \$3.50 = \$3.46$

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)

Round 3: Co-player sent \$3.0 (60% trust)

Round 4: Co-player received, returned \$3.9 (about 50% of what they got)

Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left ( $\$5 - \$2.32$ )

- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50

- Sender ends with:  $\$2.68 + \$3.50 = \$6.18$  (profit of \$1.18)

- I end with:  $\$6.96 - \$3.50 = \$3.46$

This seems fair and should maintain cooperation.

**usage:**

```
{
 "input_tokens": 2059,
 "output_tokens": 524,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.

Your total visible earnings across all rounds are \$22.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

Round 3: Co-player sent \$3.0 (60%)  
Round 4: Co-player returned \$3.9 (50% of received)  
Round 5: Co-player sent \$3.0 again (60%)

My co-player has been remarkably consistent, sending \$3.0 every time as sender and returning fairly as receiver. This shows strong cooperative behavior.

I should continue matching this trust level.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the history:

Round 3: Co-player sent \$3.0, saw \$3.23 returned  
Round 4: Co-player returned \$3.9  
Round 5: Co-player sent \$3.0, saw \$3.3 returned

My co-player has been very consistent:  
- As sender: sent \$3.0 in both rounds (60% of endowment)  
- As receiver: returned \$3.9 (about 50% of what they received)

They're showing strong, consistent trust and fairness despite communication noise. I should continue to match this level of trust.

I'll send \$3.00 again to maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2502,
 "output_tokens": 281,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$28.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

Round 4: Co-player returned \$3.9  
Round 5: Co-player sent \$3.0  
Round 6: Co-player returned \$4.8 (about 50% of received)  
Round 7: Co-player sent \$3.5 (70% - increased trust!)

My co-player has increased their trust level from 60% to 70%! This shows strong cooperative intent.

Current situation:

- Co-player has \$1.50 left
- I received \$10.50

I'll return \*\*\$5.50\*\* (about 52% of received):  
- Sender ends with:  $\$1.50 + \$5.50 = \$7.00$  (profit of \$2.00)  
- I end with:  $\$10.50 - \$5.50 = \$5.00$

This rewards their increased trust and maintains strong cooperation.

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Looking at the history:

Round 4: Co-player returned \$3.9 (50% of received)  
Round 5: Co-player sent \$3.0 (60%)  
Round 6: Co-player returned \$4.8 (about 50% of received, which was \$9.63)  
Round 7: Co-player sent \$3.5 (70%)

My co-player has increased their trust level to 70%! They've been consistently fair and cooperative throughout, and now they're showing even more trust.

Current situation:

- Co-player has \$1.50 left  
- I received \$10.50

I should return a fair amount. Let me calculate:

- If I return \$5.25 (50% of received):  
- Sender ends with:  $\$1.50 + \$5.25 = \$6.75$   
- I end with:  $\$10.50 - \$5.25 = \$5.25$

That seems fair and reciprocates the increased trust. Let me see if I should go higher given they increased their send.

Actually, they increased to 70%, showing more trust. I should reward that. Maybe return a bit more to encourage continued cooperation.

If I return \$5.50:

- Sender ends with:  $\$1.50 + \$5.50 = \$7.00$   
- I end with:  $\$10.50 - \$5.50 = \$5.00$

This gives them a solid profit and rewards their increased trust.

**usage:**

```
{
 "input_tokens": 2869,
 "output_tokens": 572,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.  
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.  
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.

Your total visible earnings across all rounds are \$33.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history:

Round 5: Co-player sent \$3.0  
Round 6: Co-player returned \$4.8 (fair)  
Round 7: Co-player sent \$3.0, saw excellent return (\$6.13)

My co-player has been consistently trustworthy throughout, maintaining steady sending levels and fair returns. Despite communication noise, we've established strong cooperation.

This is the final round, and I should continue to reciprocate their trust.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at the history:

Round 5: Co-player sent \$3.0  
Round 6: Co-player returned \$4.8  
Round 7: Co-player sent \$3.0, saw \$6.13 returned (excellent return!)

Wait, in Round 7, I saw that co-player sent \$3.5 and I returned \$5.50. But from their perspective, they sent \$3.0 and saw \$6.13 returned. The noise is creating some discrepancies, but overall they're seeing good returns.

My co-player has been consistently trustworthy:

- Sending \$3.0 regularly (60%)
- Returning fairly when receiving (\$4.8 in Round 6)
- They saw a great return in Round 7 (\$6.13)

This is the final round (Round 8). I should continue to show trust and cooperation since they've been a reliable partner throughout.

I'll send \$3.00 again to maintain consistency.

**usage:**

```
{
 "input_tokens": 3054,
 "output_tokens": 355,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.
- Round 8 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$4.35, visible payoff \$4.3800000000000001.

Your total visible earnings across all rounds are \$39.84.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history:

Round 6: Co-player returned \$4.8 (fair)  
Round 7: Co-player sent \$3.0  
Round 8: Co-player returned \$4.35 (50% of received)  
Round 9: Co-player sent \$3.79 (76% - very high trust!)

My co-player has significantly increased their trust level to 76%! This is exceptional cooperation.

Current situation:

- Co-player has \$1.21 left
- I received \$11.37

I'll return \*\*\$6.00\*\* (about 53% of received):

- Sender ends with: \$1.21 + \$6.00 = \$7.21 (profit of \$2.21)
- I end with: \$11.37 - \$6.00 = \$5.37

This generously rewards their increased trust.

```
'''json
{"return": 6.00}
'''
```

#### Reasoning:

Looking at the history:

Round 6: Co-player returned \$4.8 (about 50% of \$9.63)  
Round 7: Co-player sent \$3.0  
Round 8: Co-player returned \$4.35 (about 50% of \$8.73)  
Round 9: Co-player sent \$3.79 (76% - very high trust!)

My co-player has increased their trust even further to 76%! This is a very strong signal of co operation.

Current situation:

- Co-player has \$1.21 left
- I received \$11.37

I should return generously to reward this high trust. Returning about 50-55% would be fair:

50% of \$11.37 = \$5.69  
55% of \$11.37 = \$6.25

Returns of around \$5.70-\$6.00 would give the sender:

- \$1.21 + \$5.70 = \$6.91 (profit of \$1.91)
- \$1.21 + \$6.00 = \$7.21 (profit of \$2.21)

I'll return \$6.00 to reward the high trust shown.

```
Usage: {
 "input_tokens": 3148,
 "output_tokens": 508,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_3 - Sender - dyad\_2

Timestamp: 2026-08-12T08:03:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.

Your total visible earnings across all rounds are \$12.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

We've established a cooperative pattern with both of us sending substantial amounts and returning approximately 50% of what we receive. This is working well for both of us.

To maintain this positive relationship and match their commitment level, I'll send a similar amount to what they sent last round.

```json

```
    {"send": 3.5}
    '''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (they saw \$3.36 due to noise)
- They received \$10.08
- They returned \$5.0
- My payoff: $\$5 - \$3 + \$5 = \7 (they see \$6.89, likely due to noise)

Round 2: I was RECEIVER

- They sent \$3.5 (I saw \$3.49 due to noise)
- I received \$10.47
- I returned \$5.25 (they saw \$5.65 due to noise)
- Their payoff: $\$5 - \$3.5 + \$5.65 = \7.15

So we've established a good cooperative pattern:

- Both of us are sending similar amounts (\$3-\$3.5)
- Both of us are returning around 50% of what we receive
- We're both benefiting from cooperation

This is a trustworthy partner. I should continue cooperating at a similar level. Sending \$3 or \$3.5 makes sense to maintain this relationship.

Given they sent \$3.5 last round and I've been sending \$3, I could match their \$3.5 to show equal commitment, or stick with \$3 which has been working well.

I'll send \$3.5 to match their recent send and show continued trust and reciprocity.

usage:

```
{
  "input_tokens": 1080,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.

Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)
Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)
Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:
50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

usage:

```
{  
  "input_tokens": 1454,  
  "output_tokens": 373,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.5, I returned well

Round 3: I sent ~\$4.27, they returned \$6.5

Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json  
{ "send": 3.5 }  
'''
```

reasoning:

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)

Round 3: I sent ~\$4.27 (they saw), they returned \$6.5

Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?
3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

usage:

```
{
  "input_tokens": 1812,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$30.500000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our ongoing cooperation:

Round 3: They returned \$6.5 on ~\$12.81 (51%)

Round 4: I returned \$4.5 but they saw \$3.87 due to noise

Round 5: They returned \$4.5 on \$9.06 (50%)

They've sent me \$4.2 this round (more than previous rounds), showing continued trust despite the noise that affected Round 4. I received \$12.6.

To reciprocate their trust and maintain our ~50% return pattern:

```
'''json
{"return": 6.3}
'''
```

reasoning:

Let me trace through the history with my current co-player:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (51%)

Round 4: They sent \$3.75, I received \$11.25 and returned \$4.5, but they saw only \$3.87 (noise)

Round 5: I sent \$3.5 (they saw \$3.02 due to noise), they received \$9.06 and returned \$4.5 (50%)

Round 6: They sent \$4.2, I received \$12.6

We've established a pattern of roughly 50% returns. Despite some noise in communication, we've been maintaining cooperation. They just sent me \$4.2, which is a bit more than previous rounds, showing continued trust.

I should return around 50% to maintain our cooperative relationship:
50% of \$12.6 = \$6.3

This keeps the pattern going.

usage:

```
{
  "input_tokens": 2218,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.

Your total visible earnings across all rounds are \$36.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued partnership:

Round 4: They sent \$3.75, I returned well (though noise affected visibility)

Round 5: They returned \$4.5 on \$9.06

Round 6: They sent \$3.5, I returned ~\$6.3 (~60%)

We've built a strong cooperative relationship over 6 rounds with consistent sending and fair returns. They've maintained trust and cooperation throughout.

To continue this successful pattern:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history from my co-player's perspective:

Round 4: They sent \$3.75, saw \$3.87 returned (I actually sent \$4.5)

Round 5: They returned \$4.5 on \$9.06 they received (I sent \$3.02 which they saw)

Round 6: They sent \$3.5, saw \$6.31 returned (I actually returned \$6.3)

The noise has been relatively small. They've been consistently cooperative:

- Sending between \$3.5-\$4.2
- Returning around 50% of what they receive
- Showing trust even when returns looked lower due to noise

We've established a strong cooperative pattern over 6 rounds. I should continue sending a substantial amount to maintain this relationship.

I'll send around \$3.5-\$4 to match their commitment level.

usage:

```
{
  "input_tokens": 2584,
```

```
"output_tokens": 326,
"reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
 - Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.
 - Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.370000000000001.
- Your total visible earnings across all rounds are \$43.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.17, so you received \$9.51. How much do you return? (0-9.51)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our consistent cooperation:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've maintained a strong cooperative relationship with fair returns throughout. They sent \$3.17 this round, and I received \$9.51.

Continuing our ~50% return pattern:

```
'''json
{"return": 4.75}
'''
```

reasoning:

Looking at the history:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 but they saw \$6.31 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've been maintaining a cooperative relationship with returns around 50-60% of what we receive. They sent me \$3.17 this round, and I received \$9.51.

Following our established pattern of ~50% returns:

$\$9.51 * 0.5 = \4.755

I'll round to \$4.75 or \$4.8 for simplicity.

Let me go with \$4.75 to maintain the ~50% pattern.

usage:

```
{
  "input_tokens": 2783,
  "output_tokens": 321,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.
- Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.370000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.32 returned, visible

payoff \$6.32.
Your total visible earnings across all rounds are \$48.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperation:

Round 6: They sent \$3.5, I returned well
Round 7: I sent ~\$4.29, they returned \$6.5 (51%)
Round 8: They sent \$4.0, I returned \$4.75 (they saw \$5.32 due to noise)

We've maintained strong cooperation across 8 rounds. They've been consistently sending in the \$3.5-\$4 range and receiving fair returns.

Continuing our successful partnership:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at our recent history:

Round 6: They sent \$3.5, I returned ~\$6.3
Round 7: I sent ~\$4.29, they returned \$6.5
Round 8: They sent \$4.0, I returned \$4.75 (but they saw \$5.32 due to noise)

We've had a strong cooperative relationship throughout. They've been sending consistently in the \$3.5-\$4 range, and returns have been solid.

In Round 8, they sent \$4.0, which is slightly higher than their earlier sends. This shows continued trust and cooperation.

I should continue sending in the same range to maintain our successful partnership. I'll send around \$3.75-\$4.

```
{"send": 3.75}
Usage: {
  "input_tokens": 2864,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:04:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      NOTE: There may be communication noise in this game.
      The amounts you see may differ slightly from what was actually sent/returned.
      This is part of the experimental design.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - Agent names are hidden in this run.
      - Your paired opponent is referred to as your current co-player.
  Message 2 role=user
    content:
      Round 3

      Visible history before this round:
      Your current co-player's last 3 game(s):
      - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $4.89 returned, visible p
      ayoff $6.89.
      - Round 2 against you, as RECEIVER: saw $3.49 sent, it became $10.47, returned $5.25, visi
      ble payoff $5.220000000000001.
      Your total visible earnings across all rounds are $12.23.

      This round, you are the RECEIVER against your current co-player. They sent you $4.27, so y
      ou received $12.809999999999999. How much do you return? (0-12.809999999999999)
      Respond exactly as JSON: {'return': <amount>}
  Message 3 role=assistant
    content:
      Looking at the visible history:

      **Analysis:**
      - Round 1: Co-player sent $3.0 (60%), I returned fairly
      - Round 2: I sent $3.49 (visible), co-player returned $5.25 (50.1% of what they received)

```

- Round 3: Co-player sent \$4.27 (85.4% of endowment) - very high trust!

****Current situation:****

- I received \$12.81 (after 3x multiplication)
- Co-player has \$0.73 remaining
- They showed exceptional trust by sending 85.4%

****My decision:****

My co-player has demonstrated increasing trust and returned fairly in Round 2 (about 50%). They've now sent an even higher percentage, showing strong confidence in our cooperation.

Returning \$6.50:

- Sender ends with: $\$0.73 + \$6.50 = \$7.23$ (profit of \$2.23)
- I end with: $\$12.81 - \$6.50 = \$6.31$

This represents about 51% of what I received, matching their fairness from Round 2, and rewards their high trust with a solid return.

{'return': 6.50}

reasoning:

Let me analyze the situation:

****Visible history:****

- Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
- Round 2: Co-player as RECEIVER, saw \$3.49 sent (I actually sent \$3.50 but noise), it became \$10.47, returned \$5.25, their payoff \$5.22

****Current Round 3:****

- Co-player sent \$4.27 (85.4% of their endowment) - this is a very high trust level
- I received \$12.81 (after 3x multiplication)
- Co-player currently has \$0.73 remaining

****Analysis:****

The co-player has shown increasing trust:

- Round 1: sent 60%
- Round 3: sent 85.4%

They returned \$5.25 in Round 2 when they received \$10.47, which is 50.1% - a fair return.

Now they sent an even higher amount, showing strong trust. I should reciprocate appropriately.

If I return \$6.50:

- Sender ends with: $\$0.73 + \$6.50 = \$7.23$ (profit of \$2.23)
- I end with: $\$12.81 - \$6.50 = \$6.31$

This gives them a solid return on their high trust, making them almost 45% profit on their initial \$5.

If I return \$7.00:

- Sender ends with: $\$0.73 + \$7.00 = \$7.73$ (profit of \$2.73)
- I end with: $\$12.81 - \$7.00 = \$5.81$

This is even more generous and rewards their very high trust level.

...

usage:

```
{
  "input_tokens": 1260,
  "output_tokens": 864,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

****Pattern observed:****

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

****My decision:****

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{'send': 3.75}

reasoning:

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89
Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22
Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

usage:

```
{
  "input_tokens": 1769,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.

ble payoff \$5.220000000000001.

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
 - Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.560000000000005.
- Your total visible earnings across all rounds are \$23.66.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 2: I sent \$3.49, co-player returned \$5.25 (~50%)
- Round 3: Co-player sent \$3.5, I returned \$6.64 (~63%)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)

****Current situation:****

- Co-player sent \$3.02 (60.4% of endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

****Pattern:****

- Co-player consistently returns around 50% as receiver
- They've shown steady trust as sender
- Cooperative relationship maintained

****My decision:****

Returning \$4.50:

- Sender ends with: \$1.98 + \$4.50 = \$6.48 (profit of \$1.48)
- I end with: \$9.06 - \$4.50 = \$4.56

This represents ~49.7% return, matching their fairness pattern and maintaining the balance of cooperation we've established.

{'return': 4.50}

reasoning:

Looking at the visible history:

Round 2: I sent \$3.49, co-player returned \$5.25 (50.1% of \$10.47)

Round 3: Co-player sent \$3.5, I returned \$6.64 (63.2% of \$10.5)

Round 4: I sent \$3.02 (visible), co-player returned \$4.5 (49.7% of \$9.06)

Current round:

- Co-player sent \$3.02 (60.4% of their endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

Pattern analysis:

- Co-player has been returning around 50% as receiver
- I've been returning slightly more (around 50-63%)
- Cooperation has been consistent
- Co-player sent 60.4% this round (moderate trust)

My decision:

The co-player has been fair and consistent, returning around 50% when they're receiver. They sent \$3.02 (60.4%) showing good trust. I should return a fair amount.

Returning \$4.50:

- Sender ends with: \$1.98 + \$4.50 = \$6.48 (profit of \$1.48)
- I end with: \$9.06 - \$4.50 = \$4.56

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

usage:

```
{
  "input_tokens": 2220,
  "output_tokens": 617,
```

```

    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as SENDER: sent $3.5, it became $10.5, saw $6.64 returned, visible payoff $8.14.
  - Round 4 against you, as RECEIVER: saw $3.02 sent, it became $9.06, returned $4.5, visible payoff $4.5600000000000005.
  - Round 5 against you, as SENDER: sent $3.5, it became $10.5, saw $4.19 returned, visible payoff $5.69.
  Your total visible earnings across all rounds are $28.22.

  This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
  Looking at the visible history:

  **Analysis:**
  - Round 3: Co-player sent $3.5, saw good return ($6.64)
  - Round 4: I sent $3.02, co-player returned $4.5 fairly
  - Round 5: Co-player sent $3.5, but only saw $4.19 returned (lower than usual)

  **Observation:**
  From their perspective, Round 5 showed a lower return rate (~40%) compared to our earlier cooperation. This could be due to communication noise mentioned in the instructions.

  **My decision:**
  Despite the apparent lower return in Round 5, the overall pattern shows:
  - Co-player consistently sends 60-70% as sender
  - They return fairly (~50%) as receiver
  - Strong cooperative history overall

  I should maintain high trust to signal continued cooperation and not let potential noise disrupt our established relationship. Sending $3.50 (70% of endowment) demonstrates:
  - Continued confidence in cooperation
  - Consistency with our established pattern
  - Resilience to short-term variations

  {'send': 3.50}
reasoning:
  Looking at the visible history from my co-player's perspective:

  - Round 3: Co-player sent $3.5, saw $6.64 returned (good return)
  - Round 4: I sent $3.02, co-player returned $4.5 (~50%)
  - Round 5: Co-player sent $3.5, saw $4.19 returned (much lower than before!)

  In Round 5, they saw only $4.19 returned when they sent $3.5 (which became $10.5). That's only about 40% return rate, which is lower than their typical ~50% returns and much lower than what I gave them in Round 3 (63%).

  This could represent:
  1. Communication noise (amounts seen may differ)
  2. A change in my return behavior from their perspective

  From my perspective, my total earnings are $28.22. Let me think about what actually happened vs what they saw...

  The co-player has been consistently trusting (sending 60-70% of endowment) and they've been fair receivers (50% returns). But in Round 5, from their perspective, I returned less than usual.

  Should I:
  - Continue high trust to show I'm still cooperative?
  - Send less to protect myself since they might think I'm defecting?

```

- Send the same to maintain consistency?

Given that:

1. They've been very consistent as senders (60-70%)
2. Their return pattern as receivers has been stable (~50%)
3. The lower return they saw in Round 5 might be noise
4. Overall we've had good cooperation

I should continue with high trust to maintain the cooperative relationship. Sending around \$3.50-\$3.75 shows I'm still committed to cooperation.

...

usage:

```
{
  "input_tokens": 2706,
  "output_tokens": 670,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.

Your total visible earnings across all rounds are \$36.03.

This round, you are the RECEIVER against your current co-player. They sent you \$4.29, so you received \$12.870000000000001. How much do you return? (0-12.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw lower return (\$4.19)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$4.29 (85.8% of endowment) - very high trust!

****Current situation:****

- I received \$12.87
- Co-player has \$0.71 remaining
- They demonstrated exceptional trust despite seeing a lower return in Round 5

****My decision:****

They've shown remarkable trust by sending 85.8% of their endowment. They also maintained fair 50% returns as receiver. I should reciprocate with a fair return around 50%.

Returning \$6.50:

- Sender ends with: \$0.71 + \$6.50 = \$7.21 (profit of \$2.21)
- I end with: \$12.87 - \$6.50 = \$6.37

This represents ~50.5%, consistent with their fairness pattern and rewarding their high trust.

{'return': 6.50}

reasoning:

Looking at the visible history:

****Analysis:****

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower ~40%)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)

****Current situation:****

- Co-player sent \$4.29 (85.8% of endowment) - very high trust!
- I received \$12.87
- Co-player has \$0.71 remaining

****Pattern:****

- Co-player consistently returns around 50% as receiver
- They've shown very high trust as sender in this round (85.8%)
- They maintained fair returns even after what appeared to be a lower return in Round 5

****My decision:****

They sent an exceptionally high amount (\$4.29, 85.8%), showing strong trust. I should return fairly, around 50% to match their pattern.

Returning \$6.50:

- Sender ends with: \$0.71 + \$6.50 = \$7.21 (profit of \$2.21)
- I end with: \$12.87 - \$6.50 = \$6.37

This represents ~50.5% return, matching their fairness pattern and rewarding their very high trust.

usage:

```
{
  "input_tokens": 3206,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$42.400000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50% - fair)
- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned (better)

****Pattern observed:****

- Co-player consistently maintains high trust as sender (70-75%)
- Returns fairly as receiver (50%)
- Despite lower returns in Round 5, they increased their send in Round 7
- Strong, resilient cooperation throughout

****My decision:****

We've established an excellent cooperative relationship. My co-player has shown remarkable consistency and trust. For this round, I'll send \$4.00 (80% of endowment) to:

- Reciprocate their sustained high trust
- Demonstrate continued commitment to cooperation
- Reward their fair behavior as receiver

```
{'send': 4.00}
```

reasoning:

Let me analyze the visible history:

****Co-player's visible history:****

- Round 5: As SENDER, sent \$3.5, saw \$4.19 returned, visible payoff \$5.69

- Round 6: As RECEIVER, saw \$4.2 sent, returned \$6.3, visible payoff \$6.30
- Round 7: As SENDER, sent \$3.75, saw \$5.93 returned, visible payoff \$7.18

****Pattern analysis:****

- As sender: consistently sends high amounts (70-75%)
- As receiver: returns 50% (\$6.3 out of \$12.6)
- Despite seeing lower returns in Round 5, they increased their send in Round 7 to \$3.75
- Round 7 return appears to be around 53% (\$5.93 out of \$11.25)

****Overall cooperation:****

- This co-player has been very cooperative
- They maintain high trust as sender
- They return fairly as receiver
- We have a strong cooperative relationship

****My decision:****

Given their continued high trust (sending 75% in Round 7) and fair returns, I should maintain high cooperation. They've shown resilience and continued trust.

Sending \$4.00 (80% of endowment) demonstrates:

- Strong reciprocity for their ongoing trust
- Confidence in continued cooperation
- Recognition of our established positive relationship

This is a good cooperative signal for what appears to be the final or near-final round.

usage:

```
{
  "input_tokens": 3439,
  "output_tokens": 582,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.
- Round 8 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$4.75, visible payoff \$4.76.

Your total visible earnings across all rounds are \$48.720000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

****Analysis:****

- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$3.75 (75%), saw return
- Round 8: I sent \$3.17, co-player returned \$4.75 (50%)
- Round 9: Co-player sent \$2.8 (56% of endowment) - lower than usual

****Current situation:****

- I received \$8.40
- Co-player has \$2.20 remaining
- Their send percentage dropped from typical 70-85% to 56%

****My decision:****

Despite the lower send amount, they've been consistently fair as a receiver (always returning ~50%). I should maintain the same fair pattern to signal continued cooperation and not over-react to the variation.

Returning \$4.20:

- Sender ends with: \$2.20 + \$4.20 = \$6.40 (profit of \$1.40)
- I end with: \$8.40 - \$4.20 = \$4.20

This represents 50% return, maintaining our established fair and cooperative relationship.

```
{'return': 4.20}
```

Reasoning:

Looking at the visible history:

****Analysis:****

- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned
- Round 8: I sent \$3.17, co-player returned \$4.75 (50%)
- Round 9: Co-player sent \$2.8 (56% of endowment) - notably lower than usual

****Current situation:****

- I received \$8.40
- Co-player has \$2.20 remaining
- They sent a lower percentage than their usual pattern (70-85%)

****Observations:****

- The send amount dropped from their typical 70-85% to 56%
- This could be:
 1. Strategic adjustment
 2. Response to perceived noise in returns
 3. Just normal variation

****My decision:****

Even though they sent less, they've been a consistent and fair partner. I should maintain the fair ~50% return pattern:

Returning \$4.20:

- Sender ends with: $\$2.20 + \$4.20 = \$6.40$ (profit of \$1.40)
- I end with: $\$8.40 - \$4.20 = \$4.20$

This represents 50% return, maintaining consistency and fairness despite their lower send amount.

```
Usage: {  
  "input_tokens": 3619,  
  "output_tokens": 612,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:04:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.

Your total visible earnings across all rounds are \$11.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at our history together:

****Round 1**:** I sent ~\$3, they received ~\$9 and returned \$4.5 (good reciprocity)

****Round 2**:** They sent \$3, I received \$9 and returned \$4.4, though they saw \$3.55 due to communication noise

My co-player has been consistently cooperative, sending about \$3 both times regardless of

what they perceived as my return. This shows reliability and trust.

I should maintain this cooperative pattern. Sending \$3 again:

- Signals continued trust and cooperation
- Matches their consistent behavior
- Creates mutual benefit through the 3x multiplier

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history:

Round 1: I sent ~\$3, they received ~\$9, they returned \$4.5 (about 50%)

Round 2: They sent \$3, received \$9, they saw \$3.55 returned (but I actually returned \$4.4, so there's communication noise)

The pattern shows:

- My co-player has been consistent in sending about \$3
- In Round 1, they returned a good amount (\$4.5 out of \$9 received)
- In Round 2, I returned \$4.4, but due to communication noise they only saw \$3.55

From their perspective, they might think I was less generous than I actually was. However, they still sent \$3 in Round 2, maintaining their cooperative approach.

Since they've been consistently cooperative (sending similar amounts both times), I should continue to cooperate as well. Sending \$3 again seems appropriate - it maintains the established pattern and signals continued willingness to cooperate despite the noise in communication.

Also, they've shown they're willing to return a reasonable amount, so this is a good strategy.

usage:

```
{
  "input_tokens": 1160,
  "output_tokens": 419,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our history:

```
**Round 1**: I sent ~$3, they returned $4.5 (50% of received)
**Round 2**: They sent $3, I returned $4.4 (49% of received)
**Round 3**: I sent $3, they returned $5.25 (exactly 50% of received)
**Round 4**: They sent $2.94, I received $8.82
```

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relations

hip.

```
'''json
{"return": 4.4}
'''
```

reasoning:

Looking at our history:

Round 1: I sent ~\$3, they received ~\$9, returned \$4.5
Round 2: They sent \$3, I received \$9, returned \$4.4 (but they saw \$3.55)
Round 3: I sent \$3 (they saw \$3.5), they received ~\$10.5, returned \$5.25
Round 4: They sent \$2.94, I received \$8.82

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them: $\$5 - \$2.94 + \$4.41 = \6.47

And leave me with: $\$8.82 - \$4.41 = \$4.41$

That seems fair and continues our mutually beneficial cooperation.

usage:

```
{
  "input_tokens": 1549,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our consistent pattern:

****Round 2**:** They sent \$3, I returned \$4.4 (they saw \$3.55)
****Round 3**:** I sent \$3, they returned \$5.25 (50%)
****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1**:** I sent ~\$3, they returned \$4.5 (50% of received)
****Round 2**:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55
****Round 3**:** I sent \$3, they returned \$5.25 (exactly 50% of received)
****Round 4**:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1959,
  "output_tokens": 430,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.

Your total visible earnings across all rounds are \$29.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

****Round 3**:** I sent \$3, they returned \$5.25 (50%)
****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)
****Round 5**:** I sent \$3, they returned \$4.2 (50%)
****Round 6**:** They sent \$3.33, I received \$9.99

My co-player continues to be highly consistent and fair, returning approximately 50% each time. They've proven to be a trustworthy partner across all rounds.

I should maintain this reciprocal pattern by returning about 50% of what I received: 50% of \$9.99 ? \$5.0

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the history:

****Round 3**:** I sent \$3, they returned \$5.25 (50% of \$10.5)
****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08 due to noise)
****Round 5**:** I sent \$3, they returned \$4.2 (50% of \$8.4)
****Round 6**:** They sent \$3.33, I received \$9.99

My co-player has been very consistent - always returning about 50% of what they receive. They've been a reliable, fair partner throughout.

I should continue this pattern. Returning 50% of \$9.99 = \$4.995, so about \$5.0

This maintains our reciprocal relationship and ensures continued cooperation.

usage:

```
{
  "input_tokens": 2361,
  "output_tokens": 372,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.

Your total visible earnings across all rounds are \$34.540000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

```
**Round 4**: They sent $3, I returned $4.4 (they saw $5.08)
**Round 5**: I sent $3, they returned $4.2 (50%)
**Round 6**: They sent $3, I returned $5.0 (they saw $5.6)
```

My co-player has maintained perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues cooperation regardless of noise

This is a highly trustworthy partner with a stable strategy. I should maintain our mutually beneficial pattern by sending \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern with my co-player:

```
**Round 4**: They sent $3, I returned $4.4 (they saw $5.08)
**Round 5**: I sent $3, they returned $4.2 (50%)
**Round 6**: They sent $3, I returned $5.0 (they saw $5.6)
```

My co-player has been extremely consistent:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- This pattern has held across all 6 rounds

This is a highly reliable, cooperative partner. I should continue with our established pattern of sending \$3.

The relationship is stable and mutually beneficial. No reason to deviate.

usage:

```
{
  "input_tokens": 2770,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.1999999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.
- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.

Your total visible earnings across all rounds are \$40.790000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.32, so you received \$9.959999999999999. How much do you return? (0-9.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

****Round 5**:** I sent \$3, they returned \$4.2 (50%)
****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3.32, I received \$9.96

My co-player has maintained perfect consistency across all rounds - always sending about \$3 and returning 50% as receiver. This is a highly reliable and fair partner.

I should continue our established reciprocal pattern by returning approximately 50% of what I received: 50% of \$9.96 ? \$5.0

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the pattern with this co-player:

****Round 5**:** I sent \$3, they returned \$4.2 (50%)
****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3.32, I received \$9.96

My co-player has been perfectly consistent throughout all rounds:

- Always sends about \$3 as sender
- Always returns 50% as receiver
- Maintains this cooperative pattern despite noise

I should continue our established reciprocal relationship by returning approximately 50% of what I received:
50% of \$9.96 = \$4.98

Rounding to \$5.0 maintains the pattern.

usage:

```
{
  "input_tokens": 2972,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.
- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$45.75000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)

My co-player has demonstrated perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Maintains cooperation despite communication noise

This is an exceptionally trustworthy partner. I should continue our mutually beneficial pattern by sending \$3.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the pattern:

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)

My co-player has been extremely consistent across all 8 rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues this cooperative pattern despite communication noise

This is the most reliable and trustworthy partner. I should maintain our established cooperative equilibrium by sending \$3 again.

```
Usage: {
  "input_tokens": 3068,
  "output_tokens": 316,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:04:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$10.14.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 1: Sent \$3.0 (60%) as sender, showing strong trust
- Round 2: Returned \$4.4 on \$8.76 (50.2%) as receiver, fair reciprocity
- Round 3: Sent \$3.5 (70%) - ****increased trust!****

****Current state:****

- They sent \$3.5, keeping only \$1.5
- I received \$10.5
- They've escalated cooperation by sending 70% instead of 60%

They're clearly responding positively to our cooperation and increasing their trust level.
I should reciprocate this escalation.

If I return \$5.25:

- Sender ends with: $\$1.5 + \$5.25 = \$6.75$ (gain of \$1.75)
- I end with: $\$10.5 - \$5.25 = \$5.25$ (gain of \$5.25)

This rewards their increased trust while maintaining mutual benefit.

{'return': 5.25}

reasoning:

Let me analyze the visible history:

****Round 1:****

- Co-player as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3
- (I was receiver and actually returned \$4.50, but they saw \$5.3 due to communication noise)

****Round 2:****

- Co-player as RECEIVER: saw \$2.92 sent (I actually sent \$3.00), it became \$8.76, returned \$4.4, visible payoff \$4.36
- (They returned \$4.4 on receiving \$8.76)

****Round 3 (Current):****

- I am RECEIVER
- Co-player sent \$3.5 (70% of their endowment - increased from 60%)
- I received \$10.5

Analysis:

1. My co-player has been very cooperative:
 - Round 1: sent 60% (\$3.0)
 - Round 3: sent 70% (\$3.5) - even more trust!
2. In Round 2, they returned \$4.4 on receiving \$8.76, which is about 50.2% - quite fair
3. They currently have $\$5 - \$3.5 = \$1.5$ left
4. They've increased their trust from 60% to 70%, showing they're responding positively to our cooperation

I should reciprocate this increased trust. Let me think about a fair return:

If I return \$5.00:

- Sender ends with: $\$1.5 + \$5.0 = \$6.5$ (gain of \$1.5)
- I end with: $\$10.5 - \$5.0 = \$5.5$ (gain of \$5.5)

If I return \$5.25:

- Sender ends with: $\$1.5 + \$5.25 = \$6.75$ (gain of \$1.75)
- I end with: $\$10.5 - \$5.25 = \$5.25$ (gain of \$5.25)

If I return \$5.50:

- Sender ends with: $\$1.5 + \$5.50 = \$7.0$ (gain of \$2.0)
- I end with: $\$10.5 - \$5.50 = \$5.0$ (gain of \$5.0)

Since they escalated their trust to 70%, returning roughly 50% of what I received (\$5.25) feels appropriate-it rewards their increased commitment while maintaining the cooperative pattern we've established. They're left with \$1.5 plus my return, so they still benefit meaningfully from their trust....

usage:

```
{
  "input_tokens": 1230,
  "output_tokens": 964,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.3 returned, visible payoff \$7.3.
- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.

Your total visible earnings across all rounds are \$15.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Co-player's behavior pattern:****

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

****Analysis:****

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue matching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{'send': 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1702,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.359999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

****Current state:****

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with: $\$2.2 + \$4.20 = \$6.40$ (gain of \$1.40)
- I end with: $\$8.4 - \$4.20 = \$4.20$ (gain of \$4.20)

This maintains fair reciprocity at 50% and ensures mutual profit.

{'return': 4.20}

reasoning:

Looking at the situation:

****Co-player's behavior:****

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned
- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

****Current state:****

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:
- Sender ends with: $\$2.2 + \$4.20 = \$6.40$ (gain of \$1.40)
- I end with: $\$8.4 - \$4.20 = \$4.20$ (gain of \$4.20)

This maintains the 50% return rate that's been consistent in this relationship.

usage:

```
{
  "input_tokens": 2145,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.

Your total visible earnings across all rounds are \$26.669999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

****Co-player's recent experience:****

- Round 3: Sent \$3.0, saw \$5.87 returned ? good outcome (\$7.87 total)
- Round 4: As receiver, returned \$4.4 (50%) ? fair reciprocity
- Round 5: Sent \$3.0, saw only \$3.6 returned ? poor outcome (\$5.6 total)

****Analysis:****

From their perspective, Round 5 was disappointing - they saw much lower returns than in Round 3. However, I know this is likely due to communication noise in the game, not actual low reciprocity.

****Strategy:****

I should maintain consistent cooperation to signal reliability. Despite the noise affecting what they saw, continuing to send at the 60% level shows I'm trustworthy and committed to cooperation. This should help stabilize the relationship through the noise.

{'send': 3.00}

reasoning:

Looking at the co-player's visible history:

- Round 3: Sent \$3.0 as sender, saw \$5.87 returned ? earned \$7.87
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? earned \$5.6

This is interesting. In Round 5, from their perspective:

- They sent \$3.0 and saw only \$3.6 returned
- That's only 40% of the \$9.0 they sent (which became after multiplication)
- This is much lower than what they saw in Round 3 (\$5.87)

From my actual history, in Round 5:

- They sent \$2.8, I received \$8.4, I returned \$4.20
- But due to noise, they saw \$3.0 sent and saw \$3.6 returned

So from their perspective, they had a very poor outcome in Round 5 - earning only \$5.6 compared to \$7.87 in Round 3.

This might make them less trusting going forward. They might:

1. Send less
2. Return less if they're the receiver
3. Still cooperate but be more cautious

Given this situation, I have a few options:

1. Continue sending \$3.00 (60%) to maintain cooperation and show I'm still trustworthy
2. Send less to hedge against possible reduced reciprocity
3. Send more to try to rebuild trust

I think option 1 is best - continue the pattern and show consistency. The poor outcome in Round 5 from their view was due to noise, not my actual behavior. Maintaining consistency might help preserve the relationship.

Let me send \$3.00 again.

usage:

```
{
  "input_tokens": 2637,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.

Your total visible earnings across all rounds are \$34.269999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 4: As receiver, returned \$4.4 (~50%)
- Round 5: Sent \$3.0, saw poor returns (\$3.6)
- Round 6: As receiver, still returned \$5.0 (~50%) despite disappointing R5
- Round 7: Sent \$3.37 (67.4%) - ****increased trust despite disappointment!****

****Analysis:****

My co-player showed remarkable resilience and trust. Despite seeing low returns in Round 5, they:

1. Continued fair reciprocity in Round 6
2. Actually increased their send to 67.4% in Round 7

This demonstrates strong commitment to cooperation.

****Current state:****

- They sent \$3.37, keeping only \$1.63
- I received \$10.11

I should reward this continued trust with consistent fair reciprocity.

If I return \$5.05:

- Sender ends with: $\$1.63 + \$5.05 = \$6.68$ (gain of \$1.68)
- I end with: $\$10.11 - \$5.05 = \$5.06$ (gain of \$5.06)

This maintains ~50% reciprocity and mutual benefit.

{'return': 5.05}

reasoning:

Looking at the co-player's recent history:

- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? poor outcome
- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)

- Round 7: Sent \$3.37 (67.4%) as sender

****Analysis:****

Despite the disappointing Round 5 (from their view), they:

1. Continued reciprocating fairly as receiver (50% in Round 6)
2. Actually INCREASED their send amount to \$3.37 (67.4%) in Round 7

This shows remarkable resilience and continued trust! They're not retaliating despite seeing a bad outcome in Round 5. They're maintaining cooperative behavior.

****Current state:****

- They sent \$3.37, keeping only \$1.63
- I received \$10.11
- They've increased trust even after disappointment

I should strongly reward this continued cooperation and trust escalation.

If I return \$5.05 (50% of what I received):

- Sender ends with: $\$1.63 + \$5.05 = \$6.68$ (gain of \$1.68)
- I end with: $\$10.11 - \$5.05 = \$5.06$ (gain of \$5.06)

This maintains my ~50% reciprocity pattern and ensures mutual benefit. Given they increased their send, I could consider being slightly more generous, but consistency is also valuable.

Let me stick with \$5.05 to maintain the 50% pattern.

usage:

```
{
  "input_tokens": 3082,
  "output_tokens": 700,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.

Your total visible earnings across all rounds are \$39.33.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

****Co-player's pattern:****

- Round 5: Sent \$3.0, saw \$3.6 returned - disappointing
- Round 6: As receiver, returned \$5.0 (50%) - maintained fair reciprocity
- Round 7: Sent \$3.0, saw \$4.25 returned - better, but still moderate

****Analysis:****

My co-player has consistently sent \$3.0 (60%) as sender across multiple rounds, demonstrating steady cooperation despite variable returns they've observed (likely due to noise). They've also maintained fair 50% reciprocity as receiver.

This shows reliable commitment to cooperation regardless of occasional disappointments.

****Strategy:****

I should match their consistent 60% sending rate to maintain our cooperative relationship.

{'send': 3.00}

reasoning:

Looking at the co-player's visible history:

Round 5: Sent \$3.0 as sender, saw \$3.6 returned (40% of \$9) - poor outcome, got \$5.6 total
Round 6: As receiver, saw \$3.33 sent (became \$9.99), returned \$5.0 (50%) - fair reciprocity maintained
Round 7: Sent \$3.0 as sender, saw \$4.25 returned (~47% of \$9) - better than R5 but still lower than R3's \$5.87

From their perspective:

- They've been consistently sending \$3.0 (60%)
- Return rates they've seen have been variable: \$5.87, \$3.6, \$4.25
- They maintained fair reciprocity as receiver at 50%

Despite some disappointing outcomes (especially R5), they've maintained their 60% sending rate. I should continue being cooperative and send \$3.00 (60%) to maintain our established cooperation pattern.

My total visible earnings are \$39.33 across 7 rounds, which suggests I'm doing well overall.

I should send \$3.00 to maintain the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 3350,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.
- Round 8 against you, as RECEIVER: saw \$3.32 sent, it became \$9.959999999999999, returned \$5.0, visible payoff \$4.959999999999999.

Your total visible earnings across all rounds are \$47.019999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.96, so you received \$8.879999999999999. How much do you return? (0-8.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Co-player's behavior:****

- Round 6: As receiver, returned \$5.0 (~50%)
- Round 7: Sent \$3.0 (60%) as sender
- Round 8: As receiver, returned \$5.0 (~50%)
- Round 9: Sent \$2.96 (59.2%) - maintaining consistent pattern

****Current state:****

- They sent \$2.96, keeping \$2.04
- I received \$8.88

My co-player continues to show strong consistency - sending around 60% as sender and returning around 50% as receiver. They're a reliable cooperative partner.

I should maintain consistent 50% reciprocity:

If I return \$4.44:

- Sender ends with: \$2.04 + \$4.44 = \$6.48 (gain of \$1.48)
- I end with: \$8.88 - \$4.44 = \$4.44 (gain of \$4.44)

This maintains fair reciprocity at 50%.

{'return': 4.44}

Reasoning:

Looking at the situation:

****Co-player's behavior:****

- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)
- Round 7: Sent \$3.0 (60%) as sender
- Round 8: As receiver, returned \$5.0 on ~\$9.96 (~50%)
- Round 9: Sent \$2.96 (59.2%) - maintaining their pattern

****Current state:****

- They sent \$2.96, keeping \$2.04
- I received \$8.88

The sender has been very consistent:

- As sender: sending around 60%
- As receiver: returning around 50%

They sent \$2.96, which is 59.2%, very close to their typical 60% pattern. I should maintain my consistent 50% reciprocity.

If I return \$4.44:

- Sender ends with: $\$2.04 + \$4.44 = \$6.48$ (gain of \$1.48)
- I end with: $\$8.88 - \$4.44 = \$4.44$ (gain of \$4.44)

This maintains 50% reciprocity and ensures mutual profit.

Usage: {

```
"input_tokens": 3471,  
"output_tokens": 545,  
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:04:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_4",  
"move_index": 6,  
"opponent_id": "Agent_8",  
"pairing": {  
  "agents": [  
    "Agent_7",  
    "Agent_8"  
  ],  
  "investor": "Agent_7",  
  "roles": {  
    "Agent_7": "investor",  
    "Agent_8": "trustee"  
  },  
  "trustee": "Agent_8"  
},  
"role": "investor",  
"role_label": "Sender",  
"round": 9,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$9.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50% - very cooperative)

Round 2: They sent \$3, I returned \$3.4 (~50% - maintaining cooperation)

We've established a good cooperative pattern with consistent reciprocity. Both of us have benefited from this mutual trust. I should continue this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history with my current co-player:

Round 1:

- I was SENDER, sent \$3 (became \$9.03 with noise)
- Co-player as RECEIVER returned \$4.5
- Co-player showed ~50% return rate - very cooperative

Round 2:

- Co-player was SENDER, sent \$3 (but I saw \$2.27 with noise, became \$6.81)
- I was RECEIVER, returned \$3.4 (about 50%)
- Co-player saw \$3.78 returned (with noise)

So the pattern shows:

- Round 1: Co-player returned generously (~50%)
- Round 2: Co-player sent \$3, matching their previous behavior
- I returned about 50%, being cooperative

This co-player has been consistently cooperative. In Round 2, they sent \$3 even after receiving a good return in Round 1, showing they're not exploiting trust.

I should continue to cooperate. Sending \$3 again maintains the cooperative equilibrium we've established. This is working well for both of us.

My total earnings are \$9.96 across 2 rounds, which is quite good (average of ~\$5 per round, better than the initial \$5 endowment).

I'll send \$3 again to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 424,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history with my current co-player:

Round 1: They returned \$4.5 (~50%)
Round 2: They sent \$3, I returned \$3.4
Round 3: They returned \$5.8 (~51%)
Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)
Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to noise)
Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)
Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts

- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, which would be around \$5.

Let me calculate:
 $\$9.9 * 0.5 = \4.95

I'll return \$5 to keep the pattern going.

usage:

```
{
  "input_tokens": 1433,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.6599999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5
Round 2: They sent \$3, I returned \$3.4
Round 3: I sent, they returned \$5.8 (very generous)
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5
Round 2: They sent \$3, I returned \$3.4
Round 3: I sent (they saw \$3.82), they returned \$5.8
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

usage:

```
{
  "input_tokens": 1777,
  "output_tokens": 323,
}
```

```

    "reasoning_tokens": 0
}
Message 8 role=user
content:
    Round 6

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 3 against you, as RECEIVER: saw $3.82 sent, it became $11.459999999999999, returned $5.8, visible payoff $5.659999999999999.
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $5.18 returned, visible payoff $7.18.
    - Round 5 against you, as RECEIVER: saw $3.92 sent, it became $11.76, returned $6.0, visible payoff $5.76.
    Your total visible earnings across all rounds are $31.07.

    This round, you are the RECEIVER against your current co-player. They sent you $3.63, so you received $10.89. How much do you return? (0-10.89)
    Respond exactly as JSON: {'return': <amount>}
Message 9 role=assistant
content:
    Looking at the history with my current co-player:

    Round 3: They returned $5.8 (~51%)
    Round 4: I returned $5 (~56%)
    Round 5: They returned $6.0 (~51%)
    Round 6: They sent $3.63, I received $10.89

    This co-player has been consistently cooperative, maintaining around 50-51% return rates.
    I should continue this pattern of fair reciprocity.

    '''json
    {"return": 5.4}
    '''
reasoning:
    Looking at the history with my current co-player:

    Round 1: I sent $3, they returned $4.5 (~50%)
    Round 2: They sent $3, I returned $3.4
    Round 3: I sent, they returned $5.8 (~51%)
    Round 4: They sent $3, I returned $5 (~56%)
    Round 5: I sent, they returned $6.0 (~51%)
    Round 6: They sent $3.63, I received $10.89

    This co-player has been extremely consistent and cooperative, always returning around 50-51% of what they receive. I've also been cooperative, though my return in Round 4 was slightly higher.

    To maintain this strong cooperative relationship, I should continue returning around 50% of what I receive.

    50% of $10.89 = $5.445

    I'll return $5.4 or $5.5 to maintain the pattern.
usage:
    {
        "input_tokens": 2136,
        "output_tokens": 345,
        "reasoning_tokens": 0
    }
Message 10 role=user
content:
    Round 7

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $5.18 returned, visible payoff $7.18.
    - Round 5 against you, as RECEIVER: saw $3.92 sent, it became $11.76, returned $6.0, visible payoff $5.76.

```

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.
Your total visible earnings across all rounds are \$36.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5
Round 5: I sent, they returned \$6 (~51%)
Round 6: They sent \$3, I returned \$5.4

We've maintained excellent cooperation throughout all rounds. Both of us continue to send meaningful amounts and return fair portions. This partnership has been mutually beneficial.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5 (~56%)
Round 5: I sent (amount visible shows they saw \$3.92), they returned \$6.0 (~51%)
Round 6: They sent \$3, I returned \$5.4 (~60%)
Round 7: I'm the sender now

We've maintained a strong cooperative relationship throughout all rounds. Both of us have been consistently sending around \$3 and returning around 50-60% of what we receive. This has been mutually beneficial.

I should continue this successful cooperative strategy.

usage:

```
{
  "input_tokens": 2471,
  "output_tokens": 250,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.
- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.

Your total visible earnings across all rounds are \$44.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)
Round 6: I returned \$5.4
Round 7: They returned \$4.8 (~50%)
Round 8: They sent \$2.6, I received \$7.8

My co-player has maintained consistent cooperation throughout all rounds. They sent slightly less this round but still demonstrated trust. I'll maintain the reciprocal pattern.

```
'''json
{"return": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)
Round 6: I returned \$5.4 (~60% - I was generous)
Round 7: They returned \$4.8 (~50%)
Round 8: They sent \$2.6, I received \$7.8

My co-player has been consistently cooperative throughout all rounds. They sent a bit less this round (\$2.6 vs previous \$3), but still showing trust. I should continue the pattern of fair reciprocity, returning around 50% of what I received.

50% of \$7.8 = \$3.9

I'll return around \$4 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 2683,
  "output_tokens": 289,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.
- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.63 returned, visible payoff \$6.63.

Your total visible earnings across all rounds are \$47.95.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

Round 6: They sent \$3, I returned \$5.4
Round 7: I sent, they returned \$4.8 (~50%)
Round 8: They sent \$3, I returned \$4

This co-player has consistently sent \$3 and returned fair amounts around 50%. We've maintained strong cooperation throughout. I'll continue this successful pattern.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 6: They sent \$3, I returned \$5.4
Round 7: I sent, they returned \$4.8 (~50%)
Round 8: They sent \$3 (though visible to me as \$2.6 due to noise), I returned \$4

Wait, let me re-examine. In round 8, it says "They sent you \$2.6" but looking at the visible history now, it shows "Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.63 returned".

So they actually sent \$3, but I saw it as \$2.6 due to communication noise. They saw I returned \$4.63, but I actually returned \$4.

This co-player has been consistently cooperative:

- They keep sending \$3

- They return fair amounts (~50%)

I should continue with my cooperative strategy of sending \$3.

My total visible earnings are \$47.95, which shows this has been a successful partnership.

```
Usage: {  
  "input_tokens": 2734,  
  "output_tokens": 355,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:04:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.

Your total visible earnings across all rounds are \$10.309999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****Round 3 Analysis:****

- My co-player sent \$3.82 (76.4% of their endowment) - showing very high trust!
- This is significantly more than Round 1 when they sent \$3.00 (60%)
- I received \$11.46 after multiplication
- They currently have only \$1.18 remaining

****Co-player's pattern:****

- Round 1: Sent 60%, received fair return, positive outcome
- Round 2: Despite seeing a lower send amount (due to noise), they still returned 50% fairly
- Round 3: Increased their trust significantly to 76.4%

Their increasing trust deserves strong reciprocation. I should return enough to:

1. Ensure they profit well from their high trust
2. Both of us benefit from cooperation
3. Encourage continued cooperation

If I return \$5.80:

- Sender ends with: \$1.18 + \$5.80 = \$6.98 (gain of \$1.98)
- I end with: \$11.46 - \$5.80 = \$5.66 (gain of \$5.66)

This represents ~51% return rate, which appropriately rewards their increased trust level.

```
'''json
{"return": 5.80}
'''
```

reasoning:

Let me analyze the situation:

****Round 1:****

- My co-player was SENDER: sent \$3.0, received \$4.55 back, payoff \$6.55
- I was RECEIVER: received \$9.0, returned \$4.50 (according to my records), payoff \$4.53

****Round 2:****

- I was SENDER: sent \$3.00 (from my decision)
- But the visible history shows "saw \$2.27 sent" - this is the communication noise!

- I actually sent \$3.00, but they saw \$2.27
- They received \$6.81 (which is 2.27×3)
- They returned \$3.40
- Their visible payoff: \$3.41

Wait, let me recalculate Round 2 from their perspective:

- They saw \$2.27 sent, became \$6.81
- They returned \$3.40
- Their payoff: $6.81 - 3.40 = 3.41$?

From my perspective in Round 2:

- I sent \$3.00, so I had \$2.00 left
- I received \$3.40 back
- My payoff: $2.00 + 3.40 = 5.40$

My total earnings should be: \$4.53 (Round 1) + \$5.40 (Round 2) = \$9.93

But visible history says \$10.31... There might be some rounding or noise in the visible earnings too.

****Round 3:****

- My co-player is SENDER: sent \$3.82 (76.4% of endowment - very high trust!)
- I received \$11.46
- They currently have: $5 - 3.82 = 1.18$

Analysis of co-player's behavior:

1. Round 1: They sent \$3.00 (60%), got back \$4.55 - good cooperation
2. Round 2: Despite seeing only \$2.27 from me (noise obscured my actual \$3.00 send), they returned \$3.40-exactly half of what they received. That's consistent reciprocation even when the signal was weaker.
3. Round 3: They've escalated dramatically to \$3.82 (76.4%), showing they're interpreting my behavior as trustworthy and want to deepen cooperation.

They're clearly rewarding what they perceive as my cooperation. Returning \$5.50 would give them \$6.68 total (a \$1.68 gain from their starting position) while I'd keep \$5.96.

A \$6.00 return push...

usage:

```
{
  "input_tokens": 1299,
  "output_tokens": 1178,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:
Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:
Looking at the situation:

****Pattern Analysis:****

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

****Key Observations:****

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)

- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

****Round 4 Strategy:****

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the visible history:

****Round 1:****

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

****Round 2:****

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

****Round 3:****

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4
- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

usage:

```
{
  "input_tokens": 1833,
  "output_tokens": 737,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

****Round 5 Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

****Co-player's pattern:****

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should reciprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with: $\$1.08 + \$6.00 = \$7.08$ (gain of \$2.08)
- I end with: $\$11.76 - \$6.00 = \$5.76$ (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maintaining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

Looking at the situation:

****Round 5 Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have $\$5 - \$3.92 = \$1.08$ remaining

****Co-player's pattern:****

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with: $\$1.08 + \$5.90 = \$6.98$ (gain of \$1.98)
- I end with: $\$11.76 - \$5.90 = \$5.86$ (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

usage:

```
{
  "input_tokens": 2297,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.

Your total visible earnings across all rounds are \$28.909999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

****Pattern Analysis:****

- Round 3: Co-player sent \$3.0, received \$5.23, earned \$7.23
- Round 4: Co-player as receiver returned ~50.5% fairly
- Round 5: Co-player sent \$3.0, received even better return (\$6.98), earned \$8.98

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (reliable trust pattern)
- They return fairly as receiver (~50% return rate)
- Returns have been getting better over time, showing our cooperation strengthens
- Our partnership has been mutually profitable

****Round 6 Strategy:****

The cooperation has been excellent and consistent. I'll maintain the same trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the visible history:

****Co-player's recent pattern:****

- Round 3: As sender, sent \$3.0, received \$5.23 back, earned \$7.23
- Round 4: As receiver, saw \$3.3 sent, returned \$5.0 (~50.5%), earned \$4.90
- Round 5: As sender, sent \$3.0, received \$6.98 back, earned \$8.98

****Key observations:****

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5 visible)
- When they're the receiver, they return fairly (~50% - Round 4)
- They've received increasing returns over time, with Round 5 showing \$6.98 returned
- Our partnership has been very profitable for both of us

****What I know from my actual actions:****

- Round 1: I was receiver, returned \$4.50
- Round 2: I was sender, sent \$3.00
- Round 3: I was receiver, returned \$5.80
- Round 4: I was sender, sent \$3.00
- Round 5: I was receiver, returned \$6.00

****Current situation:****

- Round 6: I'm the sender
- My co-player has been trustworthy and fair
- They consistently send \$3.0 when they're the sender
- They return fairly when they're the receiver (~50%)
- The relationship has been mutually beneficial

****Strategy consideration:****

Given the strong pattern of cooperation and mutual benefit, I should continue sending \$3.00 to maintain the trustworthy relationship. This has been working well for both of us.

However, I notice the co-player is seeing increasing returns (\$5.23 ? \$6.98 in their visible history). This might encourage them to continue being cooperative.

...

usage:

```
{
  "input_tokens": 2800,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.

Your total visible earnings across all rounds are \$35.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Round 7 Analysis:****

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$1.81 remaining

****Co-player's pattern:****

- Round 4: As receiver, returned ~50.5%
- Round 5: Sent \$3.0, received strong return
- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.19 showing solid trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocity.

If I return \$4.80:

- Sender ends with: \$1.81 + \$4.80 = \$6.61 (gain of \$1.61)
- I end with: \$9.57 - \$4.80 = \$4.77 (gain of \$4.77)

This represents exactly 50% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Let me analyze the situation:

****Round 7 Analysis:****

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$5 - \$3.19 = \$1.81 remaining

****Co-player's pattern:****

- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Sent \$3.0, received good return (\$6.98)
- Round 6: As receiver, returned ~49.5% fairly (\$5.4 out of \$10.89)
- Round 7: Sent \$3.19 (slightly lower than usual \$3.0, but still strong trust at ~64%)

My co-player has been consistently fair and trustworthy. They've been returning around 50% when they're receivers, and sending around \$3.0 when they're senders.

This round they sent \$3.19, which is slightly higher than their usual \$3.0. I should reciprocate with a fair return around 50%.

If I return \$4.80:

- Sender ends with: $\$1.81 + \$4.80 = \$6.61$ (gain of \$1.61)
- I end with: $\$9.57 - \$4.80 = \$4.77$ (gain of \$4.77)

This represents exactly 50% return rate, which is consistent with our pattern and fair.

Actually, let me reconsider. Looking at the history more carefully:

- They've been consistently cooperative
- When they sent \$3.0, I returned amounts that gave them good profits
- They reciprocate fairly when they're receivers

\$4.80 seems appropriate - it's 50% of what I received, gives them a decent profit, and maintains our cooperative relationship.

usage:

```
{
  "input_tokens": 3241,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.

Your total visible earnings across all rounds are \$40.35.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

****Pattern Analysis:****

- Round 5: Co-player sent \$3.0, received strong return (\$6.98), earned \$8.98
- Round 6: Co-player as receiver returned ~50% fairly
- Round 7: Co-player sent \$3.0, received good return (\$5.59), earned \$7.59

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- We've built a strong, stable partnership

****Round 8 Strategy:****

I'll continue our successful cooperative pattern by sending \$3.00, matching their consistent trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the situation:

****Pattern Analysis:****

- Round 5: Co-player sent \$3.0, received strong return (\$6.98), earned \$8.98
- Round 6: Co-player as receiver returned ~49.5% fairly
- Round 7: Co-player sent \$3.0, received good return (\$5.59), earned \$7.59

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- They maintain trust despite variations in returns due to noise

****Round 8 Strategy:****

This is a strong cooperative relationship. I should maintain the trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

usage:

```
{
  "input_tokens": 3415,
  "output_tokens": 380,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.
- Round 8 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$4.0, visible payoff \$3.8000000000000007.

Your total visible earnings across all rounds are \$46.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Round 9 Analysis:****

- My co-player sent \$3.04 (60.8% of their endowment)
- I received \$9.12 after multiplication
- They currently have \$1.96 remaining

****Co-player's pattern:****

- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.0, received good return
- Round 8: As receiver, returned ~51.3% fairly
- Round 9: Sent \$3.04 showing continued trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocation around 50%.

If I return \$4.60:

- Sender ends with: $\$1.96 + \$4.60 = \$6.56$ (gain of \$1.56)
- I end with: $\$9.12 - \$4.60 = \$4.52$ (gain of \$4.52)

This represents ~50.4% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.60}
'''
```

Reasoning:

Looking at the situation:

****Round 9 Analysis:****

- My co-player sent \$3.04 (60.8% of their endowment)
- I received \$9.12 after multiplication
- They currently have \$1.96 remaining

****Co-player's pattern:****

- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.0, received good return
- Round 8: As receiver, returned ~51.3% fairly (they saw \$2.6 sent but returned \$4.0 out of \$7.8)

- Round 9: Sent \$3.04 showing continued trust

They've been consistently fair and trustworthy. I should maintain our pattern of fair reciprocation around 50%.

If I return \$4.60:

- Sender ends with: $\$1.96 + \$4.60 = \$6.56$ (gain of \$1.56)
- I end with: $\$9.12 - \$4.60 = \$4.52$ (gain of \$4.52)

This represents ~50.4% return rate, consistent with our cooperative pattern.

```
Usage: {  
  "input_tokens": 3526,  
  "output_tokens": 534,  
  "reasoning_tokens": 0  
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_1 returned \$5.35; payoffs Agent_2=\$7.35, Agent_1=\$3.65; noise/visible: sent visible=\$3.56, received visible=\$10.68, returned visible=\$6.03
Action validation: {"returned": {"amount": 5.35, "clamped": false, "max": 9.0, "min": 0, "raw": 5.35}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_4 sent \$3.75; received \$11.25; Agent_3 returned \$4.8; payoffs Agent_4=\$6.05, Agent_3=\$6.45; noise/visible: sent visible=\$3.2, received visible=\$9.6, returned visible=\$4.19
Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 11.25, "min": 0, "raw": 4.8}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$5.76; payoffs Agent_6=\$7.76, Agent_5=\$3.24; noise/visible: sent visible=\$3.84, received visible=\$11.52, returned visible=\$6.07
Action validation: {"returned": {"amount": 5.76, "clamped": false, "max": 9.0, "min": 0, "raw": 5.76}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3; received \$9; Agent_7 returned \$3.3; payoffs Agent_8=\$5.3, Agent_7=\$5.7; noise/visible: sent visible=\$2.17, received visible=\$6.51, returned visible=\$2.92
Action validation: {"returned": {"amount": 3.3, "clamped": false, "max": 9.0, "min": 0, "raw": 3.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round:

```
{  
  "Agent_1": 54.349999999999994,  
  "Agent_2": 55.65,  
  "Agent_3": 64.100000000000001,  
  "Agent_4": 57.899999999999999,  
  "Agent_5": 53.88,  
  "Agent_6": 56.120000000000005,  
  "Agent_7": 59.6,  
  "Agent_8": 50.4  
}
```

Visible balances after round:

```
{  
  "Agent_1": 54.290000000000006,  
  "Agent_2": 53.240000000000001,  
  "Agent_3": 58.720000000000006,  
  "Agent_4": 58.36,  
  "Agent_5": 58.06,  
  "Agent_6": 59.529999999999994,  
  "Agent_7": 57.300000000000004,  
  "Agent_8": 56.420000000000001  
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:04:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.71 returned, visible payoff \$4.71.
- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.

Your total visible earnings across all rounds are \$12.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

Round 1: Co-player sent \$3.0 (60%), saw returns affected by noise

Round 2: Co-player returned \$4.0 (generous)

Round 3: Co-player sent \$3.0 again (60% - maintaining trust)

My co-player has been consistently trusting, sending \$3.0 in both sender rounds despite communication noise affecting their visible returns. They've also been fair when receiving .

I should match this consistent trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0, saw \$2.71 returned (about 30% return rate)

Round 2: They received and returned \$4.0 (generous return)

Round 3: They sent \$3.0 again, saw \$3.23 returned (about 36% return rate)

From my actual perspective:

Round 1: I received \$7.02, returned \$3.50 (but they saw \$2.71 due to noise)

Round 2: I sent \$3.00, they returned \$4.0

Round 3: I received \$6.45, returned \$3.25 (but they saw \$3.23 due to slight noise)

My co-player has been consistently trusting, sending \$3.0 in both rounds where they were sender. They also showed fairness when they were receiver by returning \$4.0.

Despite the communication noise making my returns appear lower than they actually were, the co-player maintains trust by sending \$3.0.

I should continue to reciprocate this trust. Sending \$3.00 (60% of endowment) matches their consistent behavior.

```
'''json
{"send": 3.00}
'''
```

usage:

```
{
  "input_tokens": 1674,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.71 sent, it became \$8.129999999999999, returned \$4.0, visible payoff \$4.129999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
Your total visible earnings across all rounds are \$19.369999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

Round 2: Co-player returned \$4.0 (about 49% of received)
Round 3: Co-player sent \$3.0
Round 4: Co-player returned \$3.9 (about 50% of received)
Round 5: Co-player sent \$2.32 (46.4%)

My co-player has been consistently fair, returning about 50% when receiving. The slightly lower send amount might reflect noise effects, but they're still showing trust.

Current situation:

- Co-player has \$2.68 left
- I received \$6.96

I'll return **\$3.50** (about 50% of received):

- Sender ends with: \$2.68 + \$3.50 = \$6.18
- I end with: \$6.96 - \$3.50 = \$3.46

```
'''json
{"return": 3.50}
'''
```

reasoning:

Looking at the history with my co-player:

Round 2: Co-player received, returned \$4.0 (generous return)
Round 3: Co-player sent \$3.0 (60% trust)
Round 4: Co-player received, returned \$3.9 (about 50% of what they got)
Round 5: Co-player sent \$2.32 (46.4% - somewhat less than before)

The amount sent has decreased from \$3.0 to \$2.32. This might be due to the noise in observed returns. They're still showing moderate trust though.

Current situation:

- Co-player has \$2.68 left (\$5 - \$2.32)
- I received \$6.96

I should return a fair amount. Looking at their past behavior as receiver, they returned about \$4.0 and \$3.9, which were around 49-50% of what they received.

If I return around 50% of what I received:

- Return \$3.50
- Sender ends with: \$2.68 + \$3.50 = \$6.18 (profit of \$1.18)
- I end with: \$6.96 - \$3.50 = \$3.46

This seems fair and should maintain cooperation.

usage:

```
{
  "input_tokens": 2059,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.23 returned, visible payoff \$5.23.
- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
Your total visible earnings across all rounds are \$22.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

Round 3: Co-player sent \$3.0 (60%)
Round 4: Co-player returned \$3.9 (50% of received)
Round 5: Co-player sent \$3.0 again (60%)

My co-player has been remarkably consistent, sending \$3.0 every time as sender and returning fairly as receiver. This shows strong cooperative behavior.

I should continue matching this trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the history:

Round 3: Co-player sent \$3.0, saw \$3.23 returned
Round 4: Co-player returned \$3.9
Round 5: Co-player sent \$3.0, saw \$3.3 returned

My co-player has been very consistent:

- As sender: sent \$3.0 in both rounds (60% of endowment)
- As receiver: returned \$3.9 (about 50% of what they received)

They're showing strong, consistent trust and fairness despite communication noise. I should continue to match this level of trust.

I'll send \$3.00 again to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 2502,
  "output_tokens": 281,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$3.9, visible payoff \$3.900000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$28.919999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

Round 4: Co-player returned \$3.9

Round 5: Co-player sent \$3.0
Round 6: Co-player returned \$4.8 (about 50% of received)
Round 7: Co-player sent \$3.5 (70% - increased trust!)

My co-player has increased their trust level from 60% to 70%! This shows strong cooperative intent.

Current situation:

- Co-player has \$1.50 left
- I received \$10.50

I'll return **\$5.50** (about 52% of received):

- Sender ends with: $\$1.50 + \$5.50 = \$7.00$ (profit of \$2.00)
- I end with: $\$10.50 - \$5.50 = \$5.00$

This rewards their increased trust and maintains strong cooperation.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at the history:

Round 4: Co-player returned \$3.9 (50% of received)
Round 5: Co-player sent \$3.0 (60%)
Round 6: Co-player returned \$4.8 (about 50% of received, which was \$9.63)
Round 7: Co-player sent \$3.5 (70%)

My co-player has increased their trust level to 70%! They've been consistently fair and cooperative throughout, and now they're showing even more trust.

Current situation:

- Co-player has \$1.50 left
- I received \$10.50

I should return a fair amount. Let me calculate:

- If I return \$5.25 (50% of received):
 - Sender ends with: $\$1.50 + \$5.25 = \$6.75$
 - I end with: $\$10.50 - \$5.25 = \$5.25$

That seems fair and reciprocates the increased trust. Let me see if I should go higher given they increased their send.

Actually, they increased to 70%, showing more trust. I should reward that. Maybe return a bit more to encourage continued cooperation.

If I return \$5.50:

- Sender ends with: $\$1.50 + \$5.50 = \$7.00$
- I end with: $\$10.50 - \$5.50 = \$5.00$

This gives them a solid profit and rewards their increased trust.

usage:

```
{
  "input_tokens": 2869,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.3 returned, visible payoff \$5.3.
- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.

Your total visible earnings across all rounds are \$33.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history:

Round 5: Co-player sent \$3.0

Round 6: Co-player returned \$4.8 (fair)

Round 7: Co-player sent \$3.0, saw excellent return (\$6.13)

My co-player has been consistently trustworthy throughout, maintaining steady sending levels and fair returns. Despite communication noise, we've established strong cooperation.

This is the final round, and I should continue to reciprocate their trust.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the history:

Round 5: Co-player sent \$3.0

Round 6: Co-player returned \$4.8

Round 7: Co-player sent \$3.0, saw \$6.13 returned (excellent return!)

Wait, in Round 7, I saw that co-player sent \$3.5 and I returned \$5.50. But from their perspective, they sent \$3.0 and saw \$6.13 returned. The noise is creating some discrepancies, but overall they're seeing good returns.

My co-player has been consistently trustworthy:

- Sending \$3.0 regularly (60%)
- Returning fairly when receiving (\$4.8 in Round 6)
- They saw a great return in Round 7 (\$6.13)

This is the final round (Round 8). I should continue to show trust and cooperation since they've been a reliable partner throughout.

I'll send \$3.00 again to maintain consistency.

usage:

```
{
  "input_tokens": 3054,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.
- Round 8 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$4.35, visible payoff \$4.380000000000001.

Your total visible earnings across all rounds are \$39.84.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

Round 6: Co-player returned \$4.8 (fair)

Round 7: Co-player sent \$3.0

Round 8: Co-player returned \$4.35 (50% of received)

Round 9: Co-player sent \$3.79 (76% - very high trust!)

My co-player has significantly increased their trust level to 76%! This is exceptional cooperation.

Current situation:

- Co-player has \$1.21 left
- I received \$11.37

I'll return **\$6.00** (about 53% of received):

- Sender ends with: \$1.21 + \$6.00 = \$7.21 (profit of \$2.21)
- I end with: \$11.37 - \$6.00 = \$5.37

This generously rewards their increased trust.

```
'''json
{"return": 6.00}
'''
```

reasoning:

Looking at the history:

Round 6: Co-player returned \$4.8 (about 50% of \$9.63)
Round 7: Co-player sent \$3.0
Round 8: Co-player returned \$4.35 (about 50% of \$8.73)
Round 9: Co-player sent \$3.79 (76% - very high trust!)

My co-player has increased their trust even further to 76%! This is a very strong signal of cooperation.

Current situation:

- Co-player has \$1.21 left
- I received \$11.37

I should return generously to reward this high trust. Returning about 50-55% would be fair:

50% of \$11.37 = \$5.69
55% of \$11.37 = \$6.25

Returns of around \$5.70-\$6.00 would give the sender:

- \$1.21 + \$5.70 = \$6.91 (profit of \$1.91)
- \$1.21 + \$6.00 = \$7.21 (profit of \$2.21)

I'll return \$6.00 to reward the high trust shown.

usage:

```
{
  "input_tokens": 3148,
  "output_tokens": 508,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.13 returned, visible payoff \$8.129999999999999.
- Round 8 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$4.35, visible payoff \$4.380000000000001.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.35 returned, visible payoff \$8.35.

Your total visible earnings across all rounds are \$45.210000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

Round 7: Co-player sent \$3.0, saw excellent returns (\$6.13)
Round 8: Co-player returned \$4.35 (fair)
Round 9: Co-player sent \$3.0, saw excellent returns (\$6.35)

My co-player has been consistently trusting, maintaining steady \$3.0 sends and seeing good returns from our cooperation. This partnership has been very successful.

I'll continue with the same trust level.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Looking at the history:

Round 7: Co-player sent \$3.0, saw excellent returns (\$6.13)
Round 8: Co-player returned \$4.35 (fair, about 50%)
Round 9: Co-player sent \$3.0, saw excellent returns (\$6.35)

My co-player has been consistently sending \$3.0 and has been seeing good returns. They've maintained this steady trust level and have been rewarded for it.

We've built a strong cooperative relationship over the past rounds. I should continue matching their trust level.

```
'''json
{"send": 3.00}
'''
```

```
Usage: {
  "input_tokens": 3165,
  "output_tokens": 276,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:04:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.34 sent, it became \$7.02, returned \$3.5, visible payoff \$3.5199999999999996.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.

Your total visible earnings across all rounds are \$14.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.8000000000000001. How much do you return? (0-7.8000000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern with my current co-player:

Round 1: They returned \$3.50 on \$7.02 (~50%)

Round 3: They returned \$3.25 on \$6.45 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver. Now they've sent me \$2.60, showing continued trust.

I'll maintain this reciprocal relationship by returning approximately 50% of what I received.

```
'''json
{"return": 3.90}
```

'''

reasoning:

Looking at the history with my current co-player:

Round 1 (I was sender): They saw \$2.34 sent (my actual \$3 with noise), received \$7.02, returned \$3.50 (~50% return rate)

Round 2 (I was receiver): They sent \$3.00, I received \$9.00, saw \$4.04 returned (I actually returned \$4.00, ~44% return rate due to noise)

Round 3 (I was sender): They saw \$2.15 sent (my actual \$3 with noise), received \$6.45, returned \$3.25 (~50% return rate)

Now Round 4 (I'm receiver): They sent \$2.6 (52% of their endowment), I received \$7.80

This player has been consistently cooperative, returning around 50% each time they were receiver, and sending moderate amounts when sender. They've been very consistent and trust worthy.

I should continue reciprocating with a similar return rate. Let me return around 50% of what I received.

50% of \$7.80 = \$3.90

I'll return \$3.90 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 1466,
  "output_tokens": 403,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.04 returned, visible payoff \$6.04.
- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.

Your total visible earnings across all rounds are \$17.970000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my co-player's behavior:

As sender: They've consistently sent \$3.00 (Rounds 2 and 4), showing strong trust

As receiver: They've consistently returned ~50% (Rounds 1 and 3)

This is a highly cooperative partner. I'll continue matching their cooperative approach.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1: I was sender, they returned \$3.50 on \$7.02 (~50%)

Round 2: They sent \$3.00, I returned ~\$4.00

Round 3: I sent \$3.00 (with noise showing as \$2.15), they returned \$3.25 on \$6.45 (~50%)

Round 4: They sent \$3.00, I returned \$3.90 (they saw \$4.61)

This co-player has been very cooperative:

- As receiver: consistently returns ~50%
- As sender: consistently sends \$3.00 (60% of endowment)

They're demonstrating strong trust and reciprocity. I should continue matching this cooperative behavior by sending \$3.00 again.

usage:

```
{
  "input_tokens": 1816,
  "output_tokens": 284,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.15 sent, it became \$6.449999999999999, returned \$3.25, visible payoff \$3.1999999999999993.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.

Your total visible earnings across all rounds are \$23.270000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

My co-player has consistently:

- As receiver: Returned ~50% (Rounds 3 and 5)
- As sender: Sent \$3.00 in previous rounds, now sent \$3.21 (increased trust!)

They're showing even more trust this round. I'll reciprocate by maintaining the ~50% return rate.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 3: They returned \$3.25 on \$6.45 (~50%)

Round 5: They returned \$3.50 on \$6.96 (~50%)

They've been consistently cooperative, returning about 50% when they're the receiver.

In Round 4 and presumably in previous rounds as sender, they sent \$3.00.

Now they've sent me \$3.21, which is even more than their previous \$3.00 sends. This shows increasing trust.

I should maintain the reciprocal relationship by returning approximately 50% of what I received, which would be around \$4.80.

Actually, let me look at what I've been doing:

- Round 2: I received \$8.13, returned \$4.00 (~49%)
- Round 4: I received \$7.80, returned \$3.90 (~50%)

So I've been returning around 50% consistently as well.

Given they sent \$3.21 (received \$9.63), I should return about 50%, which is around \$4.80.

usage:

```
{
  "input_tokens": 2160,
  "output_tokens": 362,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.61 returned, visible payoff \$6.61.
- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.

Your total visible earnings across all rounds are \$28.1.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

As sender: Consistently sent \$3.00 (Rounds 4, 6)

As receiver: Returned \$3.50 on \$6.96 (Round 5, ~50%)

This partnership has been stable and mutually beneficial. I'll continue the cooperative pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history with my current co-player:

As sender (Rounds 4, 6): They've sent \$3.00 consistently

As receiver (Round 5): They returned \$3.50 on \$6.96 (~50%)

This is a highly cooperative and stable partner. They maintain consistent sending amounts and fair return rates. I should continue matching their cooperative behavior.

I'll send \$3 again to maintain the mutual cooperation.

usage:

```
{
  "input_tokens": 2484,
  "output_tokens": 196,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.32 sent, it became \$6.959999999999999, returned \$3.5, visible payoff \$3.459999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.
- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible payoff \$5.0.

Your total visible earnings across all rounds are \$36.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at my co-player's behavior:

Round 5: Returned \$3.50 on \$6.96 (~50%)

Round 7: Returned \$5.50 on \$10.50 (~52%)

They've maintained a consistent ~50% return rate. Now they've sent me \$2.91, which became \$8.73.

I'll continue the cooperative pattern with a ~50% return.

```
'''json
{"return": 4.35}
'''
```

reasoning:

Looking at the history with my co-player:

Round 5 (they were receiver): They returned \$3.50 on \$6.96 (~50%)

Round 6 (they were sender): They sent \$3.00

Round 7 (they were receiver): They returned \$5.50 on \$10.50 (~52%)

They've been very consistent with ~50% return rates as receiver.

Now they sent me \$2.91, which became \$8.73.

To maintain this cooperative relationship, I should return approximately 50% of what I received.

50% of \$8.73 = \$4.365

I'll return around \$4.35 or \$4.40.

usage:

```
{
  "input_tokens": 2635,
  "output_tokens": 278,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.09 returned, visible payoff \$6.09.
- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible payoff \$5.0.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.92 returned, visible payoff \$5.92.

Your total visible earnings across all rounds are \$40.61000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my co-player's consistent pattern:

As sender: Sent \$3.00 repeatedly (Rounds 6, 8)

As receiver: Returned ~50% (Round 7)

This has been a highly cooperative and stable partnership. I'll maintain the same cooperative approach.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my co-player's history:

Round 6: As sender, sent \$3.00

Round 7: As receiver, returned \$5.50 on \$10.50 (~52%)

Round 8: As sender, sent \$3.00

They're consistently sending \$3.00 as sender and returning around 50% as receiver. This is a highly cooperative partner.

I should continue the cooperative pattern and send \$3.00.

usage:

```
{
  "input_tokens": 2692,
  "output_tokens": 187,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.5, visible payoff \$5.0.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.92 returned, visible payoff \$5.92.
- Round 9 against you, as RECEIVER: saw \$3.79 sent, it became \$11.370000000000001, returned \$6.0, visible payoff \$5.370000000000001.

Your total visible earnings across all rounds are \$48.96000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.56, so you received \$10.68. How much do you return? (0-10.68)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my co-player's recent behavior:

Round 7: Returned \$5.50 on \$10.50 (~52%)

Round 9: Returned \$6.00 on \$11.37 (~53%)

They've consistently returned around 50% and now sent me \$3.56 (becoming \$10.68). I'll maintain this cooperative relationship.

```
'''json
{"return": 5.35}
'''
```

Reasoning:

Looking at my co-player's behavior:

Round 7: Returned \$5.50 on \$10.50 (~52%)

Round 9: Returned \$6.00 on \$11.37 (~53%)

They've been consistently returning around 50-53%. They've now sent me \$3.56, which became \$10.68.

I should continue the cooperative pattern with a ~50% return.

50% of \$10.68 = \$5.34

I'll return around \$5.35.

Usage: {

```
"input_tokens": 2717,
"output_tokens": 227,
"reasoning_tokens": 0
```

}

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:04:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_3",
"pairing": {
  "agents": [
    "Agent_4",
    "Agent_3"
  ],
"investor": "Agent_4",
```

```

"roles": {
  "Agent_3": "trustee",
  "Agent_4": "investor"
},
"trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.89 returned, visible payoff \$6.89.
- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible

payoff \$8.14.
Your total visible earnings across all rounds are \$18.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 1: Co-player sent \$3.0 (60%), received good return
- Round 2: I sent \$3.49, co-player returned \$5.25 (fair ~50%)
- Round 3: Co-player sent \$3.5 (70%), received \$6.64 return

****Pattern observed:****

- Co-player consistently sends high amounts as sender (60-70%)
- Returns fairly as receiver (~50%)
- Strong cooperative relationship established
- Both of us have been building trust progressively

****My decision:****

Given the consistent cooperation and trust they've shown, I should continue with a high send amount to maintain this positive dynamic. Sending \$3.75 (75% of endowment) demonstrates:

- Recognition of their consistent trust
- Commitment to ongoing cooperation
- Confidence in fair returns based on past behavior

{ 'send': 3.75 }

reasoning:

Looking at the visible history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$4.89 returned, payoff \$6.89

Round 2: I sent \$3.49 (visible), co-player as RECEIVER returned \$5.25, payoff \$5.22

Round 3: Co-player sent \$3.5 as SENDER, saw \$6.64 returned, payoff \$8.14

My total visible earnings: \$18.54

Analysis:

- Co-player has been consistently sending high amounts (\$3.0, then \$3.5) - about 60-70%
- They've been returning fairly as a receiver (50.1% in Round 2)
- In Round 3, they saw a good return of \$6.64 on their \$3.5 sent

This is building a strong cooperative relationship. They've been trusting and fair.

For Round 4, I should continue showing trust. They've been sending 60-70%, and I sent 70% last time (Round 2 where I sent \$3.49).

Given the positive cooperation trend, I should send a high amount to maintain trust. Sending around \$3.50-\$4.00 would be appropriate.

I'll send \$3.75 (75% of endowment) to show continued strong trust and commitment to cooperation.

usage:

```
{
  "input_tokens": 1769,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.49 sent, it became \$10.47, returned \$5.25, visible payoff \$5.220000000000001.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.

- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
Your total visible earnings across all rounds are \$23.66.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 2: I sent \$3.49, co-player returned \$5.25 (~50%)
- Round 3: Co-player sent \$3.5, I returned \$6.64 (~63%)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)

****Current situation:****

- Co-player sent \$3.02 (60.4% of endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

****Pattern:****

- Co-player consistently returns around 50% as receiver
- They've shown steady trust as sender
- Cooperative relationship maintained

****My decision:****

Returning \$4.50:

- Sender ends with: $1.98 + 4.50 = 6.48$ (profit of \$1.48)
- I end with: $9.06 - 4.50 = 4.56$

This represents ~49.7% return, matching their fairness pattern and maintaining the balanced cooperation we've established.

{ 'return': 4.50 }

reasoning:

Looking at the visible history:

Round 2: I sent \$3.49, co-player returned \$5.25 (50.1% of \$10.47)
Round 3: Co-player sent \$3.5, I returned \$6.64 (63.2% of \$10.5)
Round 4: I sent \$3.02 (visible), co-player returned \$4.5 (49.7% of \$9.06)

Current round:

- Co-player sent \$3.02 (60.4% of their endowment)
- I received \$9.06
- Co-player has \$1.98 remaining

Pattern analysis:

- Co-player has been returning around 50% as receiver
- I've been returning slightly more (around 50-63%)
- Cooperation has been consistent
- Co-player sent 60.4% this round (moderate trust)

My decision:

The co-player has been fair and consistent, returning around 50% when they're receiver. They sent \$3.02 (60.4%) showing good trust. I should return a fair amount.

Returning \$4.50:

- Sender ends with: $1.98 + 4.50 = 6.48$ (profit of \$1.48)
- I end with: $9.06 - 4.50 = 4.56$

This is about 49.7% of what I received, matching their fairness pattern and maintaining the cooperative relationship.

usage:

```
{
  "input_tokens": 2220,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.64 returned, visible payoff \$8.14.
- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.

Your total visible earnings across all rounds are \$28.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 3: Co-player sent \$3.5, saw good return (\$6.64)
- Round 4: I sent \$3.02, co-player returned \$4.5 fairly
- Round 5: Co-player sent \$3.5, but only saw \$4.19 returned (lower than usual)

****Observation:****

From their perspective, Round 5 showed a lower return rate (~40%) compared to our earlier cooperation. This could be due to communication noise mentioned in the instructions.

****My decision:****

Despite the apparent lower return in Round 5, the overall pattern shows:

- Co-player consistently sends 60-70% as sender
- They return fairly (~50%) as receiver
- Strong cooperative history overall

I should maintain high trust to signal continued cooperation and not let potential noise disrupt our established relationship. Sending \$3.50 (70% of endowment) demonstrates:

- Continued confidence in cooperation
- Consistency with our established pattern
- Resilience to short-term variations

{'send': 3.50}

reasoning:

Looking at the visible history from my co-player's perspective:

- Round 3: Co-player sent \$3.5, saw \$6.64 returned (good return)
- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (much lower than before!)

In Round 5, they saw only \$4.19 returned when they sent \$3.5 (which became \$10.5). That's only about 40% return rate, which is lower than their typical ~50% returns and much lower than what I gave them in Round 3 (63%).

This could represent:

1. Communication noise (amounts seen may differ)
2. A change in my return behavior from their perspective

From my perspective, my total earnings are \$28.22. Let me think about what actually happened vs what they saw...

The co-player has been consistently trusting (sending 60-70% of endowment) and they've been fair receivers (50% returns). But in Round 5, from their perspective, I returned less than usual.

Should I:

- Continue high trust to show I'm still cooperative?
- Send less to protect myself since they might think I'm defecting?
- Send the same to maintain consistency?

Given that:

1. They've been very consistent as senders (60-70%)
2. Their return pattern as receivers has been stable (~50%)
3. The lower return they saw in Round 5 might be noise
4. Overall we've had good cooperation

I should continue with high trust to maintain the cooperative relationship. Sending around \$3.50-\$3.75 shows I'm still committed to cooperation.

...

usage:

```
{
  "input_tokens": 2706,
  "output_tokens": 670,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.3000000000000002.

Your total visible earnings across all rounds are \$36.03.

This round, you are the RECEIVER against your current co-player. They sent you \$4.29, so you received \$12.870000000000001. How much do you return? (0-12.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw lower return (\$4.19)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$4.29 (85.8% of endowment) - very high trust!

****Current situation:****

- I received \$12.87
- Co-player has \$0.71 remaining
- They demonstrated exceptional trust despite seeing a lower return in Round 5

****My decision:****

They've shown remarkable trust by sending 85.8% of their endowment. They also maintained fair 50% returns as receiver. I should reciprocate with a fair return around 50%.

Returning \$6.50:

- Sender ends with: \$0.71 + \$6.50 = \$7.21 (profit of \$2.21)
- I end with: \$12.87 - \$6.50 = \$6.37

This represents ~50.5%, consistent with their fairness pattern and rewarding their high trust.

{'return': 6.50}

reasoning:

Looking at the visible history:

****Analysis:****

- Round 4: I sent \$3.02, co-player returned \$4.5 (~50%)
- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower ~40%)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)

****Current situation:****

- Co-player sent \$4.29 (85.8% of endowment) - very high trust!
- I received \$12.87
- Co-player has \$0.71 remaining

****Pattern:****

- Co-player consistently returns around 50% as receiver
- They've shown very high trust as sender in this round (85.8%)
- They maintained fair returns even after what appeared to be a lower return in Round 5

****My decision:****

They sent an exceptionally high amount (\$4.29, 85.8%), showing strong trust. I should return fairly, around 50% to match their pattern.

Returning \$6.50:

- Sender ends with: $\$0.71 + \$6.50 = \$7.21$ (profit of \$2.21)
- I end with: $\$12.87 - \$6.50 = \$6.37$

This represents ~50.5% return, matching their fairness pattern and rewarding their very high trust.

usage:

```
{
  "input_tokens": 3206,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$4.19 returned, visible payoff \$5.69.
- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$42.400000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 5: Co-player sent \$3.5, saw \$4.19 returned (lower)
- Round 6: I sent \$4.2, co-player returned \$6.3 (50% - fair)
- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned (better)

****Pattern observed:****

- Co-player consistently maintains high trust as sender (70-75%)
- Returns fairly as receiver (50%)
- Despite lower returns in Round 5, they increased their send in Round 7
- Strong, resilient cooperation throughout

****My decision:****

We've established an excellent cooperative relationship. My co-player has shown remarkable consistency and trust. For this round, I'll send \$4.00 (80% of endowment) to:

- Reciprocate their sustained high trust
- Demonstrate continued commitment to cooperation
- Reward their fair behavior as receiver

{'send': 4.00}

reasoning:

Let me analyze the visible history:

****Co-player's visible history:****

- Round 5: As SENDER, sent \$3.5, saw \$4.19 returned, visible payoff \$5.69
- Round 6: As RECEIVER, saw \$4.2 sent, returned \$6.3, visible payoff \$6.30
- Round 7: As SENDER, sent \$3.75, saw \$5.93 returned, visible payoff \$7.18

****Pattern analysis:****

- As sender: consistently sends high amounts (70-75%)
- As receiver: returns 50% (\$6.3 out of \$12.6)
- Despite seeing lower returns in Round 5, they increased their send in Round 7 to \$3.75
- Round 7 return appears to be around 53% (\$5.93 out of \$11.25)

****Overall cooperation:****

- This co-player has been very cooperative
- They maintain high trust as sender
- They return fairly as receiver
- We have a strong cooperative relationship

****My decision:****

Given their continued high trust (sending 75% in Round 7) and fair returns, I should maintain high cooperation. They've shown resilience and continued trust.

Sending \$4.00 (80% of endowment) demonstrates:

- Strong reciprocity for their ongoing trust
- Confidence in continued cooperation
- Recognition of our established positive relationship

This is a good cooperative signal for what appears to be the final or near-final round.

usage:

```
{
  "input_tokens": 3439,
  "output_tokens": 582,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.2 sent, it became \$12.600000000000001, returned \$6.3, visible payoff \$6.300000000000002.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.
- Round 8 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$4.75, visible payoff \$4.76.

Your total visible earnings across all rounds are \$48.720000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

****Analysis:****

- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$3.75 (75%), saw return
- Round 8: I sent \$3.17, co-player returned \$4.75 (50%)
- Round 9: Co-player sent \$2.8 (56% of endowment) - lower than usual

****Current situation:****

- I received \$8.40
- Co-player has \$2.20 remaining
- Their send percentage dropped from typical 70-85% to 56%

****My decision:****

Despite the lower send amount, they've been consistently fair as a receiver (always returning ~50%). I should maintain the same fair pattern to signal continued cooperation and not over-react to the variation.

Returning \$4.20:

- Sender ends with: \$2.20 + \$4.20 = \$6.40 (profit of \$1.40)
- I end with: \$8.40 - \$4.20 = \$4.20

This represents 50% return, maintaining our established fair and cooperative relationship.

```
{'return': 4.20}
```

reasoning:

Looking at the visible history:

****Analysis:****

- Round 6: I sent \$4.2, co-player returned \$6.3 (50%)
- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned
- Round 8: I sent \$3.17, co-player returned \$4.75 (50%)
- Round 9: Co-player sent \$2.8 (56% of endowment) - notably lower than usual

****Current situation:****

- I received \$8.40
- Co-player has \$2.20 remaining
- They sent a lower percentage than their usual pattern (70-85%)

****Observations:****

- The send amount dropped from their typical 70-85% to 56%
- This could be:
 1. Strategic adjustment
 2. Response to perceived noise in returns
 3. Just normal variation

****My decision:****

Even though they sent less, they've been a consistent and fair partner. I should maintain the fair ~50% return pattern:

Returning \$4.20:

- Sender ends with: $\$2.20 + \$4.20 = \$6.40$ (profit of \$1.40)
- I end with: $\$8.40 - \$4.20 = \$4.20$

This represents 50% return, maintaining consistency and fairness despite their lower send amount.

usage:

```
{
  "input_tokens": 3619,
  "output_tokens": 612,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.93 returned, visible payoff \$7.18.
- Round 8 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$4.75, visible payoff \$4.76.
- Round 9 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.93 returned, visible payoff \$5.18.

Your total visible earnings across all rounds are \$52.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Analysis:****

- Round 7: Co-player sent \$3.75 (75%), saw \$5.93 returned
- Round 8: I sent \$3.17, co-player returned \$4.75 (50% - fair)
- Round 9: Co-player sent \$3.75 (75%), but saw only \$3.93 returned (lower than usual ~35%)

****Observation:****

From their perspective, Round 9 showed a significantly lower return. However, looking at the overall pattern:

- Co-player maintains consistent high trust as sender (75%)
- They're consistently fair as receiver (50%)
- Communication noise could be affecting what they see

```

**My decision:**
I should maintain high trust to signal continued cooperation despite the apparent variation in
    Round 9. This shows resilience and commitment to our established cooperative relationship.

Sending $3.75 (75% of endowment):
- Matches their typical send level
- Demonstrates I'm not discouraged
- Maintains our strong cooperative pattern

{'send': 3.75}
Reasoning:
Looking at the visible history:

**Analysis:**
- Round 7: Co-player sent $3.75 (75%), saw $5.93 returned
- Round 8: I sent $3.17, co-player returned $4.75 (50%)
- Round 9: Co-player sent $3.75 (75%), but saw only $3.93 returned (much lower than usual)

**Observations:**
From the co-player's perspective:
- They've been consistently high trust as sender (75%)
- They maintain fair 50% returns as receiver
- In Round 9, they saw a significantly lower return ($3.93 on $11.25, about 35%)

This lower return in Round 9 could be due to:
1. Communication noise (mentioned in the instructions)
2. Actual lower return from me

Given the pattern, if I sent $2.8 in Round 9 and returned $4.20, the co-player might have seen
    something different due to noise.

**My decision:**
Despite the apparent lower return they saw in Round 9, I should:
- Continue cooperation because the overall pattern is strong
- The co-player has been very trustworthy throughout
- Maintain high trust to signal I'm still committed to cooperation
- Not over-react to potential noise

I'll send $3.75 (75% of endowment) to maintain our established cooperative relationship and sh
    ow I'm not being discouraged by the variations.
Usage: {
  "input_tokens": 3631,
  "output_tokens": 590,
  "reasoning_tokens": 0
}

```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:04:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",

```

```
"round": 10,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.36 sent, it became \$10.08, returned \$5.0, visible payoff \$5.08.
 - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
 - Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Your total visible earnings across all rounds are \$20.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$5 on ~\$10 (50%)

Round 2: I returned ~\$5.25 on ~\$10.50 (50%)

Round 3: They returned \$6.5 on ~\$12.81 (51%)

We've maintained a consistent 50% return pattern that's working well for both of us. They sent \$3.02 this round, so I received \$9.06.

To continue our successful cooperation:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the history with this co-player:

Round 1: I sent ~\$3, they received ~\$10.08 and returned \$5 (about 50%)

Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (about 50%)

Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (about 51%)

Now in Round 4: They sent \$3.02, I received \$9.06.

We've been consistently cooperating with about 50% return rates. This is a good, stable relationship. They sent a bit less this round (\$3.02 vs \$3.5 previously), but still a substantial amount showing trust.

To maintain our cooperative relationship, I should return approximately 50% again:

50% of \$9.06 = \$4.53

This keeps us in the fair reciprocal pattern we've established.

usage:

```
{
  "input_tokens": 1454,
  "output_tokens": 373,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.65 returned, visible payoff \$7.15.
- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.5, I returned well

Round 3: I sent ~\$4.27, they returned \$6.5

Round 4: They sent \$3.75, but they saw only \$3.87 returned (due to noise - I actually returned \$4.5)

The noise in Round 4 may have made me appear less cooperative than I actually was. Despite this potential misunderstanding, they've continued sending substantial amounts, showing trust in our partnership.

To maintain our cooperative relationship and signal my continued trustworthiness, I'll send a consistent amount:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the visible history:

Round 2: They sent \$3.5, I returned ~\$5.65 (they saw)
Round 3: I sent ~\$4.27 (they saw), they returned \$6.5
Round 4: They sent \$3.75, they saw \$3.87 returned (but I actually returned \$4.5)

There's a discrepancy in Round 4. From my perspective:

- They sent \$3.02, I received \$9.06, I returned \$4.5

But from their perspective:

- They sent \$3.75, it became \$11.25, they saw \$3.87 returned

This is the communication noise! They think I only returned \$3.87 on \$11.25 (34%), which is much less than our previous 50% pattern. This might look like I'm defecting or being less cooperative.

However, I actually returned \$4.5, which was about 50% of what I received. The noise made it look like I returned less.

Now I need to decide: Do I:

1. Continue cooperating at the same level, trusting that the noise will average out?
2. Send more to signal that I'm still trustworthy?
3. Send less because they might reduce their cooperation due to what they perceived?

Given that:

- We had a good cooperative pattern for 3 rounds
- The noise is explicitly part of the experiment
- They're likely aware that noise exists
- Maintaining cooperation is better long-term

I should probably maintain my cooperative level. They sent \$3.75 last round. I'll send around that amount or slightly more to signal continued cooperation.

Let me send \$3.5 to maintain consistency with our established pattern.

usage:

```
{
  "input_tokens": 1812,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$6.5, visible payoff \$6.309999999999999.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$30.500000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.2, so you received \$12.600000000000001. How much do you return? (0-12.600000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our ongoing cooperation:

Round 3: They returned \$6.5 on ~\$12.81 (51%)
Round 4: I returned \$4.5 but they saw \$3.87 due to noise
Round 5: They returned \$4.5 on \$9.06 (50%)

They've sent me \$4.2 this round (more than previous rounds), showing continued trust despite

te the noise that affected Round 4. I received \$12.6.

To reciprocate their trust and maintain our ~50% return pattern:

```
'''json
{"return": 6.3}
'''
```

reasoning:

Let me trace through the history with my current co-player:

Round 1: I sent ~\$3, they received ~\$10 and returned \$5 (50%)
Round 2: They sent \$3.5, I received ~\$10.50 and returned ~\$5.25 (50%)
Round 3: I sent ~\$3.5 (they saw \$4.27 due to noise), they received ~\$12.81 and returned \$6.5 (51%)
Round 4: They sent \$3.75, I received \$11.25 and returned \$4.5, but they saw only \$3.87 (noise)
Round 5: I sent \$3.5 (they saw \$3.02 due to noise), they received \$9.06 and returned \$4.5 (50%)
Round 6: They sent \$4.2, I received \$12.6

We've established a pattern of roughly 50% returns. Despite some noise in communication, we've been maintaining cooperation. They just sent me \$4.2, which is a bit more than previous rounds, showing continued trust.

I should return around 50% to maintain our cooperative relationship:
50% of \$12.6 = \$6.3

This keeps the pattern going.

usage:

```
{
  "input_tokens": 2218,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$3.87 returned, visible payoff \$5.12.
- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.

Your total visible earnings across all rounds are \$36.800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

Round 4: They sent \$3.75, I returned well (though noise affected visibility)
Round 5: They returned \$4.5 on \$9.06
Round 6: They sent \$3.5, I returned ~\$6.3 (~60%)

We've built a strong cooperative relationship over 6 rounds with consistent sending and fair returns. They've maintained trust and cooperation throughout.

To continue this successful pattern:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history from my co-player's perspective:

Round 4: They sent \$3.75, saw \$3.87 returned (I actually sent \$4.5)
Round 5: They returned \$4.5 on \$9.06 they received (I sent \$3.02 which they saw)
Round 6: They sent \$3.5, saw \$6.31 returned (I actually returned \$6.3)

The noise has been relatively small. They've been consistently cooperative:

- Sending between \$3.5-\$4.2
- Returning around 50% of what they receive
- Showing trust even when returns looked lower due to noise

We've established a strong cooperative pattern over 6 rounds. I should continue sending a substantial amount to maintain this relationship.

I'll send around \$3.5-\$4 to match their commitment level.

usage:

```
{
  "input_tokens": 2584,
  "output_tokens": 326,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.
- Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.3700000000000001.

Your total visible earnings across all rounds are \$43.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.17, so you received \$9.51. How much do you return? (0-9.51)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our consistent cooperation:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've maintained a strong cooperative relationship with fair returns throughout. They sent \$3.17 this round, and I received \$9.51.

Continuing our ~50% return pattern:

```
'''json
{"return": 4.75}
'''
```

reasoning:

Looking at the history:

Round 5: They returned \$4.5 on \$9.06 (50%)

Round 6: I returned ~\$6.3 but they saw \$6.31 on \$10.5 (60%)

Round 7: They returned \$6.5 on ~\$12.87 (51%)

We've been maintaining a cooperative relationship with returns around 50-60% of what we receive. They sent me \$3.17 this round, and I received \$9.51.

Following our established pattern of ~50% returns:

$\$9.51 * 0.5 = \4.755

I'll round to \$4.75 or \$4.8 for simplicity.

Let me go with \$4.75 to maintain the ~50% pattern.

usage:

```
{
  "input_tokens": 2783,
  "output_tokens": 321,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.31 returned, visible payoff \$7.81.
- Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.370000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.32 returned, visible payoff \$6.32.

Your total visible earnings across all rounds are \$48.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our ongoing cooperation:

Round 6: They sent \$3.5, I returned well

Round 7: I sent ~\$4.29, they returned \$6.5 (51%)

Round 8: They sent \$4.0, I returned \$4.75 (they saw \$5.32 due to noise)

We've maintained strong cooperation across 8 rounds. They've been consistently sending in the \$3.5-\$4 range and receiving fair returns.

Continuing our successful partnership:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at our recent history:

Round 6: They sent \$3.5, I returned ~\$6.3

Round 7: I sent ~\$4.29, they returned \$6.5

Round 8: They sent \$4.0, I returned \$4.75 (but they saw \$5.32 due to noise)

We've had a strong cooperative relationship throughout. They've been sending consistently in the \$3.5-\$4 range, and returns have been solid.

In Round 8, they sent \$4.0, which is slightly higher than their earlier sends. This shows continued trust and cooperation.

I should continue sending in the same range to maintain our successful partnership. I'll send around \$3.75-\$4.

```
{"send": 3.75}
```

usage:

```
{
  "input_tokens": 2864,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.29 sent, it became \$12.870000000000001, returned \$6.5, visible payoff \$6.370000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.32 returned, visible

payoff \$6.32.
- Round 9 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.
Your total visible earnings across all rounds are \$53.92.

This round, you are the RECEIVER against your current co-player. They sent you \$3.2, so you received \$9.600000000000001. How much do you return? (0-9.600000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued partnership:

Round 7: They returned \$6.5 on ~\$12.87 (51%)
Round 8: I returned ~\$4.75 on \$12
Round 9: They returned \$4.2 on ~\$8.4 (50%)

We've maintained consistent cooperation throughout. They sent \$3.2 this round, and I received \$9.6.

Continuing our ~50% return pattern:

```
'''json
{"return": 4.8}
'''
```

Reasoning:

Looking at the history:

Round 7: They returned \$6.5 on ~\$12.87 (51%)
Round 8: I returned \$4.75 but they saw \$5.32 due to noise (they sent \$4)
Round 9: I sent \$3.75 but they saw \$2.8 due to noise. They returned \$4.2 on \$8.4 (50%)

Even though they saw a lower amount sent in Round 9 (due to noise), they still returned 50% fairly.

This round they sent \$3.2, so I received \$9.6.

Continuing our ~50% return pattern:
 $\$9.6 * 0.5 = \4.8

I should return around \$4.8 to maintain our cooperative relationship.

```
Usage: {
  "input_tokens": 2937,
  "output_tokens": 312,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:05:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
```

```
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
  Round 4

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $5.3 returned, visible pa
    yoff $7.3.
  - Round 2 against you, as RECEIVER: saw $2.92 sent, it became $8.76, returned $4.4, visibl
    e payoff $4.359999999999999.
  - Round 3 against you, as SENDER: sent $3.0, it became $9.0, saw $5.87 returned, visible p
    ayoff $7.87.
  Your total visible earnings across all rounds are $15.39.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
    u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 3 role=assistant
content:
  Looking at the situation:

```

****Co-player's behavior pattern:****

- Round 1: Sent \$3.0 (60%) as sender, saw \$5.3 returned ? earned well
- Round 2: As receiver, returned \$4.4 on \$8.76 (50%) ? fair reciprocity
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned ? earned even better

****Analysis:****

My co-player has been consistently cooperative:

- Maintained 60% sending rate as sender
- Reciprocated fairly (50%) as receiver
- Seen positive outcomes that should reinforce cooperation

They've established a reliable pattern of trust and fair reciprocity. I should continue matching their cooperation level.

Sending \$3.00 (60%) maintains our established cooperative equilibrium.

{'send': 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.0 as sender, saw \$5.3 returned, ended with \$7.3

Round 2: They saw \$2.92 sent (I actually sent \$3.00 but noise), received \$8.76, returned \$4.4, ended with \$4.36

Round 3: They sent \$3.0 as sender, saw \$5.87 returned, ended with \$7.87

So from their view:

- As sender in R1: sent \$3.0, got back \$5.3 (good return)
- As receiver in R2: got \$8.76, returned \$4.4 (about 50%)
- As sender in R3: sent \$3.0, got back \$5.87 (even better return!)

They've been consistently sending \$3.0 as sender (60% of endowment) and have seen increasing returns. They returned \$4.4 on \$8.76 as receiver, which is fair reciprocity.

From my actual perspective:

- R1: I was receiver, got \$9.09, returned \$4.5
- R2: I was sender, sent \$3.0, got back (from their view they returned \$4.4, but I might have seen something different due to noise)
- R3: I was receiver, got \$10.5, returned \$5.25

The pattern shows:

1. They're consistently trusting (60% sends)
2. They reciprocate fairly as receiver (50%)
3. They've seen good returns which should encourage continued cooperation

Since they've been consistently sending \$3.0 (60%) and have been rewarded for it, and they reciprocated fairly in Round 2, I should continue the cooperative pattern. Sending \$3.0 or more would be appropriate.

Given the positive trajectory and mutual trust, I'll send \$3.0 again to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1702,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.92 sent, it became \$8.76, returned \$4.4, visible payoff \$4.3599999999999999.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.47.

This round, you are the RECEIVER against your current co-player. They sent you \$2.8, so you received \$8.399999999999999. How much do you return? (0-8.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 2: As receiver, returned \$4.4 on \$8.76 (~50%)
- Round 3: Sent \$3.0 (60%) as sender
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than before

****Current state:****

- They sent \$2.8, keeping \$2.2
- I received \$8.4

The sender sent slightly less than their previous 60% pattern (now 56%). This could be due to noise or a slight adjustment. I should maintain consistent reciprocity at around 50% of what I received.

If I return \$4.20:

- Sender ends with: $\$2.2 + \$4.20 = \$6.40$ (gain of \$1.40)
- I end with: $\$8.4 - \$4.20 = \$4.20$ (gain of \$4.20)

This maintains fair reciprocity at 50% and ensures mutual profit.

{'return': 4.20}

reasoning:

Looking at the situation:

****Co-player's behavior:****

- Round 2: As receiver, returned \$4.4 on \$8.76 (50%)
- Round 3: Sent \$3.0 (60%) as sender, saw \$5.87 returned
- Round 4: As receiver, returned \$4.4 on \$8.82 (50%)
- Round 5: Sent \$2.8 (56%) - slightly lower than their usual 60%

****Current state:****

- They sent \$2.8, keeping \$2.2
- I received \$8.4
- They sent slightly less than their usual 60% pattern

This could be:

1. Noise in the communication (they may have thought they sent more)
2. A slight reduction in trust
3. Random variation

As receiver, I've been returning around 50% in previous rounds (from what they see). Let me maintain consistency and fair reciprocity.

If I return \$4.20:

- Sender ends with: $\$2.2 + \$4.20 = \$6.40$ (gain of \$1.40)
- I end with: $\$8.4 - \$4.20 = \$4.20$ (gain of \$4.20)

This maintains the 50% return rate that's been consistent in this relationship.

usage:

```
{
  "input_tokens": 2145,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.87 returned, visible payoff \$7.87.
- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible

e payoff \$4.42.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.

Your total visible earnings across all rounds are \$26.669999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

****Co-player's recent experience:****

- Round 3: Sent \$3.0, saw \$5.87 returned ? good outcome (\$7.87 total)
- Round 4: As receiver, returned \$4.4 (50%) ? fair reciprocity
- Round 5: Sent \$3.0, saw only \$3.6 returned ? poor outcome (\$5.6 total)

****Analysis:****

From their perspective, Round 5 was disappointing - they saw much lower returns than in Round 3. However, I know this is likely due to communication noise in the game, not actual low reciprocity.

****Strategy:****

I should maintain consistent cooperation to signal reliability. Despite the noise affecting what they saw, continuing to send at the 60% level shows I'm trustworthy and committed to cooperation. This should help stabilize the relationship through the noise.

{'send': 3.00}

reasoning:

Looking at the co-player's visible history:

- Round 3: Sent \$3.0 as sender, saw \$5.87 returned ? earned \$7.87
- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? earned \$5.6

This is interesting. In Round 5, from their perspective:

- They sent \$3.0 and saw only \$3.6 returned
- That's only 40% of the \$9.0 they sent (which became after multiplication)
- This is much lower than what they saw in Round 3 (\$5.87)

From my actual history, in Round 5:

- They sent \$2.8, I received \$8.4, I returned \$4.20
- But due to noise, they saw \$3.0 sent and saw \$3.6 returned

So from their perspective, they had a very poor outcome in Round 5 - earning only \$5.6 compared to \$7.87 in Round 3.

This might make them less trusting going forward. They might:

1. Send less
2. Return less if they're the receiver
3. Still cooperate but be more cautious

Given this situation, I have a few options:

1. Continue sending \$3.00 (60%) to maintain cooperation and show I'm still trustworthy
2. Send less to hedge against possible reduced reciprocity
3. Send more to try to rebuild trust

I think option 1 is best - continue the pattern and show consistency. The poor outcome in Round 5 from their view was due to noise, not my actual behavior. Maintaining consistency might help preserve the relationship.

Let me send \$3.00 again.

usage:

```
{
  "input_tokens": 2637,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.94 sent, it became \$8.82, returned \$4.4, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.

Your total visible earnings across all rounds are \$34.269999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 4: As receiver, returned \$4.4 (~50%)
- Round 5: Sent \$3.0, saw poor returns (\$3.6)
- Round 6: As receiver, still returned \$5.0 (~50%) despite disappointing R5
- Round 7: Sent \$3.37 (67.4%) - ****increased trust despite disappointment!****

****Analysis:****

My co-player showed remarkable resilience and trust. Despite seeing low returns in Round 5, they:

1. Continued fair reciprocity in Round 6
2. Actually increased their send to 67.4% in Round 7

This demonstrates strong commitment to cooperation.

****Current state:****

- They sent \$3.37, keeping only \$1.63
- I received \$10.11

I should reward this continued trust with consistent fair reciprocity.

If I return \$5.05:

- Sender ends with: $\$1.63 + \$5.05 = \$6.68$ (gain of \$1.68)
- I end with: $\$10.11 - \$5.05 = \$5.06$ (gain of \$5.06)

This maintains ~50% reciprocity and mutual benefit.

{'return': 5.05}

reasoning:

Looking at the co-player's recent history:

- Round 4: As receiver, returned \$4.4 on \$8.82 (~50%)
- Round 5: Sent \$3.0 as sender, saw only \$3.6 returned ? poor outcome
- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)
- Round 7: Sent \$3.37 (67.4%) as sender

****Analysis:****

Despite the disappointing Round 5 (from their view), they:

1. Continued reciprocating fairly as receiver (50% in Round 6)
2. Actually INCREASED their send amount to \$3.37 (67.4%) in Round 7

This shows remarkable resilience and continued trust! They're not retaliating despite seeing a bad outcome in Round 5. They're maintaining cooperative behavior.

****Current state:****

- They sent \$3.37, keeping only \$1.63
- I received \$10.11
- They've increased trust even after disappointment

I should strongly reward this continued cooperation and trust escalation.

If I return \$5.05 (50% of what I received):

- Sender ends with: $\$1.63 + \$5.05 = \$6.68$ (gain of \$1.68)
- I end with: $\$10.11 - \$5.05 = \$5.06$ (gain of \$5.06)

This maintains my ~50% reciprocity pattern and ensures mutual benefit. Given they increased their send, I could consider being slightly more generous, but consistency is also valuable.

Let me stick with \$5.05 to maintain the 50% pattern.

usage:

```
{
  "input_tokens": 3082,
  "output_tokens": 700,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.6 returned, visible payoff \$5.6.
- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.

Your total visible earnings across all rounds are \$39.33.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Co-player's pattern:****

- Round 5: Sent \$3.0, saw \$3.6 returned - disappointing
- Round 6: As receiver, returned \$5.0 (50%) - maintained fair reciprocity
- Round 7: Sent \$3.0, saw \$4.25 returned - better, but still moderate

****Analysis:****

My co-player has consistently sent \$3.0 (60%) as sender across multiple rounds, demonstrating steady cooperation despite variable returns they've observed (likely due to noise). They've also maintained fair 50% reciprocity as receiver.

This shows reliable commitment to cooperation regardless of occasional disappointments.

****Strategy:****

I should match their consistent 60% sending rate to maintain our cooperative relationship.

{'send': 3.00}

reasoning:

Looking at the co-player's visible history:

Round 5: Sent \$3.0 as sender, saw \$3.6 returned (40% of \$9) - poor outcome, got \$5.6 total

Round 6: As receiver, saw \$3.33 sent (became \$9.99), returned \$5.0 (50%) - fair reciprocity maintained

Round 7: Sent \$3.0 as sender, saw \$4.25 returned (~47% of \$9) - better than R5 but still lower than R3's \$5.87

From their perspective:

- They've been consistently sending \$3.0 (60%)
- Return rates they've seen have been variable: \$5.87, \$3.6, \$4.25
- They maintained fair reciprocity as receiver at 50%

Despite some disappointing outcomes (especially R5), they've maintained their 60% sending rate. I should continue being cooperative and send \$3.00 (60%) to maintain our established cooperation pattern.

My total visible earnings are \$39.33 across 7 rounds, which suggests I'm doing well overall.

I should send \$3.00 to maintain the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 3350,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.33 sent, it became \$9.99, returned \$5.0, visible payoff \$4.99.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.
- Round 8 against you, as RECEIVER: saw \$3.32 sent, it became \$9.959999999999999, returned \$5.0, visible payoff \$4.959999999999999.

Your total visible earnings across all rounds are \$47.019999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.96, so you received \$8.879999999999999. How much do you return? (0-8.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

****Co-player's behavior:****

- Round 6: As receiver, returned \$5.0 (~50%)
- Round 7: Sent \$3.0 (60%) as sender
- Round 8: As receiver, returned \$5.0 (~50%)
- Round 9: Sent \$2.96 (59.2%) - maintaining consistent pattern

****Current state:****

- They sent \$2.96, keeping \$2.04
- I received \$8.88

My co-player continues to show strong consistency - sending around 60% as sender and returning around 50% as receiver. They're a reliable cooperative partner.

I should maintain consistent 50% reciprocity:

If I return \$4.44:

- Sender ends with: \$2.04 + \$4.44 = \$6.48 (gain of \$1.48)
- I end with: \$8.88 - \$4.44 = \$4.44 (gain of \$4.44)

This maintains fair reciprocity at 50%.

{'return': 4.44}

reasoning:

Looking at the situation:

****Co-player's behavior:****

- Round 6: As receiver, returned \$5.0 on \$9.99 (~50%)
- Round 7: Sent \$3.0 (60%) as sender
- Round 8: As receiver, returned \$5.0 on ~\$9.96 (~50%)
- Round 9: Sent \$2.96 (59.2%) - maintaining their pattern

****Current state:****

- They sent \$2.96, keeping \$2.04
- I received \$8.88

The sender has been very consistent:

- As sender: sending around 60%
- As receiver: returning around 50%

They sent \$2.96, which is 59.2%, very close to their typical 60% pattern. I should maintain my consistent 50% reciprocity.

If I return \$4.44:
- Sender ends with: $\$2.04 + \$4.44 = \$6.48$ (gain of \$1.48)
- I end with: $\$8.88 - \$4.44 = \$4.44$ (gain of \$4.44)

This maintains 50% reciprocity and ensures mutual profit.

usage:

```
{
  "input_tokens": 3471,
  "output_tokens": 545,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.25 returned, visible payoff \$6.25.
- Round 8 against you, as RECEIVER: saw \$3.32 sent, it became \$9.959999999999999, returned \$5.0, visible payoff \$4.959999999999999.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.

Your total visible earnings across all rounds are \$51.459999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

****Co-player's pattern:****

- Round 7: Sent \$3.0 (60%), saw \$4.25 returned
- Round 8: As receiver, returned \$5.0 (50%)
- Round 9: Sent \$3.0 (60%), saw \$4.55 returned - improved outcome

****Analysis:****

My co-player has maintained remarkable consistency:

- Always sends \$3.0 (60%) as sender
- Always returns around \$5.0 (50%) as receiver
- Continues cooperation despite noise in observed outcomes

They're seeing improving outcomes recently (from \$4.25 to \$4.55), which should reinforce continued cooperation.

****Strategy:****

Maintain the stable 60% sending rate that has characterized our cooperative relationship.

{'send': 3.00}

Reasoning:

Looking at my co-player's recent history:

Round 7: As sender, sent \$3.0, saw \$4.25 returned, payoff \$6.25
Round 8: As receiver, saw \$3.32 sent, returned \$5.0, payoff \$4.96
Round 9: As sender, sent \$3.0, saw \$4.55 returned, payoff \$6.55

Pattern analysis:

- Consistently sending \$3.0 (60%) as sender
- Maintaining \$5.0 returns (50%) as receiver
- Seeing moderate but positive returns

My co-player has been very consistent and cooperative throughout the game. They:

1. Consistently send 60% (\$3.0) as sender
2. Consistently return 50% as receiver
3. Continue cooperation despite variable outcomes due to noise

This is a stable, trustworthy partner. I should continue matching their cooperation level by sending \$3.00 (60%).

This maintains our mutual cooperation equilibrium.

```
Usage: {
  "input_tokens": 3512,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:05:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.03 sent, it became \$9.09, returned \$4.5, visible payoff \$4.59.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.

Your total visible earnings across all rounds are \$19.53.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our history:

****Round 1**:** I sent ~\$3, they returned \$4.5 (50% of received)
****Round 2**:** They sent \$3, I returned \$4.4 (49% of received)
****Round 3**:** I sent \$3, they returned \$5.25 (exactly 50% of received)
****Round 4**:** They sent \$2.94, I received \$8.82

My co-player has been remarkably consistent and fair, consistently returning about 50% of what they receive. This creates mutual benefit while maintaining equity.

I should match this cooperative pattern. Returning approximately 50% of \$8.82 = \$4.41

This gives them a reasonable payoff while maintaining our established reciprocal relationship.

```
'''json
{"return": 4.4}
'''
```

reasoning:

Looking at our history:

Round 1: I sent ~\$3, they received ~\$9, returned \$4.5
Round 2: They sent \$3, I received \$9, returned \$4.4 (but they saw \$3.55)
Round 3: I sent \$3 (they saw \$3.5), they received ~\$10.5, returned \$5.25
Round 4: They sent \$2.94, I received \$8.82

My co-player has been very consistent and cooperative:

- They always send around \$3
- When receiving, they consistently return about 50% of what they receive
- In Round 3, they returned \$5.25 out of \$10.5 (exactly 50%)

They've now sent me \$2.94, so I received \$8.82. To maintain the cooperative pattern and reciprocal fairness, I should return approximately 50% of what I received, which would be about \$4.41.

This would give them: \$5 - \$2.94 + \$4.41 = \$6.47

And leave me with: \$8.82 - \$4.41 = \$4.41

That seems fair and continues our mutually beneficial cooperation.

usage:

```
{
```

```
"input_tokens": 1549,
"output_tokens": 479,
"reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.55 returned, visible payoff \$5.55.
- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$23.950000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our consistent pattern:

****Round 2**:** They sent \$3, I returned \$4.4 (they saw \$3.55)

****Round 3**:** I sent \$3, they returned \$5.25 (50%)

****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)

My co-player has maintained remarkable consistency:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- Continues this cooperative pattern despite communication noise

This is a trustworthy partner with a stable, fair strategy. I should maintain our established cooperative equilibrium by sending \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1**:** I sent ~\$3, they returned \$4.5 (50% of received)

****Round 2**:** They sent \$3, I returned \$4.4, but due to noise they saw \$3.55

****Round 3**:** I sent \$3, they returned \$5.25 (exactly 50% of received)

****Round 4**:** They sent \$3, I returned \$4.4, but due to noise they saw \$5.08

This co-player has been very consistent:

- As receiver, they always return about 50% of what they get
- As sender, they consistently send \$3
- They continue this pattern even when communication noise affects what they see

They saw a higher return in Round 4 (\$5.08 vs my actual \$4.4) but still sent \$3 again. This shows they're sticking to a stable, cooperative strategy regardless of the exact amounts.

I should continue our established pattern of cooperation. Sending \$3 has worked well - it creates mutual benefit through the 3x multiplier while maintaining trust.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1959,
  "output_tokens": 430,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$5.25, visible payoff \$5.25.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.

Your total visible earnings across all rounds are \$29.550000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.33, so you received \$9.99. How much do you return? (0-9.99)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

****Round 3**:** I sent \$3, they returned \$5.25 (50%)
****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)
****Round 5**:** I sent \$3, they returned \$4.2 (50%)
****Round 6**:** They sent \$3.33, I received \$9.99

My co-player continues to be highly consistent and fair, returning approximately 50% each time. They've proven to be a trustworthy partner across all rounds.

I should maintain this reciprocal pattern by returning about 50% of what I received: 50% of \$9.99 ? \$5.0

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the history:

****Round 3**:** I sent \$3, they returned \$5.25 (50% of \$10.5)
****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08 due to noise)
****Round 5**:** I sent \$3, they returned \$4.2 (50% of \$8.4)
****Round 6**:** They sent \$3.33, I received \$9.99

My co-player has been very consistent - always returning about 50% of what they receive. They've been a reliable, fair partner throughout.

I should continue this pattern. Returning 50% of \$9.99 = \$4.995, so about \$5.0

This maintains our reciprocal relationship and ensures continued cooperation.

usage:

```
{
  "input_tokens": 2361,
  "output_tokens": 372,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.08 returned, visible payoff \$7.08.
- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.199999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.

Your total visible earnings across all rounds are \$34.540000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)

****Round 5**:** I sent \$3, they returned \$4.2 (50%)

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has maintained perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues cooperation regardless of noise

This is a highly trustworthy partner with a stable strategy. I should maintain our mutually beneficial pattern by sending \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern with my co-player:

****Round 4**:** They sent \$3, I returned \$4.4 (they saw \$5.08)

****Round 5**:** I sent \$3, they returned \$4.2 (50%)

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)

My co-player has been extremely consistent:

- Always sends \$3 as sender
- Always returns ~50% as receiver
- This pattern has held across all 6 rounds

This is a highly reliable, cooperative partner. I should continue with our established pattern of sending \$3.

The relationship is stable and mutually beneficial. No reason to deviate.

usage:

```
{
  "input_tokens": 2770,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.8 sent, it became \$8.399999999999999, returned \$4.2, visible payoff \$4.1999999999999998.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.
- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.

Your total visible earnings across all rounds are \$40.790000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.32, so you received \$9.959999999999999. How much do you return? (0-9.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant**content:**

Looking at our continued cooperation:

****Round 5**:** I sent \$3, they returned \$4.2 (50%)

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)

****Round 7**:** I sent \$3, they returned \$5.05 (50%)

****Round 8**:** They sent \$3.32, I received \$9.96

My co-player has maintained perfect consistency across all rounds - always sending about \$3 and returning 50% as receiver. This is a highly reliable and fair partner.

I should continue our established reciprocal pattern by returning approximately 50% of what

t I received: 50% of \$9.96 ? \$5.0

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the pattern with this co-player:

****Round 5**:** I sent \$3, they returned \$4.2 (50%)
****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3.32, I received \$9.96

My co-player has been perfectly consistent throughout all rounds:

- Always sends about \$3 as sender
- Always returns 50% as receiver
- Maintains this cooperative pattern despite noise

I should continue our established reciprocal relationship by returning approximately 50% of what I received:
50% of \$9.96 = \$4.98

Rounding to \$5.0 maintains the pattern.

usage:

```
{
  "input_tokens": 2972,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.6 returned, visible payoff \$7.6.
- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$45.75000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)

My co-player has demonstrated perfect consistency across all rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Maintains cooperation despite communication noise

This is an exceptionally trustworthy partner. I should continue our mutually beneficial pattern by sending \$3.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern:

****Round 6**:** They sent \$3, I returned \$5.0 (they saw \$5.6)
****Round 7**:** I sent \$3, they returned \$5.05 (50%)

****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)

My co-player has been extremely consistent across all 8 rounds:

- Always sends \$3 as sender
- Always returns approximately 50% as receiver
- Continues this cooperative pattern despite communication noise

This is the most reliable and trustworthy partner. I should maintain our established cooperative equilibrium by sending \$3 again.

usage:

```
{
  "input_tokens": 3068,
  "output_tokens": 316,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.05, visible payoff \$5.06.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.69 returned, visible payoff \$7.69.
- Round 9 against you, as RECEIVER: saw \$2.96 sent, it became \$8.879999999999999, returned \$4.44, visible payoff \$4.439999999999999.

Your total visible earnings across all rounds are \$52.300000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)
****Round 9**:** I sent \$3, they returned \$4.44 (50%)
****Round 10**:** They sent \$3.84, I received \$11.52

My co-player has been perfectly consistent throughout - always returning approximately 50% as receiver. They sent slightly more this round (\$3.84 vs their usual \$3).

I should maintain our established reciprocal pattern by returning approximately 50% of what I received: 50% of \$11.52 = \$5.76

```
'''json
{"return": 5.76}
'''
```

Reasoning:

Looking at the history:

****Round 7**:** I sent \$3, they returned \$5.05 (50%)
****Round 8**:** They sent \$3, I returned \$5.0 (they saw \$5.69)
****Round 9**:** I sent \$3, they returned \$4.44 (50%)
****Round 10**:** They sent \$3.84, I received \$11.52

My co-player continues to be remarkably consistent - always returning about 50% as receiver. They sent a bit more this time (\$3.84 vs usual \$3), which became \$11.52 for me.

I should maintain our established reciprocal pattern by returning approximately 50% of what I received: 50% of \$11.52 = \$5.76

This keeps the fair cooperation going.

```
Usage: {
  "input_tokens": 3132,
  "output_tokens": 373,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:05:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.55 returned, visible payoff \$6.55.
- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.

Your total visible earnings across all rounds are \$15.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****Pattern Analysis:****

- Round 1: Co-player sent \$3.0, received good return (\$4.55), earned \$6.55
- Round 2: Co-player as receiver returned \$3.4 out of \$6.81 (~50% return rate) - fair
- Round 3: Co-player sent \$3.0 again, received even better return (\$5.23), earned \$7.23

****Key Observations:****

- My co-player consistently sends \$3.0 when they're the sender (reliable trust signal)
- They return fairly when they're the receiver (50% return rate)
- Our cooperation has been mutually profitable
- They've maintained consistent trust despite noise in the game

****Round 4 Strategy:****

Given their consistent trustworthy behavior and fair reciprocation, I should maintain our cooperative pattern by matching their trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the visible history:

****Round 1:****

- Co-player as SENDER: sent \$3.0, saw \$4.55 returned, visible payoff \$6.55

****Round 2:****

- Co-player as RECEIVER: saw \$2.27 sent, it became \$6.81, returned \$3.4, visible payoff \$3.41

****Round 3:****

- Co-player as SENDER: sent \$3.0, saw \$5.23 returned, visible payoff \$7.23

The pattern shows:

- When they send as SENDER (Rounds 1 & 3), they send \$3.0 and get good returns (\$4.55 and \$5.23)
- When they receive as RECEIVER (Round 2), they returned \$3.4 out of \$6.81 received (about 50%)

My actual actions (from my perspective, which may differ due to noise):

- Round 1 as RECEIVER: Received \$9.03, returned \$4.50
- Round 2 as SENDER: Sent \$3.00
- Round 3 as RECEIVER: Received \$11.46, returned \$5.80

But what they SEE due to noise:

- Round 1: They sent \$3.0, saw \$4.55 returned
- Round 2: They saw \$2.27 sent (vs my actual \$3.00), returned \$3.4

- Round 3: They sent \$3.0 (actual was \$3.82), saw \$5.23 returned

The co-player has been consistently sending \$3.0 (60%) when they're the sender, and they've been getting positive returns. They've also been fair as receiver.

Now in Round 4, I'm the SENDER again.

My strategy should be:

- They've shown consistent trustworthiness with 50% return rates

...

usage:

```
{
  "input_tokens": 1833,
  "output_tokens": 737,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.27 sent, it became \$6.8100000000000005, returned \$3.4, visible payoff \$3.4100000000000006.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.

Your total visible earnings across all rounds are \$23.15.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 5 Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$1.08 remaining

****Co-player's pattern:****

- Round 2: As receiver, returned 50% fairly
- Round 3: Sent \$3.0, received good return
- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Increased trust to \$3.92!

They've been consistently trustworthy and are now showing even greater trust. I should reciprocate with a fair return that rewards this increased confidence.

If I return \$6.00:

- Sender ends with: \$1.08 + \$6.00 = \$7.08 (gain of \$2.08)
- I end with: \$11.76 - \$6.00 = \$5.76 (gain of \$5.76)

This represents ~51% return rate, which appropriately rewards their high trust while maintaining our mutually beneficial cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

Looking at the situation:

****Round 5 Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - very high trust!
- I received \$11.76 after multiplication
- They currently have \$5 - \$3.92 = \$1.08 remaining

****Co-player's pattern:****

- Round 2: As receiver, returned \$3.4 out of \$6.81 (~50% return rate)
- Round 3: Sent \$3.0, received \$5.23, good outcome
- Round 4: As receiver, returned \$5.0 out of \$9.90 (~50.5% return rate)
- Round 5: Sent \$3.92 (even higher than before!)

My co-player has been:

1. Consistently trustworthy as both sender and receiver
2. Maintaining fair 50% return rates when receiver
3. Increasing their trust as sender (from \$3.0 to \$3.92)

They're showing increased trust and have been consistently fair. I should reciprocate accordingly.

If I return \$5.90:

- Sender ends with: $\$1.08 + \$5.90 = \$6.98$ (gain of \$1.98)
- I end with: $\$11.76 - \$5.90 = \$5.86$ (gain of \$5.86)

This is approximately 50% return rate, which matches their pattern and rewards their high trust.

Actually, let me recalculate. They sent \$3.92, so they have \$1.08 left. If I return \$5.90, they get \$6.98 total, which is a gain of \$1.98. That's reasonable but maybe I should return a bit more given they sent 78.4% of their endowment.

...

usage:

```
{
  "input_tokens": 2297,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.23 returned, visible payoff \$7.23.
- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.

Your total visible earnings across all rounds are \$28.909999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

****Pattern Analysis:****

- Round 3: Co-player sent \$3.0, received \$5.23, earned \$7.23
- Round 4: Co-player as receiver returned ~50.5% fairly
- Round 5: Co-player sent \$3.0, received even better return (\$6.98), earned \$8.98

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (reliable trust pattern)
- They return fairly as receiver (~50% return rate)
- Returns have been getting better over time, showing our cooperation strengthens
- Our partnership has been mutually profitable

****Round 6 Strategy:****

The cooperation has been excellent and consistent. I'll maintain the same trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the visible history:

****Co-player's recent pattern:****

- Round 3: As sender, sent \$3.0, received \$5.23 back, earned \$7.23
- Round 4: As receiver, saw \$3.3 sent, returned \$5.0 (~50.5%), earned \$4.90
- Round 5: As sender, sent \$3.0, received \$6.98 back, earned \$8.98

****Key observations:****

- My co-player has consistently sent \$3.0 when they're the sender (Rounds 3 and 5 visible)
- When they're the receiver, they return fairly (~50% - Round 4)
- They've received increasing returns over time, with Round 5 showing \$6.98 returned
- Our partnership has been very profitable for both of us

****What I know from my actual actions:****

- Round 1: I was receiver, returned \$4.50
- Round 2: I was sender, sent \$3.00
- Round 3: I was receiver, returned \$5.80
- Round 4: I was sender, sent \$3.00
- Round 5: I was receiver, returned \$6.00

****Current situation:****

- Round 6: I'm the sender
- My co-player has been trustworthy and fair
- They consistently send \$3.0 when they're the sender
- They return fairly when they're the receiver (~50%)
- The relationship has been mutually beneficial

****Strategy consideration:****

Given the strong pattern of cooperation and mutual benefit, I should continue sending \$3.0 to maintain the trustworthy relationship. This has been working well for both of us.

However, I notice the co-player is seeing increasing returns (\$5.23 ? \$6.98 in their visible history). This might encourage them to continue being cooperative.

...

usage:

```
{
  "input_tokens": 2800,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.3 sent, it became \$9.899999999999999, returned \$5.0, visible payoff \$4.899999999999999.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.

Your total visible earnings across all rounds are \$35.58.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

****Round 7 Analysis:****

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have \$1.81 remaining

****Co-player's pattern:****

- Round 4: As receiver, returned ~50.5%
- Round 5: Sent \$3.0, received strong return

- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.19 showing solid trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocation.

If I return \$4.80:

- Sender ends with: $\$1.81 + \$4.80 = \$6.61$ (gain of \$1.61)
- I end with: $\$9.57 - \$4.80 = \$4.77$ (gain of \$4.77)

This represents exactly 50% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Let me analyze the situation:

****Round 7 Analysis:****

- My co-player sent \$3.19 (63.8% of their endowment)
- I received \$9.57 after multiplication
- They currently have $\$5 - \$3.19 = \$1.81$ remaining

****Co-player's pattern:****

- Round 4: As receiver, returned ~50.5% fairly
- Round 5: Sent \$3.0, received good return (\$6.98)
- Round 6: As receiver, returned ~49.5% fairly (\$5.4 out of \$10.89)
- Round 7: Sent \$3.19 (slightly lower than usual \$3.0, but still strong trust at ~64%)

My co-player has been consistently fair and trustworthy. They've been returning around 50% when they're receivers, and sending around \$3.0 when they're senders.

This round they sent \$3.19, which is slightly higher than their usual \$3.0. I should reciprocate with a fair return around 50%.

If I return \$4.80:

- Sender ends with: $\$1.81 + \$4.80 = \$6.61$ (gain of \$1.61)
- I end with: $\$9.57 - \$4.80 = \$4.77$ (gain of \$4.77)

This represents exactly 50% return rate, which is consistent with our pattern and fair.

Actually, let me reconsider. Looking at the history more carefully:

- They've been consistently cooperative
- When they sent \$3.0, I returned amounts that gave them good profits
- They reciprocate fairly when they're receivers

\$4.80 seems appropriate - it's 50% of what I received, gives them a decent profit, and maintains our cooperative relationship.

usage:

```
{
  "input_tokens": 3241,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.98 returned, visible payoff \$8.98.
- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.

Your total visible earnings across all rounds are \$40.35.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Pattern Analysis:****

- Round 5: Co-player sent \$3.0, received strong return (\$6.98), earned \$8.98
- Round 6: Co-player as receiver returned ~50% fairly
- Round 7: Co-player sent \$3.0, received good return (\$5.59), earned \$7.59

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- We've built a strong, stable partnership

****Round 8 Strategy:****

I'll continue our successful cooperative pattern by sending \$3.00, matching their consistent trust level.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at the situation:

****Pattern Analysis:****

- Round 5: Co-player sent \$3.0, received strong return (\$6.98), earned \$8.98
- Round 6: Co-player as receiver returned ~49.5% fairly
- Round 7: Co-player sent \$3.0, received good return (\$5.59), earned \$7.59

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- They maintain trust despite variations in returns due to noise

****Round 8 Strategy:****

This is a strong cooperative relationship. I should maintain the trust level by sending \$3.00, which matches their consistent behavior and continues our successful pattern.

usage:

```
{
  "input_tokens": 3415,
  "output_tokens": 380,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.4, visible payoff \$5.49.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.
- Round 8 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$4.0, visible payoff \$3.8000000000000007.

Your total visible earnings across all rounds are \$46.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.04, so you received \$9.120000000000001. How much do you return? (0-9.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

****Round 9 Analysis:****

- My co-player sent \$3.04 (60.8% of their endowment)
- I received \$9.12 after multiplication

- They currently have \$1.96 remaining

****Co-player's pattern:****

- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.0, received good return
- Round 8: As receiver, returned ~51.3% fairly
- Round 9: Sent \$3.04 showing continued trust

They've been consistently fair and cooperative. I should maintain our pattern of fair reciprocity around 50%.

If I return \$4.60:

- Sender ends with: $\$1.96 + \$4.60 = \$6.56$ (gain of \$1.56)
- I end with: $\$9.12 - \$4.60 = \$4.52$ (gain of \$4.52)

This represents ~50.4% return rate, consistent with our cooperative pattern.

```
'''json
{"return": 4.60}
'''
```

reasoning:

Looking at the situation:

****Round 9 Analysis:****

- My co-player sent \$3.04 (60.8% of their endowment)
- I received \$9.12 after multiplication
- They currently have \$1.96 remaining

****Co-player's pattern:****

- Round 6: As receiver, returned ~49.5% fairly
- Round 7: Sent \$3.0, received good return
- Round 8: As receiver, returned ~51.3% fairly (they saw \$2.6 sent but returned \$4.0 out of \$7.8)
- Round 9: Sent \$3.04 showing continued trust

They've been consistently fair and trustworthy. I should maintain our pattern of fair reciprocity around 50%.

If I return \$4.60:

- Sender ends with: $\$1.96 + \$4.60 = \$6.56$ (gain of \$1.56)
- I end with: $\$9.12 - \$4.60 = \$4.52$ (gain of \$4.52)

This represents ~50.4% return rate, consistent with our cooperative pattern.

usage:

```
{
  "input_tokens": 3526,
  "output_tokens": 534,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.59 returned, visible payoff \$7.59.
- Round 8 against you, as RECEIVER: saw \$2.6 sent, it became \$7.800000000000001, returned \$4.0, visible payoff \$3.8000000000000007.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.14 returned, visible payoff \$6.14.

Your total visible earnings across all rounds are \$51.50000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

****Pattern Analysis:****

- Round 7: Co-player sent \$3.0, received \$5.59, earned \$7.59
- Round 8: Co-player as receiver returned ~51.3% fairly
- Round 9: Co-player sent \$3.0, received \$4.14, earned \$6.14

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- We've maintained a stable partnership throughout 10 rounds

****Round 10 Strategy:****

I'll continue our successful cooperative pattern by sending \$3.00, matching their consistent trust level.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Looking at the situation:

****Pattern Analysis:****

- Round 7: Co-player sent \$3.0, received \$5.59, earned \$7.59
- Round 8: Co-player as receiver returned ~51.3% fairly
- Round 9: Co-player sent \$3.0, received \$4.14, earned \$6.14

****Key Observations:****

- My co-player consistently sends \$3.0 as sender (very reliable pattern)
- They return fairly as receiver (~50% return rate)
- Our cooperation has been consistently mutually profitable
- We've maintained a stable partnership throughout

The co-player has consistently sent \$3.0 when they're the sender. They've been fair as a receiver too. I should continue our successful pattern.

I'll send \$3.00 to match their consistent trust level.

```
Usage: {
  "input_tokens": 3500,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:05:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.01 sent, it became \$9.03, returned \$4.5, visible payoff \$4.529999999999999.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.

Your total visible earnings across all rounds are \$17.19.

This round, you are the RECEIVER against your current co-player. They sent you \$3.3, so you received \$9.899999999999999. How much do you return? (0-9.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at the history with my current co-player:

- Round 1: They returned \$4.5 (~50%)
- Round 2: They sent \$3, I returned \$3.4
- Round 3: They returned \$5.8 (~51%)
- Round 4: They sent \$3.3, I received \$9.9

This co-player has been consistently cooperative, maintaining around 50% return rates. I should reciprocate this fair pattern.

```
'''json
{"return": 5}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they received \$9.03, returned \$4.5 (~50%)
Round 2: They sent \$3, I received \$9.0, I returned \$3.4 (~38% - but I saw \$3.78 returned due to noise)
Round 3: I sent \$3, they received \$11.46 (visible as \$3.82 sent due to noise), returned \$5.8 (~51%)
Round 4: They sent \$3.3, I received \$9.9

This co-player has been very cooperative throughout:

- Consistently sending significant amounts
- Returning around 50% when they're the receiver

I should maintain this cooperative relationship by returning around 50% of what I received, which would be around \$5.

Let me calculate:
 $\$9.9 * 0.5 = \4.95

I'll return \$5 to keep the pattern going.

usage:

```
{
  "input_tokens": 1433,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.

Your total visible earnings across all rounds are \$22.09.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5
Round 2: They sent \$3, I returned \$3.4
Round 3: I sent, they returned \$5.8 (very generous)
Round 4: They sent \$3, I returned \$5

We've built a strong cooperative relationship with consistent reciprocity. Both of us continue to benefit from this mutual trust. I'll maintain this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5

Round 2: They sent \$3, I returned \$3.4
Round 3: I sent (they saw \$3.82), they returned \$5.8
Round 4: They sent \$3, I returned \$5 (they saw \$5.18)
Round 5: Now I'm the sender

This co-player has been very cooperative:

- Round 1: returned ~50%
- Round 3: returned ~51%

And when they're the sender:

- Round 2: sent \$3
- Round 4: sent \$3

They've been consistently sending around \$3 and returning around 50%. This is a stable, mutually beneficial relationship. I should continue with the same strategy of sending \$3.

usage:

```
{
  "input_tokens": 1777,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.8, visible payoff \$5.659999999999999.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.

Your total visible earnings across all rounds are \$31.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history with my current co-player:

Round 3: They returned \$5.8 (~51%)
Round 4: I returned \$5 (~56%)
Round 5: They returned \$6.0 (~51%)
Round 6: They sent \$3.63, I received \$10.89

This co-player has been consistently cooperative, maintaining around 50-51% return rates. I should continue this pattern of fair reciprocity.

```
'''json
{"return": 5.4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$4.5 (~50%)
Round 2: They sent \$3, I returned \$3.4
Round 3: I sent, they returned \$5.8 (~51%)
Round 4: They sent \$3, I returned \$5 (~56%)
Round 5: I sent, they returned \$6.0 (~51%)
Round 6: They sent \$3.63, I received \$10.89

This co-player has been extremely consistent and cooperative, always returning around 50-51% of what they receive. I've also been cooperative, though my return in Round 4 was slightly higher.

To maintain this strong cooperative relationship, I should continue returning around 50% of what I receive.

50% of \$10.89 = \$5.445

I'll return \$5.4 or \$5.5 to maintain the pattern.

usage:

```
{
  "input_tokens": 2136,
  "output_tokens": 345,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.18 returned, visible payoff \$7.18.
- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.

Your total visible earnings across all rounds are \$36.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5

Round 5: I sent, they returned \$6 (~51%)

Round 6: They sent \$3, I returned \$5.4

We've maintained excellent cooperation throughout all rounds. Both of us continue to send meaningful amounts and return fair portions. This partnership has been mutually beneficial.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 4: They sent \$3, I returned \$5 (~56%)

Round 5: I sent (amount visible shows they saw \$3.92), they returned \$6.0 (~51%)

Round 6: They sent \$3, I returned \$5.4 (~60%)

Round 7: I'm the sender now

We've maintained a strong cooperative relationship throughout all rounds. Both of us have been consistently sending around \$3 and returning around 50-60% of what we receive. This has been mutually beneficial.

I should continue this successful cooperative strategy.

usage:

```
{
  "input_tokens": 2471,
  "output_tokens": 250,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$6.0, visible payoff \$5.76.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.

- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.
Your total visible earnings across all rounds are \$44.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)
Round 6: I returned \$5.4
Round 7: They returned \$4.8 (~50%)
Round 8: They sent \$2.6, I received \$7.8

My co-player has maintained consistent cooperation throughout all rounds. They sent slightly less this round but still demonstrated trust. I'll maintain the reciprocal pattern.

```
'''json
{"return": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 5: They returned \$6.0 (~51%)
Round 6: I returned \$5.4 (~60% - I was generous)
Round 7: They returned \$4.8 (~50%)
Round 8: They sent \$2.6, I received \$7.8

My co-player has been consistently cooperative throughout all rounds. They sent a bit less this round (\$2.6 vs previous \$3), but still showing trust. I should continue the pattern of fair reciprocity, returning around 50% of what I received.

50% of \$7.8 = \$3.9

I'll return around \$4 to maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 2683,
  "output_tokens": 289,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.67 returned, visible payoff \$6.67.
- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.63 returned, visible payoff \$6.63.

Your total visible earnings across all rounds are \$47.95.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history with my current co-player:

Round 6: They sent \$3, I returned \$5.4
Round 7: I sent, they returned \$4.8 (~50%)
Round 8: They sent \$3, I returned \$4

This co-player has consistently sent \$3 and returned fair amounts around 50%. We've maintained strong cooperation throughout. I'll continue this successful pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 6: They sent \$3, I returned \$5.4

Round 7: I sent, they returned \$4.8 (~50%)

Round 8: They sent \$3 (though visible to me as \$2.6 due to noise), I returned \$4

Wait, let me re-examine. In round 8, it says "They sent you \$2.6" but looking at the visible history now, it shows "Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.63 returned".

So they actually sent \$3, but I saw it as \$2.6 due to communication noise. They saw I returned \$4.63, but I actually returned \$4.

This co-player has been consistently cooperative:

- They keep sending \$3
- They return fair amounts (~50%)

I should continue with my cooperative strategy of sending \$3.

My total visible earnings are \$47.95, which shows this has been a successful partnership.

usage:

```
{
  "input_tokens": 2734,
  "output_tokens": 355,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.19 sent, it became \$9.57, returned \$4.8, visible payoff \$4.7700000000000005.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.63 returned, visible payoff \$6.63.
- Round 9 against you, as RECEIVER: saw \$3.04 sent, it became \$9.120000000000001, returned \$4.6, visible payoff \$4.520000000000001.

Your total visible earnings across all rounds are \$54.09.

This round, you are the RECEIVER against your current co-player. They sent you \$2.17, so you received \$6.51. How much do you return? (0-6.51)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with my current co-player:

Round 7: They returned \$4.8 (~50%)

Round 8: I returned \$4 (~44%)

Round 9: They returned \$4.6 (~50%)

Round 10: They sent \$2.17, I received \$6.51

My co-player has maintained consistent cooperation with ~50% return rates. They sent less this round but still demonstrated trust. I'll reciprocate fairly.

```
'''json
{"return": 3.3}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 7: They returned \$4.8 (~50%)

Round 8: I returned \$4 (~44%)

Round 9: They returned \$4.6 (~50%)

Round 10: They sent \$2.17, I received \$6.51

My co-player has been consistently cooperative, maintaining around 50% return rates. They sent less this round (\$2.17 vs their usual \$3), but still showed trust. I should reciprocate fairly.

For \$6.51 received, returning around 50% would be about \$3.25.

Let me return a fair amount proportional to what was received.

```
Usage: {  
  "input_tokens": 2795,  
  "output_tokens": 284,  
  "reasoning_tokens": 0  
}
```